Isolated Footing Design (IS 456-2000)

Design For Isolated Footing 1

Design For Isolated Footing 2

Design For Isolated Footing 3

Design For Isolated Footing 4

Design For Isolated Footing 5

Design For Isolated Footing 6

Design For Isolated Footing 37

Design For Isolated Footing 38

Design For Isolated Footing 39

Design For Isolated Footing 40

Design For Isolated Footing 41

Design For Isolated Footing 42

Design For Isolated Footing 73

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Design For Isolated Footing 113

Design For Isolated Footing 114

Design For Isolated Footing 145

Design For Isolated Footing 146

Design For Isolated Footing 147

Design For Isolated Footing 148

Design For Isolated Footing 149

Design For Isolated Footing 150

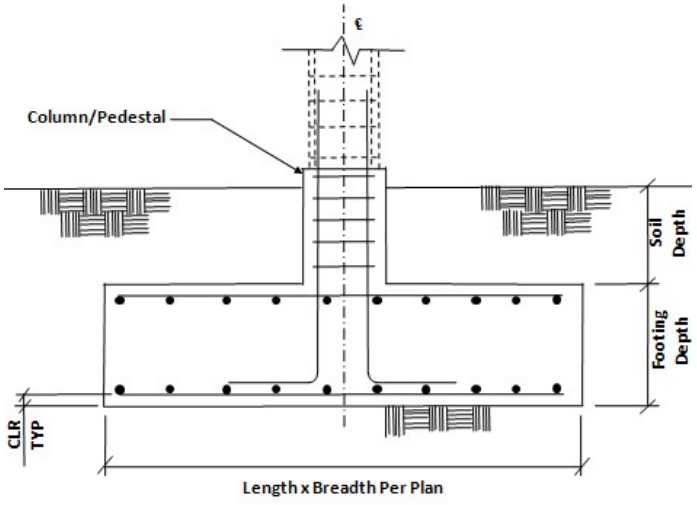
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness

1	1	3.050 m	3.050 m	0.356 m
2	2	3.600 m	3.600 m	0.456 m
3	3	3.650 m	3.650 m	0.456 m
4	4	3.650 m	3.650 m	0.456 m
5	5	3.600 m	3.600 m	0.456 m
6	6	3.400 m	3.400 m	0.406 m
37	7	3.400 m	3.400 m	0.406 m
38	8	4.050 m	4.050 m	0.507 m
39	9	4.100 m	4.100 m	0.507 m
40	10	4.100 m	4.100 m	0.507 m
41	11	4.050 m	4.050 m	0.507 m
42	12	3.850 m	3.850 m	0.507 m
73	13	3.400 m	3.400 m	0.406 m
74	14	4.050 m	4.050 m	0.507 m
75	15	4.100 m	4.100 m	0.507 m
76	16	4.100 m	4.100 m	0.507 m
77	17	4.050 m	4.050 m	0.507 m
78	18	3.850 m	3.850 m	0.507 m
109	19	3.400 m	3.400 m	0.406 m
110	20	4.050 m	4.050 m	0.507 m
111	21	4.050 m	4.050 m	0.507 m
112	22	4.050 m	4.050 m	0.507 m
113	23	4.050 m	4.050 m	0.507 m
114	24	3.800 m	3.800 m	0.507 m
145	25	3.450 m	3.450 m	0.456 m
146	26	4.000 m	4.000 m	0.507 m
147	27	4.050 m	4.050 m	0.507 m
148	28	4.050 m	4.050 m	0.507 m
149	29	4.050 m	4.050 m	0.507 m
150	30	3.850 m	3.850 m	0.507 m

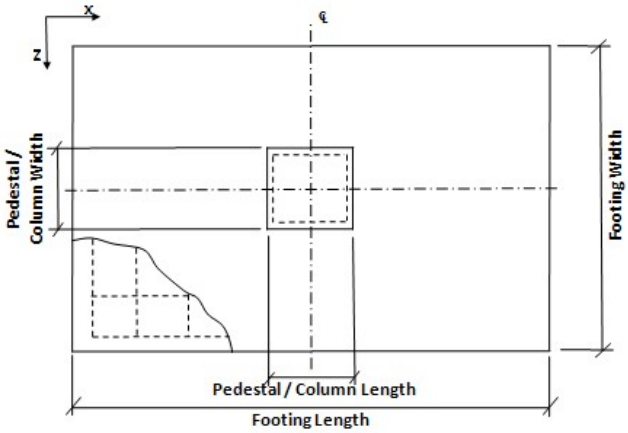
Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
1	Ø8 @ 65 mm c/c	Ø8 @ 55 mm c/c	Ø6 @ 50 mm c/c	Ø6 @ 75 mm c/c	N/A	N/A
2	Ø10 @ 75 mm c/c	Ø10 @ 65 mm c/c	Ø8 @ 50 mm c/c	Ø6 @ 75 mm c/c	N/A	N/A
3	Ø10 @ 70 mm c/c	Ø10 @ 65 mm c/c	Ø8 @ 50 mm c/c	Ø6 @ 75 mm c/c	N/A	N/A
4	Ø10 @ 70 mm c/c	Ø10 @ 65 mm c/c	Ø8 @ 50 mm c/c	Ø6 @ 75 mm c/c	N/A	N/A
5	Ø10 @ 70 mm c/c	Ø10 @ 65 mm c/c	Ø8 @ 50 mm c/c	Ø6 @ 75 mm c/c	N/A	N/A
6	Ø10 @ 70 mm c/c	Ø10 @ 60 mm c/c	Ø8 @ 60 mm c/c	Ø6 @ 70 mm c/c	N/A	N/A
37	Ø10 @ 70 mm c/c	Ø10 @ 60 mm c/c	Ø8 @ 60 mm c/c	Ø6 @ 70 mm c/c	N/A	N/A
38	Ø10 @ 60 mm c/c	Ø10 @ 55 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 60 mm c/c	N/A	N/A
39	Ø10 @ 60 mm c/c	Ø10 @ 55 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 65 mm c/c	N/A	N/A
40	Ø10 @ 60 mm c/c	Ø10 @ 55 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 65 mm c/c	N/A	N/A
41	Ø10 @ 60 mm c/c	Ø10 @ 55 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 60 mm c/c	N/A	N/A
42	Ø10 @ 65 mm c/c	Ø10 @ 60 mm c/c	Ø10 @ 65 mm c/c	Ø6 @ 75 mm c/c	N/A	N/A
73	Ø10 @ 70 mm c/c	Ø10 @ 60 mm c/c	Ø8 @ 60 mm c/c	Ø6 @ 70 mm c/c	N/A	N/A
74	Ø10 @ 60 mm c/c	Ø10 @ 55 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 60 mm c/c	N/A	N/A
75	Ø10 @ 60 mm c/c	Ø10 @ 50 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 65 mm c/c	N/A	N/A
76	Ø10 @ 60 mm c/c	Ø10 @ 50 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 65 mm c/c	N/A	N/A
77	Ø10 @ 60 mm c/c	Ø10 @ 55 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 60 mm c/c	N/A	N/A
78	Ø10 @ 65 mm c/c	Ø10 @ 60 mm c/c	Ø10 @ 65 mm c/c	Ø6 @ 75 mm c/c	N/A	N/A
109	Ø10 @ 70 mm c/c	Ø10 @ 60 mm c/c	Ø8 @ 60 mm c/c	Ø6 @ 70 mm c/c	N/A	N/A
110	Ø10 @ 60 mm c/c	Ø10 @ 55 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 60 mm c/c	N/A	N/A
111	Ø10 @ 60 mm c/c	Ø10 @ 55 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 60 mm c/c	N/A	N/A
112	Ø10 @ 60 mm c/c	Ø10 @ 55 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 60 mm c/c	N/A	N/A
113	Ø10 @ 60 mm c/c	Ø10 @ 55 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 60 mm c/c	N/A	N/A
114	Ø10 @ 65 mm c/c	Ø10 @ 60 mm c/c	Ø10 @ 65 mm c/c	Ø6 @ 75 mm c/c	N/A	N/A
145	Ø8 @ 50 mm c/c	Ø10 @ 70 mm c/c	Ø8 @ 55 mm c/c	Ø6 @ 75 mm c/c	N/A	N/A
146	Ø10 @ 60 mm c/c	Ø10 @ 55 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 60 mm c/c	N/A	N/A
147	Ø10 @ 60 mm c/c	Ø10 @ 55 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 60 mm c/c	N/A	N/A

148	Ø10 @ 60 mm c/c	Ø10 @ 55 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 60 mm c/c	N/A	N/A
149	Ø10 @ 60 mm c/c	Ø10 @ 55 mm c/c	Ø10 @ 70 mm c/c	Ø6 @ 60 mm c/c	N/A	N/A
150	Ø10 @ 65 mm c/c	Ø10 @ 60 mm c/c	Ø10 @ 65 mm c/c	Ø6 @ 75 mm c/c	N/A	N/A

Isolated Footing 1



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type : Calculate Dimension
Footing Thickness (Ft) : 305.000 mm
Footing Length - X (Fl) : 1000.000 mm
Footing Width - Z (Fw) : 1000.000 mm
Eccentricity along X (Oxd) : 0.000 mm
Eccentricity along Z (Ozd) : 0.000 mm

Column Dimensions

Column Shape : Rectangular

Column Length - X (Pl) : 0.600 m
Column Width - Z (Pw) : 0.400 m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 25.000 kN/m3
Strength of Concrete : 25.000 N/mm2
Yield Strength of Steel : 415.000 N/mm2
Minimum Bar Size : Ø6
Maximum Bar Size : Ø32
Minimum Bar Spacing : 50.000 mm
Maximum Bar Spacing : 500.000 mm
Pedestal Clear Cover (P, CL) : 50.000 mm
Footing Clear Cover (F, CL) : 50.000 mm

Soil Properties

Soil Type : Drained
Unit Weight : 22.000 kN/m3
Soil Bearing Capacity : 100.000 kN/m2
Soil Surcharge : 0.000 kN/m2
Depth of Soil above Footing : 0.000 mm
Cohesion : 0.000 kN/m2
Min Percentage of Slab : 0.000

Sliding and Overturning

Coefficient of Friction : 0.500
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**

101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL

Load Combination/s- Strength Level

Load Combination Number	Load Combination Title
201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	543.985	26.694	30.648	61.510	-67.343
102	435.188	21.355	24.519	49.208	-53.874

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	815.978	40.041	45.973	92.265	-101.015
202	522.226	25.626	29.422	59.049	-64.649
203	652.782	32.033	36.778	73.812	-80.812
204	489.587	24.025	27.584	55.359	-60.609

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 5.516 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

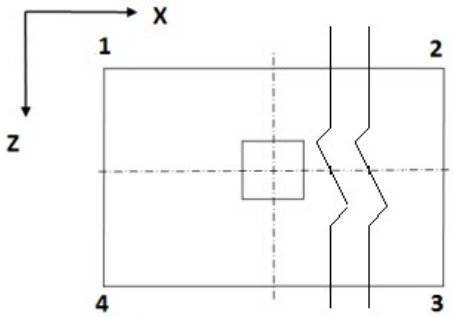
Length (L_2) = 3.050 m Governing Load Case : # 101

Width (W_2) = 3.050 m Governing Load Case : # 101

Depth (D_2) = 0.356 m Governing Load Case : # 101

Area (A_2) = 9.303 m^2

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**35.1546**	67.0809	97.0500	65.1238	0.000
101	35.1546	**67.0809**	97.0500	65.1238	0.000
101	35.1546	67.0809	**97.0500**	65.1238	0.000
101	35.1546	67.0809	97.0500	**65.1238**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**35.1546**	67.0809	97.0500	65.1238
101	35.1546	**67.0809**	97.0500	65.1238
101	35.1546	67.0809	**97.0500**	65.1238
101	35.1546	67.0809	97.0500	**65.1238**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 9.303 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	11.518	10.032	13.234	12.423
102	11.850	10.321	13.616	12.781

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 26.694 kN
Governing Restoring Force : 307.458 kN
Minimum Sliding Ratio for the Critical Load Case : 11.518
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 70.857 kNm
Governing Resisting Moment : 937.731 kNm
Minimum Overturning Ratio for the Critical Load Case : 13.234

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 30.648 kN
Governing Restoring Force : 307.458 kN
Minimum Sliding Ratio for the Critical Load Case : 10.032
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -75.485 kNm
Governing Resisting Moment : 937.731 kNm
Minimum Overturning Ratio for the Critical Load Case : 12.423

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #201

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.302 m

Governing moment (M_u) = 277.620 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 958.088562 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #201

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.302 m

Governing moment (M_u) = 240.892 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

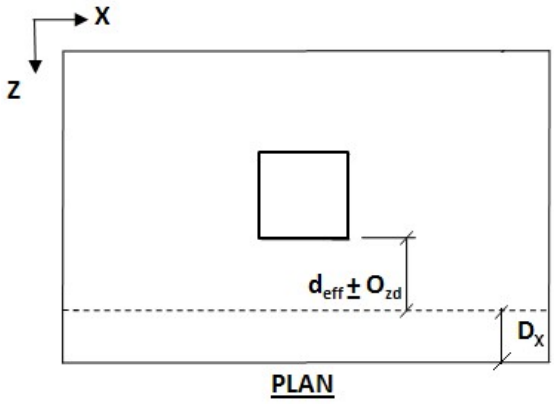
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 958.088562 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.302 m

Shear Force(S) = 320.296 kN

Shear Stress(T_v) = 347.731553 kN/m2

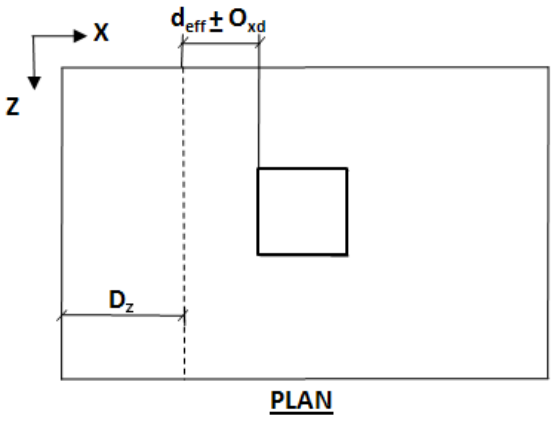
Percentage Of Steel(P_t) = 0.2503

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 364.910 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.302 m

Shear Force(S) = 293.942 kN

Shear Stress(T_v) = 319.121086 kN/m2

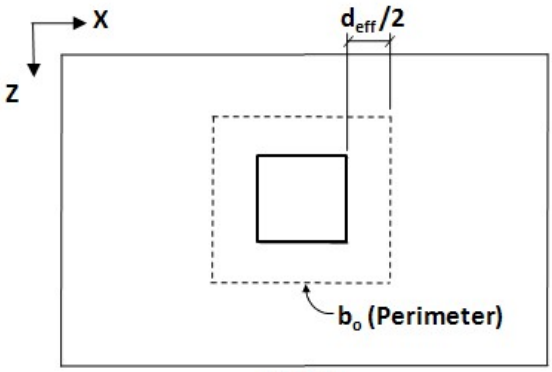
Percentage Of Steel(P_t) = 0.2904

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 389.179 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 760.436 kN

Shear Stress(T_v) = 784.913 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 25 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.007 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.175 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 25 mm

As Per IS 456 2000 Clause 26.2.1

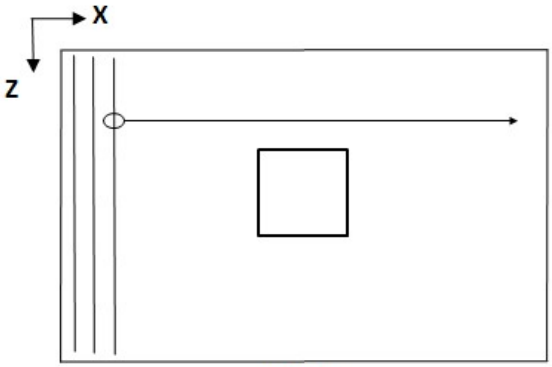
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.007 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.275 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #201

Minimum Area of Steel (A_{stmin}) = 1302.960 mm2

Calculated Area of Steel (A_{st}) = 2675.118 mm2

Provided Area of Steel ($A_{st,Provided}$) = 2675.118 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø8

Minimum spacing allowed (S_{min}) = 48.000 mm

Selected spacing (S) = 55.509 mm

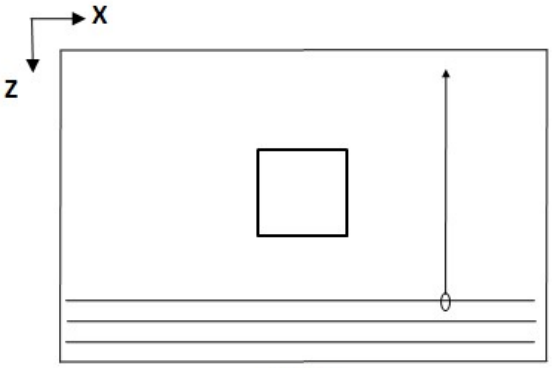
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø8 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #201

Minimum Area of Steel (A_{stmin}) = 1302.960 mm2

Calculated Area of Steel (A_{st}) = 2305.067 mm2

Provided Area of Steel ($A_{st,Provided}$) = 2305.067 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø8

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 65.378 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

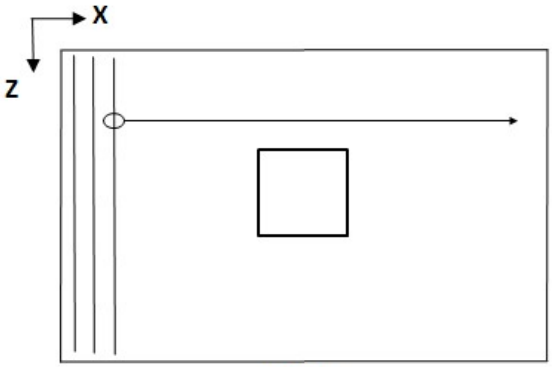
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø8 @ 65.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1302.960 mm2

Calculated Area of Steel (A_{st}) = 1116.300 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1302.960 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 22.683 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 75.487 mm

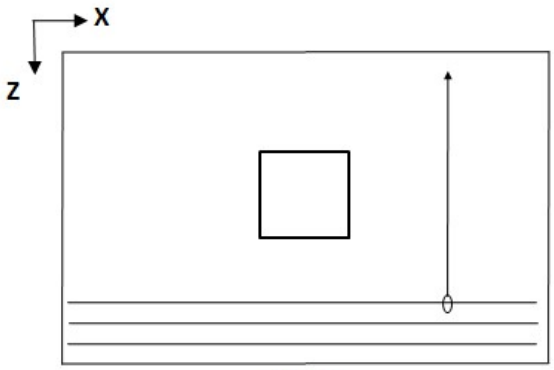
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 75 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1302.960 mm2

Calculated Area of Steel (A_{st}) = 1647.200 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1647.200 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 19.388 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 50.759 mm

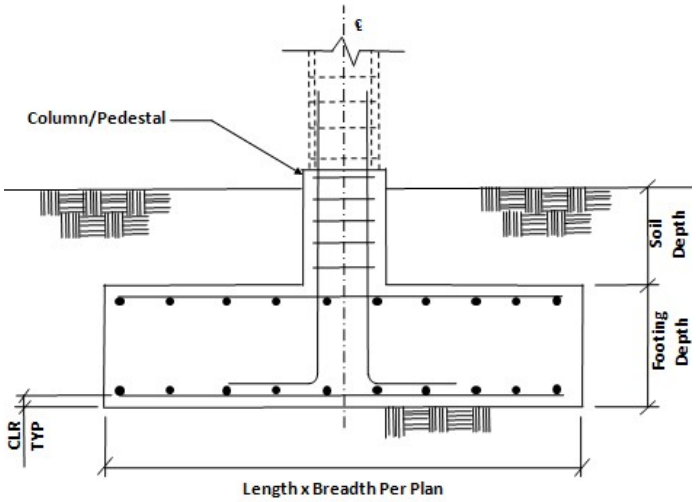
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

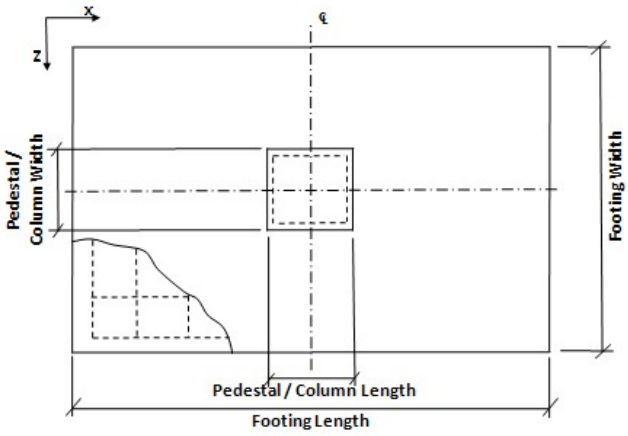
Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 50 mm o.c.

Isolated Footing 2



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	903.905	35.630	32.969	61.777	-79.001
102	723.124	28.504	26.376	49.422	-63.201

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1355.858	53.445	49.454	92.666	-118.502
202	867.749	34.205	31.651	59.306	-75.841
203	1084.687	42.756	39.563	74.133	-94.802
204	813.515	32.067	29.673	55.600	-71.101

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

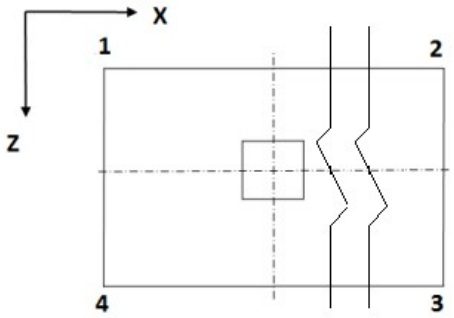
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 9.115 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	3.600	m	Governing Load Case :	# 101
Width (W_2) =	3.600	m	Governing Load Case :	# 101
Depth (D_2) =	0.456	m	Governing Load Case :	# 101
Area (A_2) =	12.960	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**56.5755**	79.6902	98.1661	75.0514	0.000
101	56.5755	**79.6902**	98.1661	75.0514	0.000
101	56.5755	79.6902	**98.1661**	75.0514	0.000
101	56.5755	79.6902	98.1661	**75.0514**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**56.5755**	79.6902	98.1661	75.0514
101	56.5755	**79.6902**	98.1661	75.0514
101	56.5755	79.6902	**98.1661**	75.0514
101	56.5755	79.6902	98.1661	**75.0514**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 12.960 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	14.071	15.207	25.126	20.084

102	14.418	15.582	25.745	20.578

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 35.630 kN
Governing Restoring Force : 501.363 kN
Minimum Sliding Ratio for the Critical Load Case : 14.071
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 71.833 kNm
Governing Resisting Moment : 1804.873 kNm
Minimum Overturning Ratio for the Critical Load Case : 25.126

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 32.969 kN
Governing Restoring Force : 501.363 kN
Minimum Sliding Ratio for the Critical Load Case : 15.207
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -89.868 kNm
Governing Resisting Moment : 1804.873 kNm
Minimum Overturning Ratio for the Critical Load Case : 20.084

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.302 m

Governing moment (M_u) = 421.629 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1130.858631 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.302 m

Governing moment (M_u) = 379.558 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

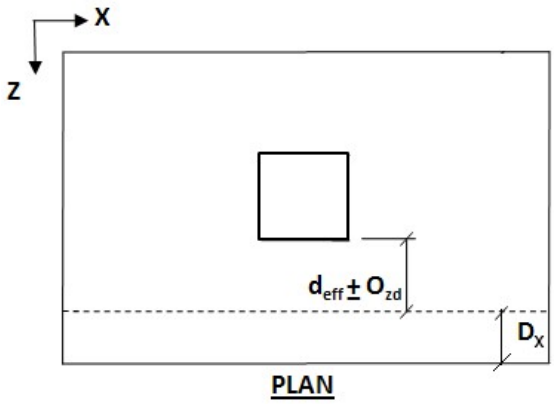
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1130.858631 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.402 m

Shear Force(S) = 491.074 kN

Shear Stress(T_v) = 339.327035 kN/m2

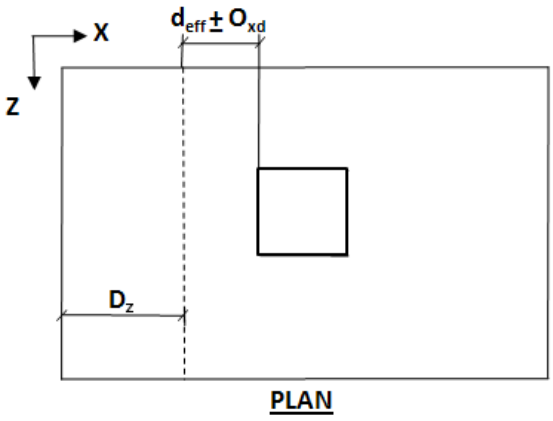
Percentage Of Steel(P_t) = 0.2350

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 355.077 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.402 m

Shear Force(S) = 461.162 kN

Shear Stress(T_v) = 318.658217 kN/m2

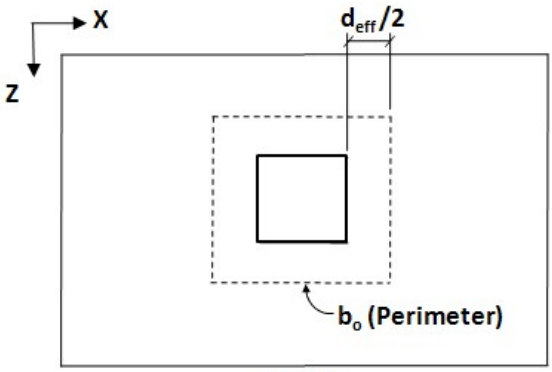
Percentage Of Steel(P_t) = 0.2623

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 372.460 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1271.786 kN

Shear Stress(T_v) = 876.842 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.450 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

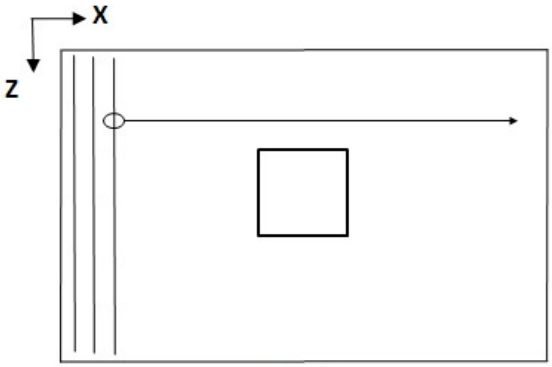
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.550 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1542.240 mm2

Calculated Area of Steel (A_{st}) = 4126.962 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4126.962 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 67.115 mm

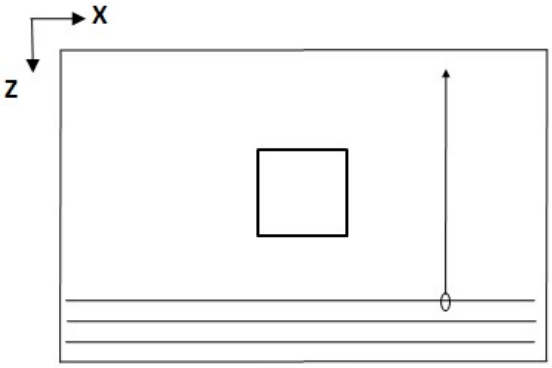
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 65.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1542.240 mm2

Calculated Area of Steel (A_{st}) = 3688.830 mm2

Provided Area of Steel ($A_{st,Provided}$) = 3688.830 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 75.870 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

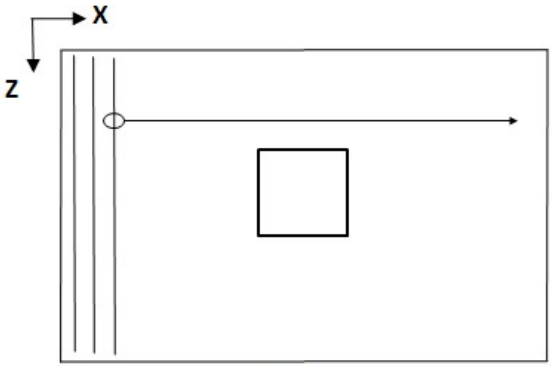
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 75.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1542.240 mm2

Calculated Area of Steel (A_{st}) = 1317.600 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1542.240 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 39.039 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 75.957 mm

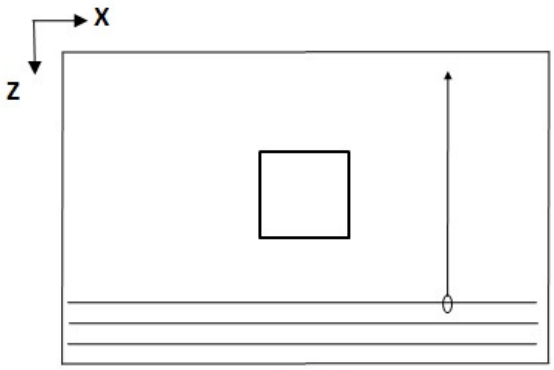
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 75 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1542.240 mm2

Calculated Area of Steel (A_{st}) = 3331.957 mm2

Provided Area of Steel ($A_{st,Provided}$) = 3331.957 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 34.312 kNm

Selected bar Size (d_b) = Ø8

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 52.909 mm

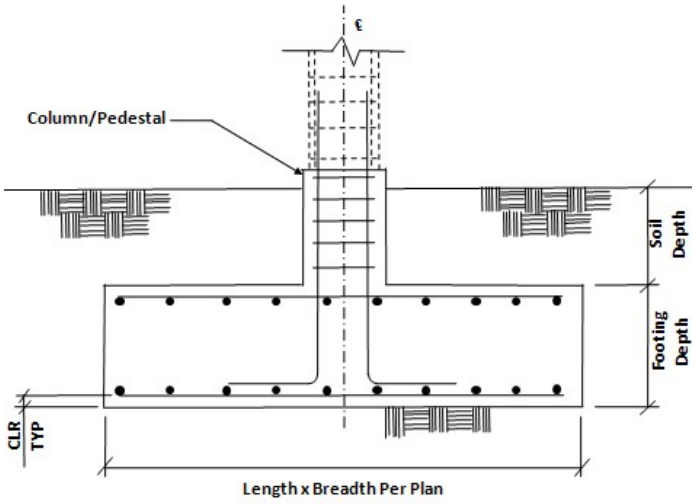
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

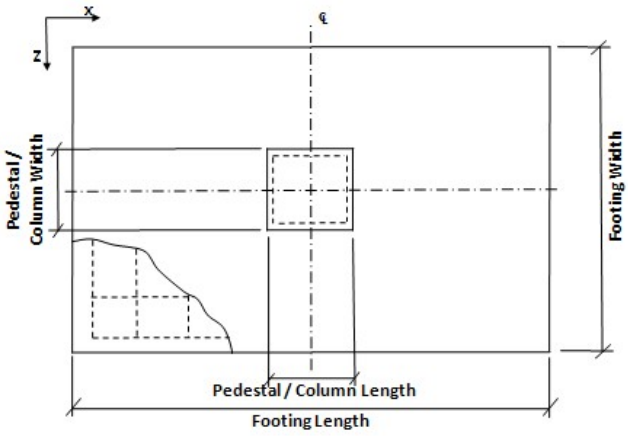
Based on spacing reinforcement increment; provided reinforcement is

Ø8 @ 50 mm o.c.

Isolated Footing 3



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	931.069	36.040	32.939	61.755	-79.462
102	744.855	28.832	26.352	49.404	-63.569

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1396.604	54.059	49.409	92.633	-119.192
202	893.826	34.598	31.622	59.285	-76.283
203	1117.283	43.248	39.527	74.107	-95.354
204	837.962	32.436	29.645	55.580	-71.515

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

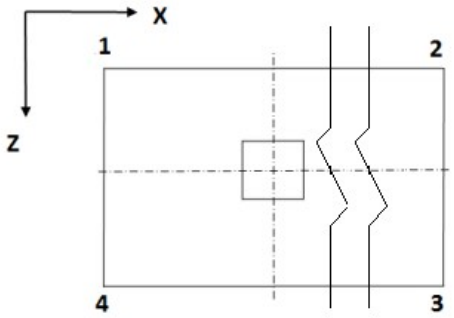
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 9.387 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	3.650	m	Governing Load Case :	# 101
Width (W_2) =	3.650	m	Governing Load Case :	# 101
Depth (D_2) =	0.456	m	Governing Load Case :	# 101
Area (A_2) =	13.323	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**57.4913**	79.8134	97.5327	75.2105	0.000
101	57.4913	**79.8134**	97.5327	75.2105	0.000
101	57.4913	79.8134	**97.5327**	75.2105	0.000
101	57.4913	79.8134	97.5327	**75.2105**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**57.4913**	79.8134	97.5327	75.2105
101	57.4913	**79.8134**	97.5327	75.2105
101	57.4913	79.8134	**97.5327**	75.2105
101	57.4913	79.8134	97.5327	**75.2105**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 13.323 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	14.327	15.675	26.247	20.835

102	14.679	16.061	26.892	21.347

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 36.040 kN
Governing Restoring Force : 516.327 kN
Minimum Sliding Ratio for the Critical Load Case : 14.327
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 71.802 kNm
Governing Resisting Moment : 1884.558 kNm
Minimum Overturning Ratio for the Critical Load Case : 26.247

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 32.939 kN
Governing Restoring Force : 516.327 kN
Minimum Sliding Ratio for the Critical Load Case : 15.675
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -90.453 kNm
Governing Resisting Moment : 1884.558 kNm
Minimum Overturning Ratio for the Critical Load Case : 20.835

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.302 m

Governing moment (M_u) = 440.198 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1146.565000 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.302 m

Governing moment (M_u) = 396.981 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

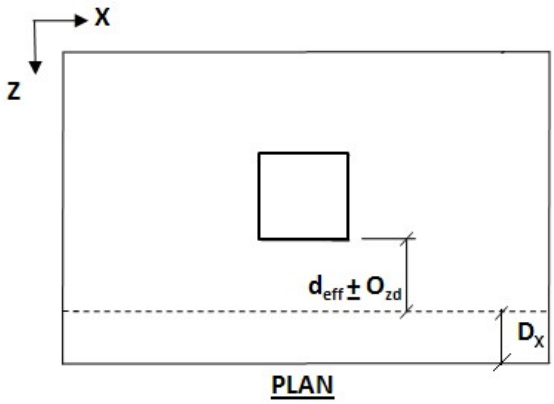
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1146.565000 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.402 m

Shear Force(S) = 507.404 kN

Shear Stress(T_v) = 345.807893 kN/m2

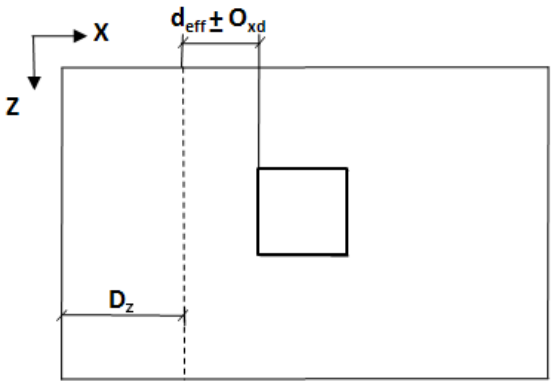
Percentage Of Steel(P_t) = 0.2428

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 360.137 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.402 m

Shear Force(S) = 477.204 kN

Shear Stress(T_v) = 325.226212 kN/m2

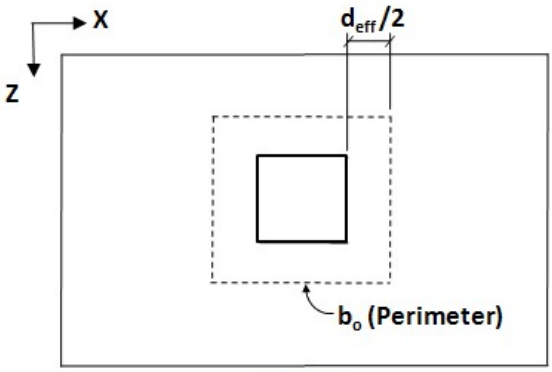
Percentage Of Steel(P_t) = 0.2705

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 377.448 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1312.362 kN

Shear Stress(T_v) = 904.817 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.475 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

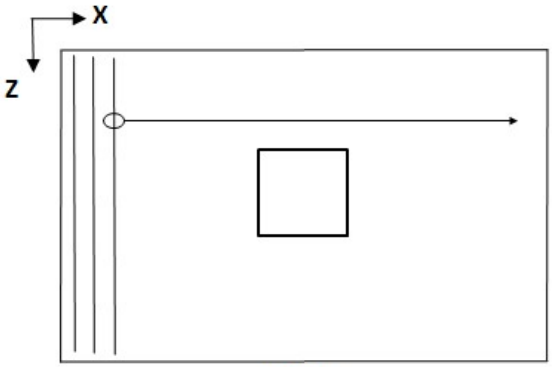
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.575 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1563.660 mm2

Calculated Area of Steel (A_{st}) = 4317.997 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4317.997 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 65.556 mm

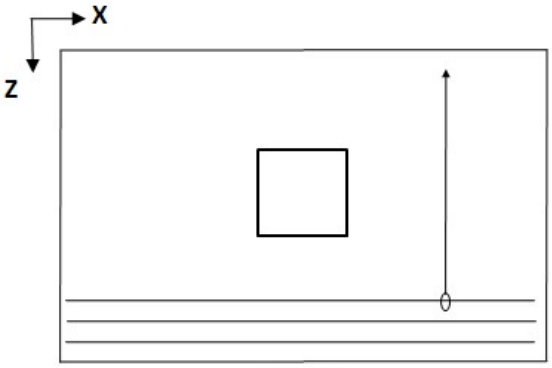
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 65.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1563.660 mm2

Calculated Area of Steel (A_{st}) = 3865.922 mm2

Provided Area of Steel ($A_{st,Provided}$) = 3865.922 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 72.245 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

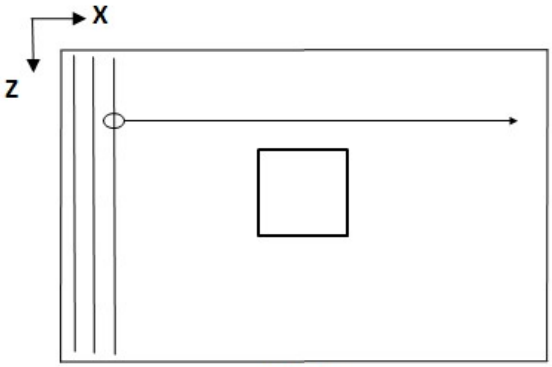
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1563.660 mm2

Calculated Area of Steel (A_{st}) = 1335.900 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1563.660 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 40.828 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 75.404 mm

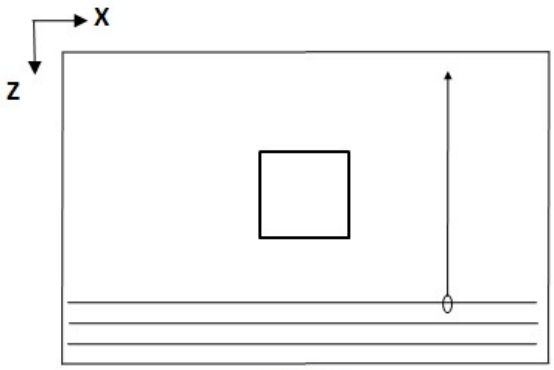
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 75 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1563.660 mm2

Calculated Area of Steel (A_{st}) = 3492.577 mm2

Provided Area of Steel ($A_{st,Provided}$) = 3492.577 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 35.958 kNm

Selected bar Size (d_b) = Ø8

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 51.333 mm

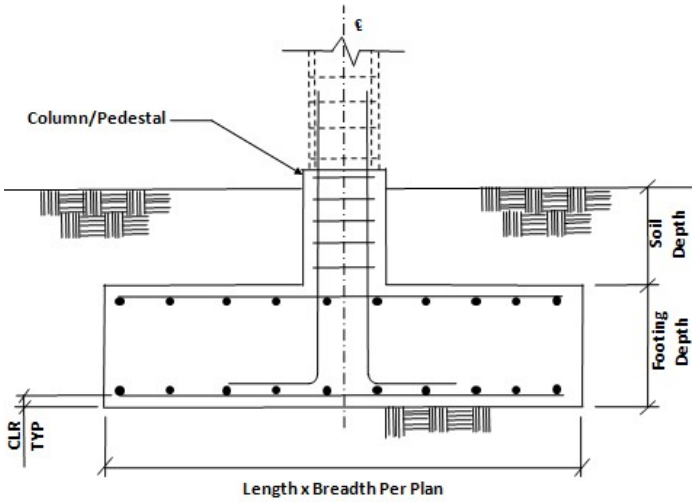
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

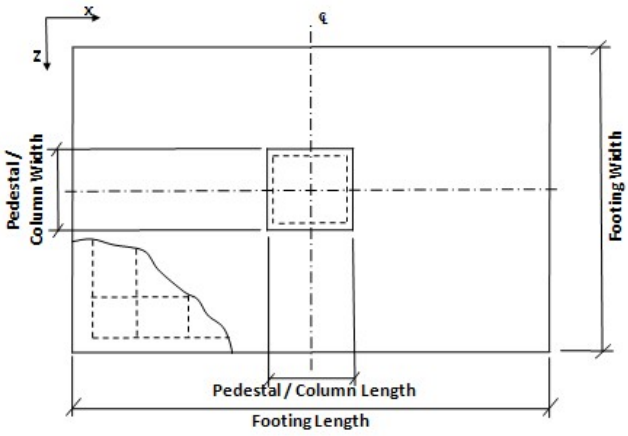
Based on spacing reinforcement increment; provided reinforcement is

Ø8 @ 50 mm o.c.

Isolated Footing 4



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	930.976	36.107	32.931	61.745	-79.515
102	744.781	28.886	26.345	49.396	-63.612

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1396.464	54.160	49.396	92.617	-119.273
202	893.737	34.663	31.614	59.275	-76.334
203	1117.171	43.328	39.517	74.093	-95.418
204	837.879	32.496	29.638	55.570	-71.564

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

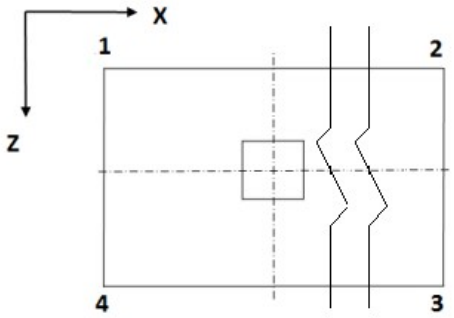
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 9.386 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	3.650	m	Governing Load Case :	# 101
Width (W_2) =	3.650	m	Governing Load Case :	# 101
Depth (D_2) =	0.456	m	Governing Load Case :	# 101
Area (A_2) =	13.323	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**57.4768**	79.8172	97.5332	75.1928	0.000
101	57.4768	**79.8172**	97.5332	75.1928	0.000
101	57.4768	79.8172	**97.5332**	75.1928	0.000
101	57.4768	79.8172	97.5332	**75.1928**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**57.4768**	79.8172	97.5332	75.1928
101	57.4768	**79.8172**	97.5332	75.1928
101	57.4768	79.8172	**97.5332**	75.1928
101	57.4768	79.8172	97.5332	**75.1928**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 13.323 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	14.299	15.678	26.249	20.816

102	14.650	16.063	26.895	21.328

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 36.107 kN
Governing Restoring Force : 516.280 kN
Minimum Sliding Ratio for the Critical Load Case : 14.299
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 71.788 kNm
Governing Resisting Moment : 1884.388 kNm
Minimum Overturning Ratio for the Critical Load Case : 26.249

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 32.931 kN
Governing Restoring Force : 516.280 kN
Minimum Sliding Ratio for the Critical Load Case : 15.678
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -90.527 kNm
Governing Resisting Moment : 1884.388 kNm
Minimum Overturning Ratio for the Critical Load Case : 20.816

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.302 m

Governing moment (M_u) = 440.151 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1146.565000 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.302 m

Governing moment (M_u) = 396.979 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

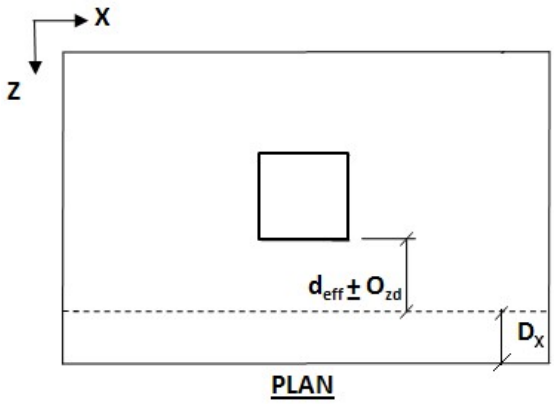
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1146.565000 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.402 m

Shear Force(S) = 507.350 kN

Shear Stress(T_v) = 345.771003 kN/m2

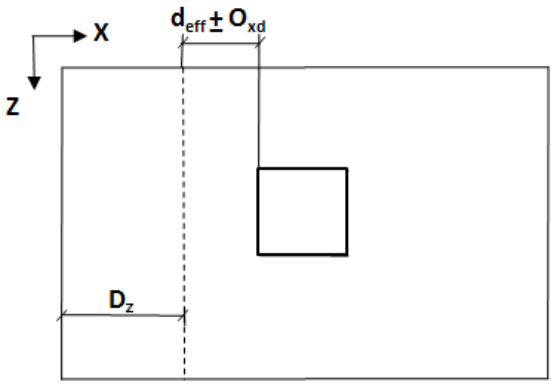
Percentage Of Steel(P_t) = 0.2428

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 360.136 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.402 m

Shear Force(S) = 477.200 kN

Shear Stress(T_v) = 325.223509 kN/m2

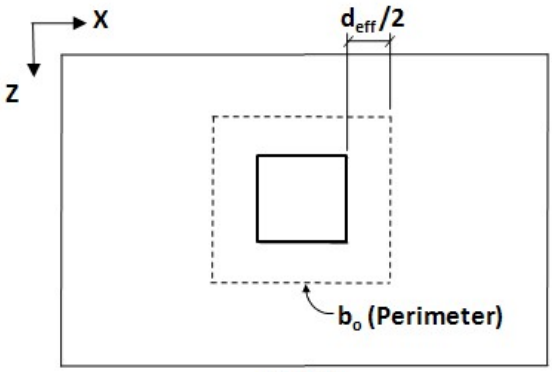
Percentage Of Steel(P_t) = 0.2705

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 377.429 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1312.231 kN

Shear Stress(T_v) = 904.727 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.475 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

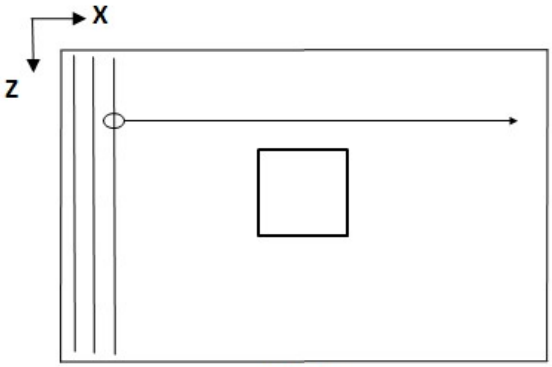
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.575 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1563.660 mm2

Calculated Area of Steel (A_{st}) = 4317.500 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4317.500 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 65.556 mm

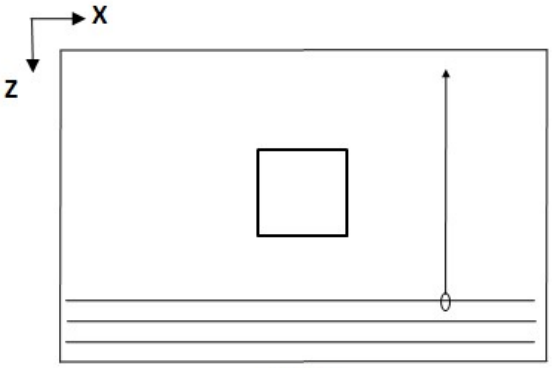
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 65.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1563.660 mm2

Calculated Area of Steel (A_{st}) = 3865.902 mm2

Provided Area of Steel ($A_{st,Provided}$) = 3865.902 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 72.245 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

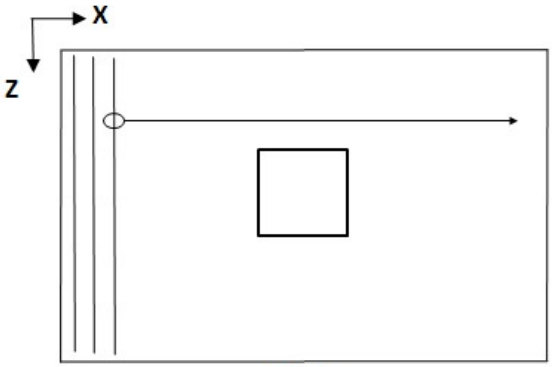
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1563.660 mm2

Calculated Area of Steel (A_{st}) = 1335.900 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1563.660 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 40.828 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 75.404 mm

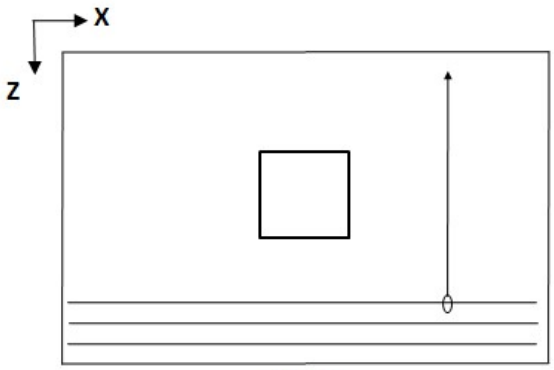
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 75 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1563.660 mm2

Calculated Area of Steel (A_{st}) = 3492.559 mm2

Provided Area of Steel ($A_{st,Provided}$) = 3492.559 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 35.958 kNm

Selected bar Size (d_b) = Ø8

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 51.333 mm

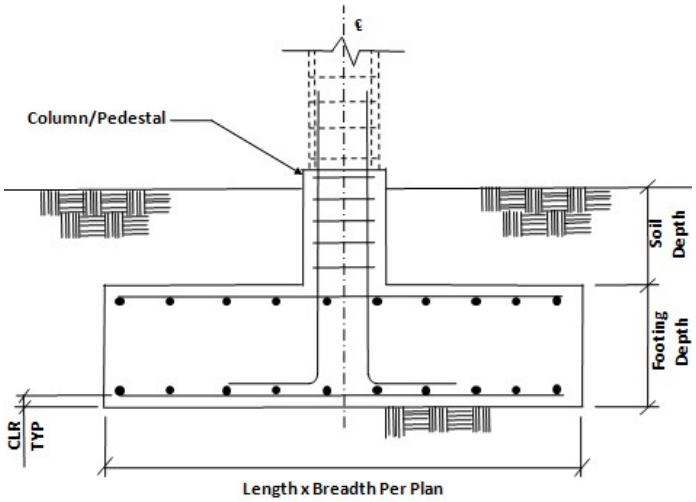
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

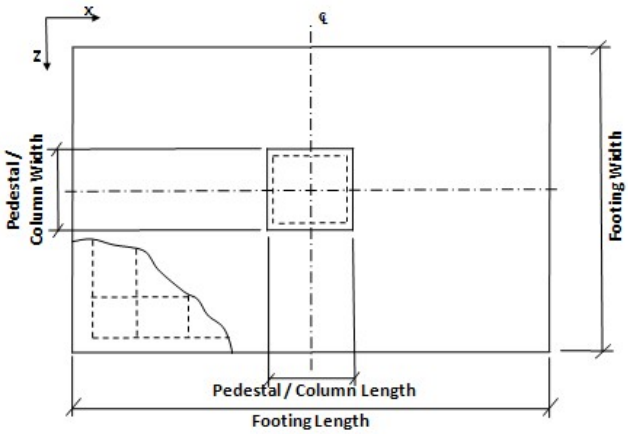
Based on spacing reinforcement increment; provided reinforcement is

Ø8 @ 50 mm o.c.

Isolated Footing 5



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	909.192	36.706	32.967	61.761	-80.089
102	727.354	29.365	26.374	49.409	-64.071

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1363.789	55.059	49.451	92.642	-120.133
202	872.825	35.238	31.648	59.291	-76.885
203	1091.031	44.047	39.561	74.114	-96.107
204	818.273	33.035	29.670	55.585	-72.080

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

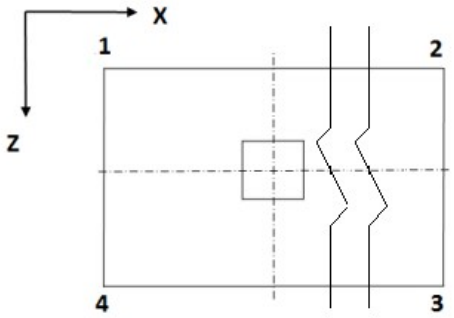
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 9.168 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	3.600	m	Governing Load Case :	# 101
Width (W_2) =	3.600	m	Governing Load Case :	# 101
Depth (D_2) =	0.456	m	Governing Load Case :	# 101
Area (A_2) =	12.960	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**56.8035**	80.2824	98.7540	75.2751	0.000
101	56.8035	**80.2824**	98.7540	75.2751	0.000
101	56.8035	80.2824	**98.7540**	75.2751	0.000
101	56.8035	80.2824	98.7540	**75.2751**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**56.8035**	80.2824	98.7540	75.2751
101	56.8035	**80.2824**	98.7540	75.2751
101	56.8035	80.2824	**98.7540**	75.2751
101	56.8035	80.2824	98.7540	**75.2751**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 12.960 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	13.731	15.288	25.264	19.876

102	14.067	15.663	25.884	20.363

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 36.706 kN
Governing Restoring Force : 504.006 kN
Minimum Sliding Ratio for the Critical Load Case : 13.731
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 71.816 kNm
Governing Resisting Moment : 1814.389 kNm
Minimum Overturning Ratio for the Critical Load Case : 25.264

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 32.967 kN
Governing Restoring Force : 504.006 kN
Minimum Sliding Ratio for the Critical Load Case : 15.288
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -91.284 kNm
Governing Resisting Moment : 1814.389 kNm
Minimum Overturning Ratio for the Critical Load Case : 19.876

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.302 m

Governing moment (M_u) = 423.876 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1130.858631 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.302 m

Governing moment (M_u) = 382.180 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

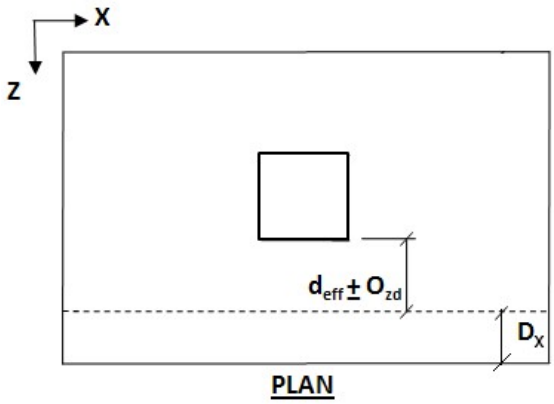
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1130.858631 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.402 m

Shear Force(S) = 493.704 kN

Shear Stress(T_v) = 341.144174 kN/m2

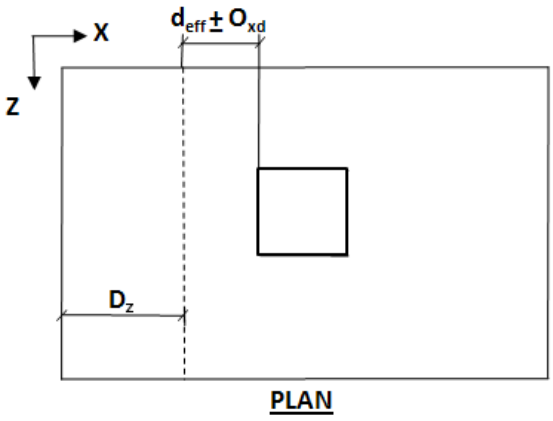
Percentage Of Steel(P_t) = 0.2367

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 356.191 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.402 m

Shear Force(S) = 464.331 kN

Shear Stress(T_v) = 320.848009 kN/m2

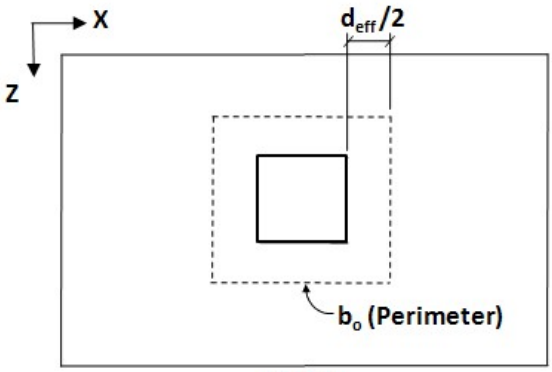
Percentage Of Steel(P_t) = 0.2638

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 373.361 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1279.225 kN

Shear Stress(T_v) = 881.971 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.450 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

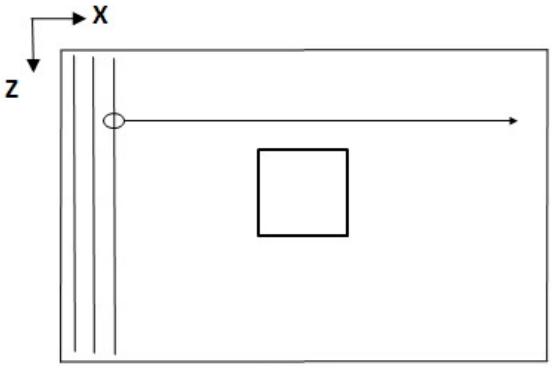
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.550 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1542.240 mm2

Calculated Area of Steel (A_{st}) = 4150.555 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4150.555 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 67.115 mm

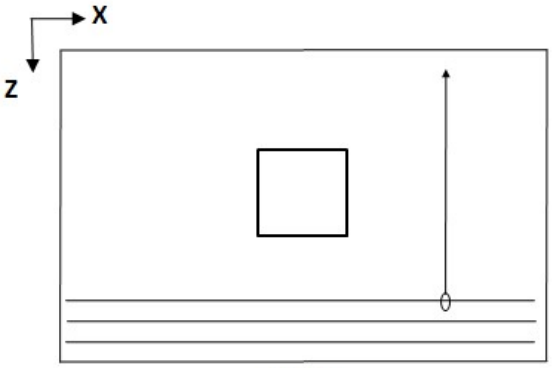
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 65.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1542.240 mm2

Calculated Area of Steel (A_{st}) = 3715.945 mm2

Provided Area of Steel ($A_{st,Provided}$) = 3715.945 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 74.255 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

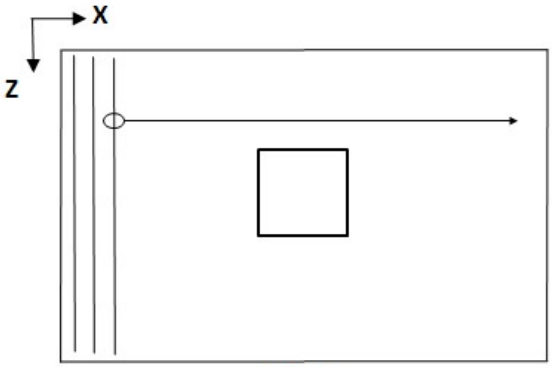
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1542.240 mm2

Calculated Area of Steel (A_{st}) = 1317.600 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1542.240 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 39.039 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 75.957 mm

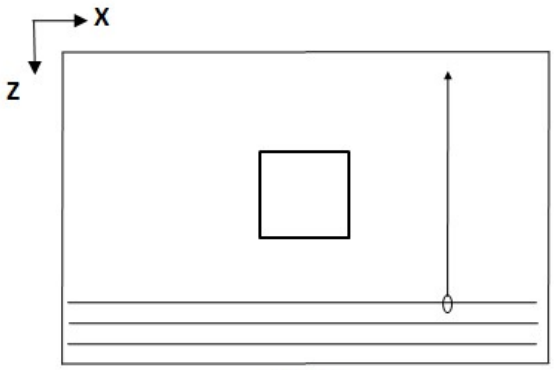
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 75 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1542.240 mm2

Calculated Area of Steel (A_{st}) = 3356.587 mm2

Provided Area of Steel ($A_{st,Provided}$) = 3356.587 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 34.312 kNm

Selected bar Size (d_b) = Ø8

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 52.909 mm

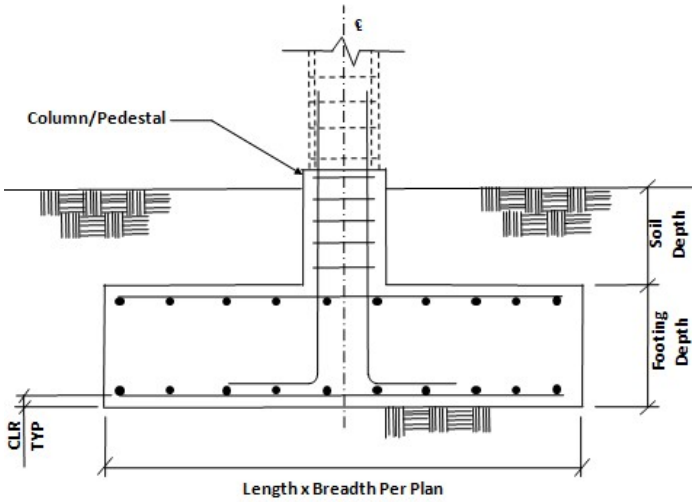
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

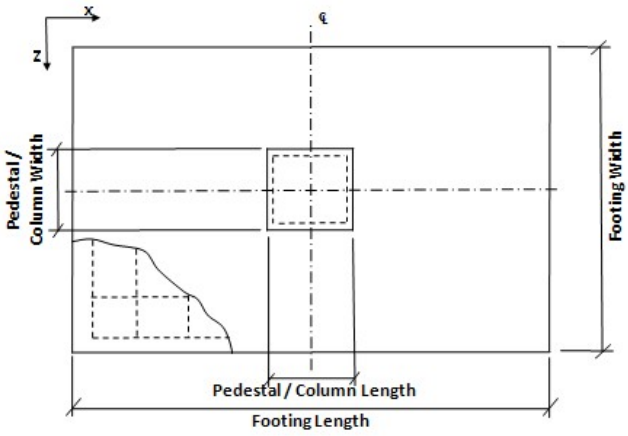
Based on spacing reinforcement increment; provided reinforcement is

Ø8 @ 50 mm o.c.

Isolated Footing 6



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	797.834	29.583	29.034	59.695	-74.688
102	638.267	23.666	23.227	47.756	-59.750

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1196.750	44.374	43.551	89.542	-112.032
202	765.920	28.399	27.873	57.307	-71.701
203	957.400	35.499	34.841	71.633	-89.626
204	718.050	26.625	26.131	53.725	-67.219

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

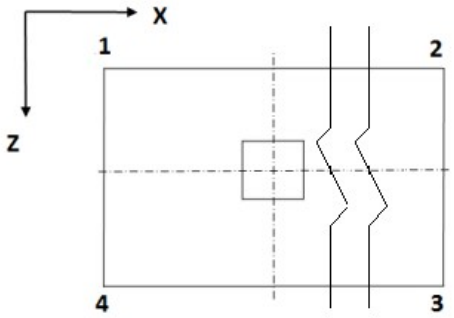
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 8.055 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	3.400	m	Governing Load Case :	# 101
Width (W_2) =	3.400	m	Governing Load Case :	# 101
Depth (D_2) =	0.406	m	Governing Load Case :	# 101
Area (A_2) =	11.560	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**53.3978**	78.9562	99.8857	74.3273	0.000
101	53.3978	**78.9562**	99.8857	74.3273	0.000
101	53.3978	78.9562	**99.8857**	74.3273	0.000
101	53.3978	78.9562	99.8857	**74.3273**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**53.3978**	78.9562	99.8857	74.3273
101	53.3978	**78.9562**	99.8857	74.3273
101	53.3978	78.9562	**99.8857**	74.3273
101	53.3978	78.9562	99.8857	**74.3273**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 11.560 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	14.975	15.258	21.971	17.992

102	15.347	15.637	22.518	18.440

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 29.583 kN
Governing Restoring Force : 442.989 kN
Minimum Sliding Ratio for the Critical Load Case : 14.975
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 68.550 kNm
Governing Resisting Moment : 1506.136 kNm
Minimum Overturning Ratio for the Critical Load Case : 21.971

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 29.034 kN
Governing Restoring Force : 442.989 kN
Minimum Sliding Ratio for the Critical Load Case : 15.258
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -83.711 kNm
Governing Resisting Moment : 1506.136 kNm
Minimum Overturning Ratio for the Critical Load Case : 17.992

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 350.709 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 743.655736 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 313.053 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

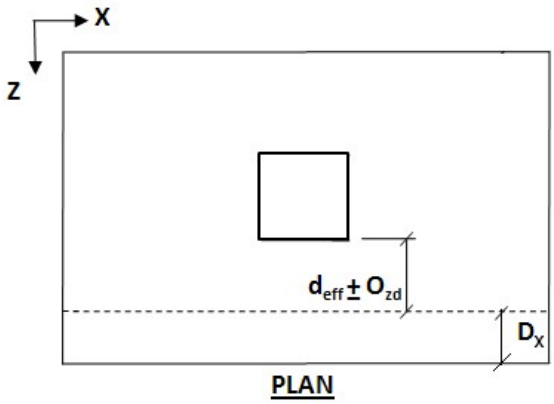
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 743.655736 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.352 m

Shear Force(S) = 444.661 kN

Shear Stress(T_v) = 371.541563 kN/m2

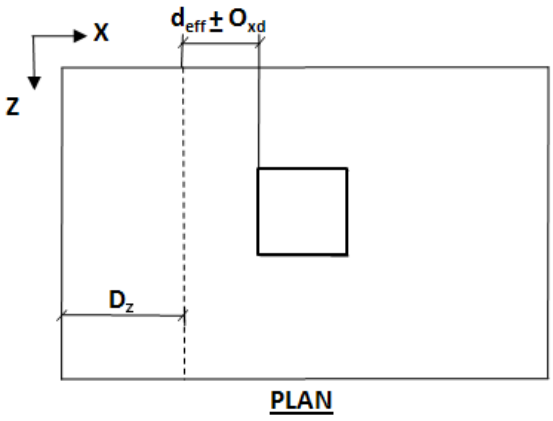
Percentage Of Steel(P_t) = 0.2693

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 376.719 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.352 m

Shear Force(S) = 416.130 kN

Shear Stress(T_v) = 347.702049 kN/m2

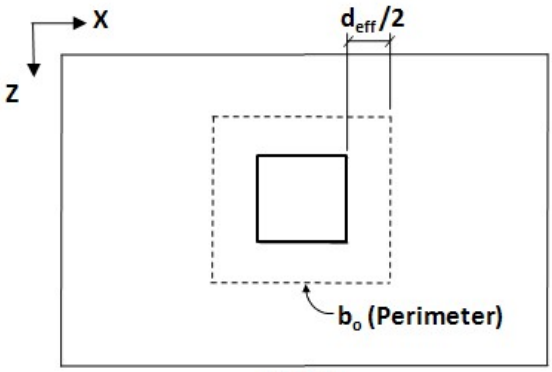
Percentage Of Steel(P_t) = 0.3035

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 396.614 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1122.636 kN

Shear Stress(T_v) = 935.830 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

$K_s \times T_c$ = 1250.0000 kN/m2

$T_v <= K_s \times T_c$ hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

$$\text{Development Length}(l_d) = \frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}} = 1.289 \text{ m}$$

$$\text{Allowable Length}(l_{db}) = \left[\frac{(B - b)}{2} - cc\right] = 1.350 \text{ m}$$

$l_{db} >= l_d$ hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

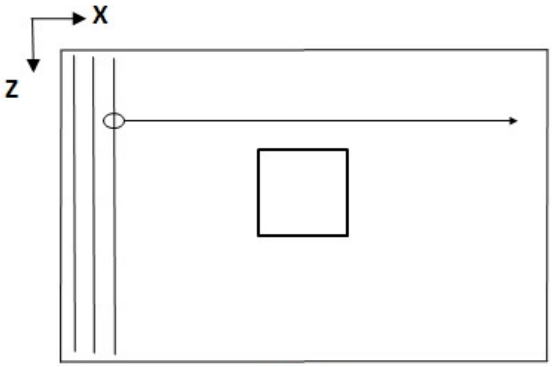
$$\text{Development Length}(l_d) = \frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}} = 1.289 \text{ m}$$

$$\text{Allowable Length}(l_{db}) = \left[\frac{(H - h)}{2} - cc\right] = 1.450 \text{ m}$$

$l_{db} >= l_d$ hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 4195.742 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4195.742 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 62.075 mm

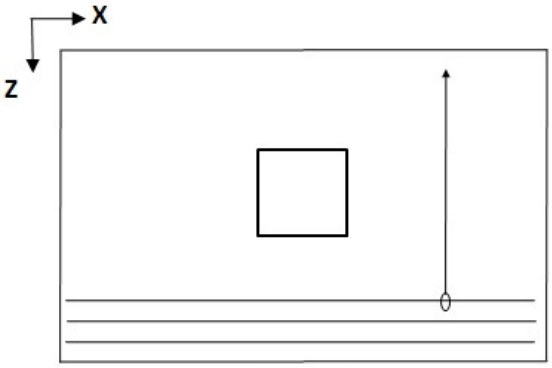
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 3707.039 mm2

Provided Area of Steel ($A_{st,Provided}$) = 3707.039 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 70.000 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

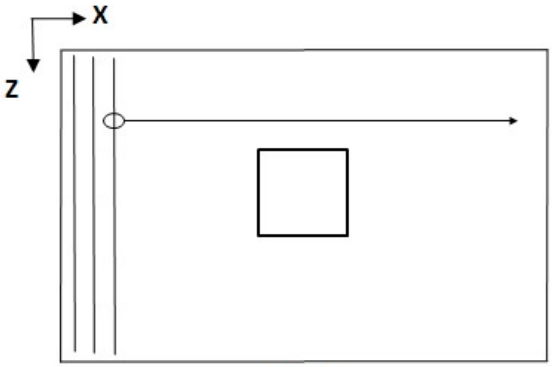
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 1244.400 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1252.560 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 32.406 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 74.864 mm

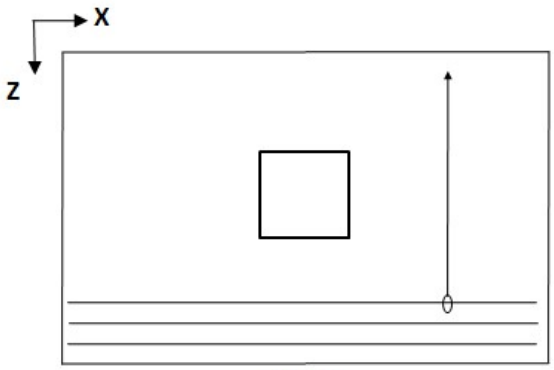
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 70 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 2724.398 mm2

Provided Area of Steel ($A_{st,Provided}$) = 2724.398 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 28.229 kNm

Selected bar Size (d_b) = Ø8

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 60.963 mm

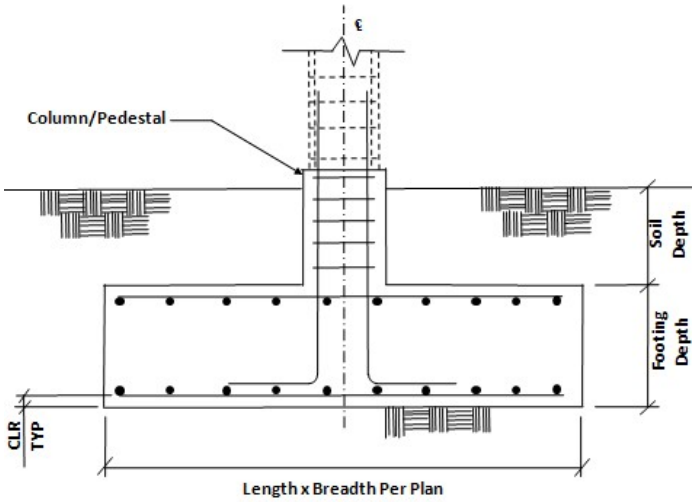
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

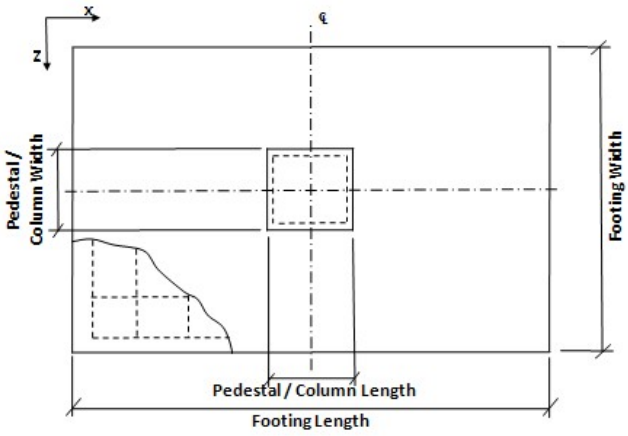
Based on spacing reinforcement increment; provided reinforcement is

Ø8 @ 60 mm o.c.

Isolated Footing 37



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	781.389	28.571	36.792	70.388	-67.193
102	625.112	22.857	29.433	56.311	-53.755

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1172.084	42.856	55.188	105.582	-100.790
202	750.134	27.428	35.320	67.573	-64.506
203	937.667	34.285	44.150	84.466	-80.632
204	703.251	25.714	33.113	63.349	-60.474

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

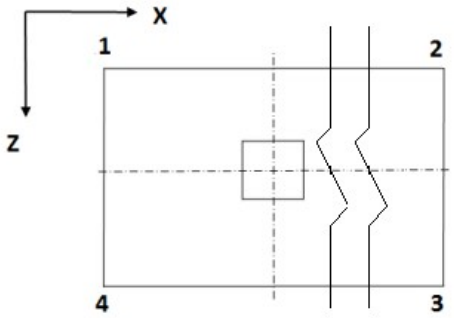
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 7.890 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	3.400	m	Governing Load Case :	# 101
Width (W_2) =	3.400	m	Governing Load Case :	# 101
Depth (D_2) =	0.406	m	Governing Load Case :	# 101
Area (A_2) =	11.560	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**51.1729**	74.3487	99.2656	76.0898	0.000
101	51.1729	**74.3487**	99.2656	76.0898	0.000
101	51.1729	74.3487	**99.2656**	76.0898	0.000
101	51.1729	74.3487	99.2656	**76.0898**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**51.1729**	74.3487	99.2656	76.0898
101	51.1729	**74.3487**	99.2656	76.0898
101	51.1729	74.3487	**99.2656**	76.0898
101	51.1729	74.3487	99.2656	**76.0898**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 11.560 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	15.217	11.817	18.113	19.474

102	15.603	12.116	18.572	19.967

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 28.571 kN
Governing Restoring Force : 434.767 kN
Minimum Sliding Ratio for the Critical Load Case : 15.217
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 81.610 kNm
Governing Resisting Moment : 1478.182 kNm
Minimum Overturning Ratio for the Critical Load Case : 18.113

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 36.792 kN
Governing Restoring Force : 434.767 kN
Minimum Sliding Ratio for the Critical Load Case : 11.817
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -75.907 kNm
Governing Resisting Moment : 1478.182 kNm
Minimum Overturning Ratio for the Critical Load Case : 19.474

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 350.644 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 743.655736 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 303.907 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

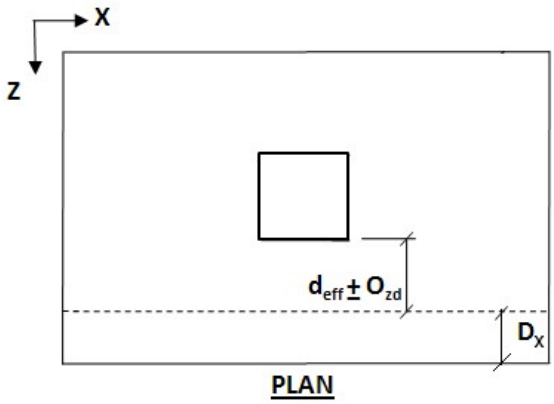
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 743.655736 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.352 m

Shear Force(S) = 444.064 kN

Shear Stress(T_v) = 371.042732 kN/m2

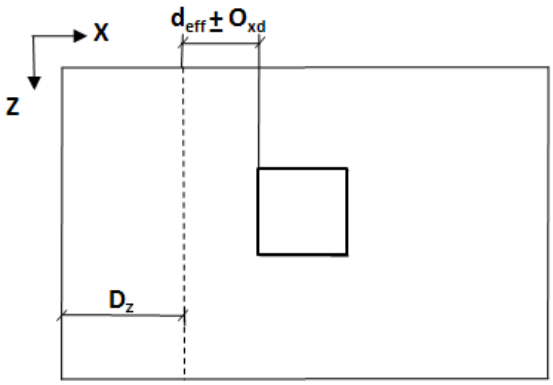
Percentage Of Steel(P_t) = 0.2611

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 371.683 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.352 m

Shear Force(S) = 404.122 kN

Shear Stress(T_v) = 337.668966 kN/m2

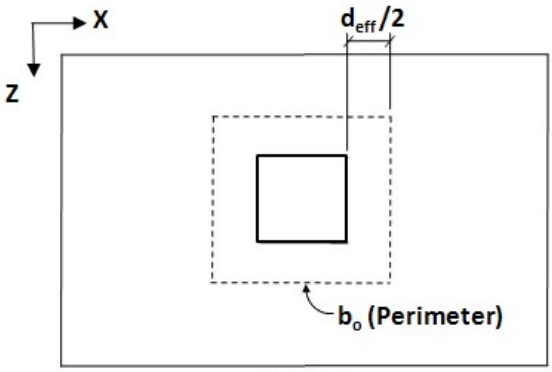
Percentage Of Steel(P_t) = 0.3035

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 396.581 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1099.498 kN

Shear Stress(T_v) = 916.541 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.350 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

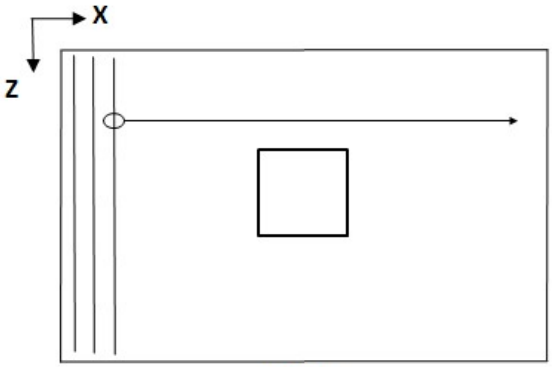
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.450 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 4194.885 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4194.885 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 62.075 mm

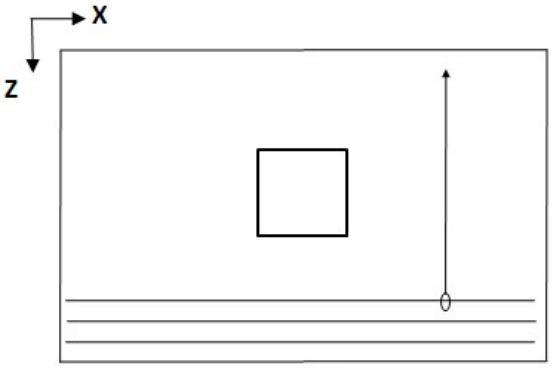
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 3589.959 mm2

Provided Area of Steel ($A_{st,Provided}$) = 3589.959 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 73.111 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

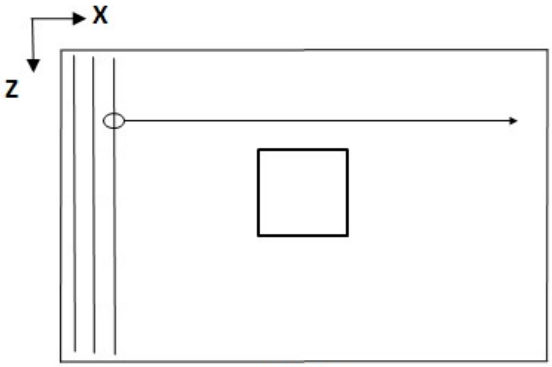
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 1244.400 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1252.560 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 32.406 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 74.864 mm

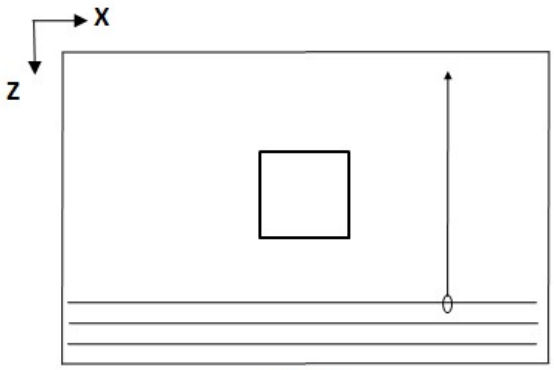
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 70 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 2640.257 mm2

Provided Area of Steel ($A_{st,Provided}$) = 2640.257 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 28.229 kNm

Selected bar Size (d_b) = Ø8

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 63.308 mm

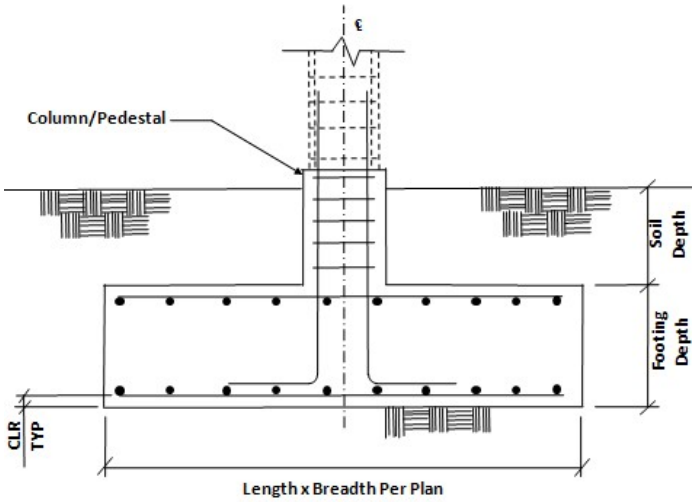
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

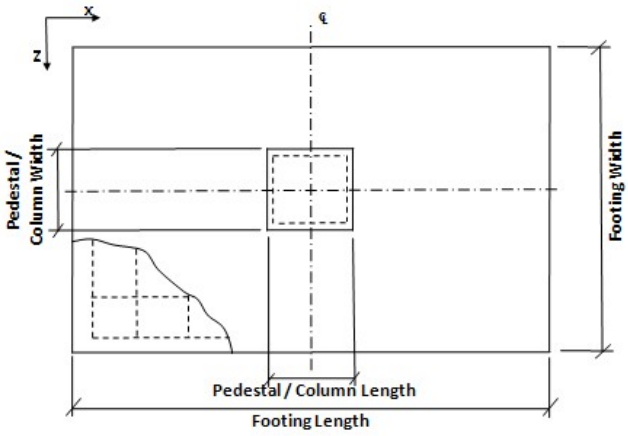
Based on spacing reinforcement increment; provided reinforcement is

Ø8 @ 60 mm o.c.

Isolated Footing 38



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1230.870	37.991	38.755	72.517	-81.512
102	984.696	30.393	31.004	58.013	-65.209

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1846.305	56.987	58.133	108.775	-122.267
202	1181.635	36.472	37.205	69.616	-78.251
203	1477.044	45.590	46.506	87.020	-97.814
204	1107.783	34.192	34.880	65.265	-73.360

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

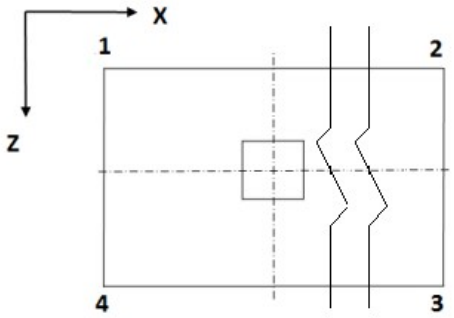
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.385 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.050	m	Governing Load Case :	# 101
Width (W_2) =	4.050	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.403	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**66.6403**	83.4580	98.6929	81.8752	0.000
101	66.6403	**83.4580**	98.6929	81.8752	0.000
101	66.6403	83.4580	**98.6929**	81.8752	0.000
101	66.6403	83.4580	98.6929	**81.8752**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**66.6403**	83.4580	98.6929	81.8752
101	66.6403	**83.4580**	98.6929	81.8752
101	66.6403	83.4580	**98.6929**	81.8752
101	66.6403	83.4580	98.6929	**81.8752**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 16.403 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	17.845	17.494	32.557	29.493

102	18.257	17.897	33.308	30.173

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 37.991 kN
Governing Restoring Force : 677.969 kN
Minimum Sliding Ratio for the Critical Load Case : 17.845
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 84.337 kNm
Governing Resisting Moment : 2745.726 kNm
Minimum Overturning Ratio for the Critical Load Case : 32.557

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 38.755 kN
Governing Restoring Force : 677.969 kN
Minimum Sliding Ratio for the Critical Load Case : 17.494
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -93.099 kNm
Governing Resisting Moment : 2745.726 kNm
Minimum Overturning Ratio for the Critical Load Case : 29.493

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 650.488 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 586.169 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

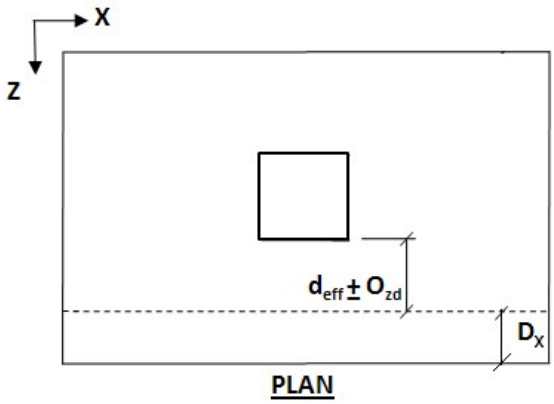
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 667.917 kN

Shear Stress(T_v) = 364.862445 kN/m2

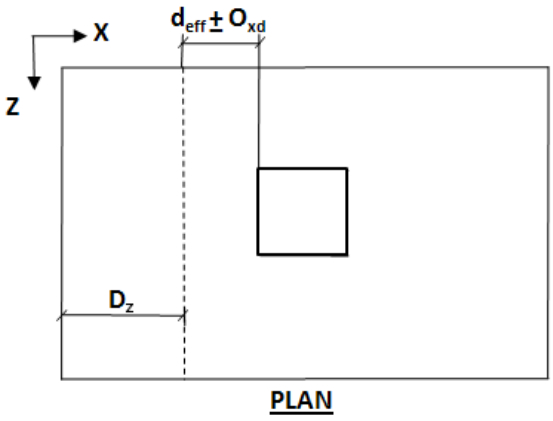
Percentage Of Steel(P_t) = 0.2562

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 368.630 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 624.922 kN

Shear Stress(T_v) = 341.375413 kN/m2

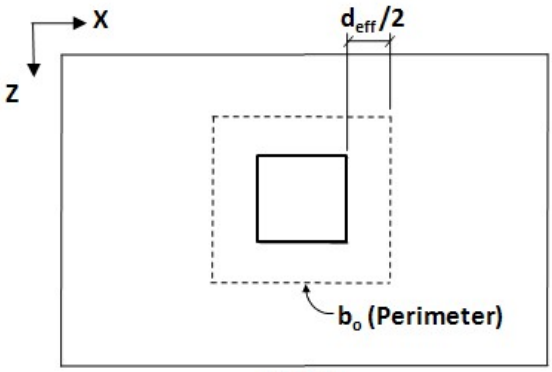
Percentage Of Steel(P_t) = 0.2857

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 386.461 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1745.414 kN

Shear Stress(T_v) = 1014.059 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

$K_s \times T_c$ = 1250.0000 kN/m2

$T_v <= K_s \times T_c$ hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.675 m

$l_{db} >= l_d$ hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

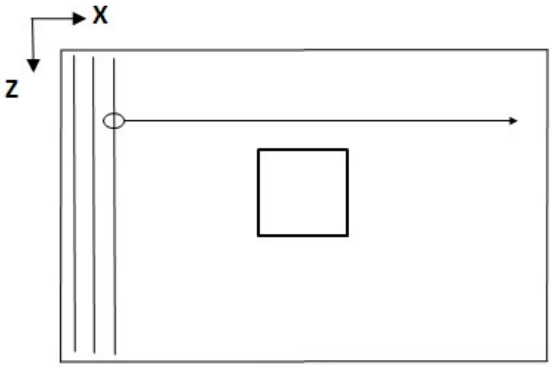
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.775 m

$l_{db} >= l_d$ hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 5466.384 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5466.384 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 57.101 mm

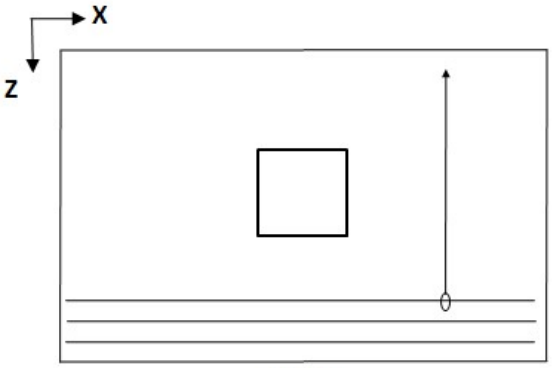
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4890.873 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4890.873 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 63.548 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

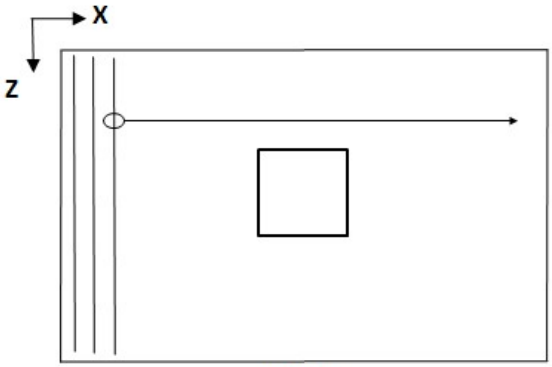
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 1725.300 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1978.020 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 57.140 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 64.656 mm

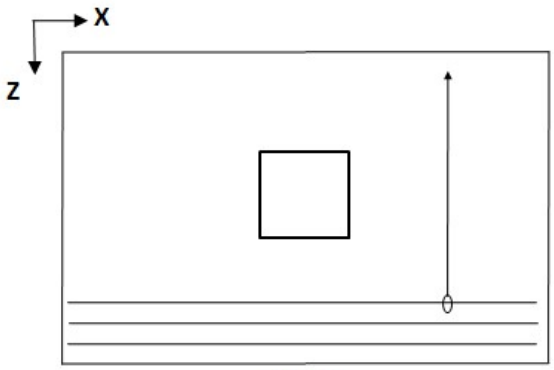
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 60 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4280.672 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4280.672 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 51.050 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 72.963 mm

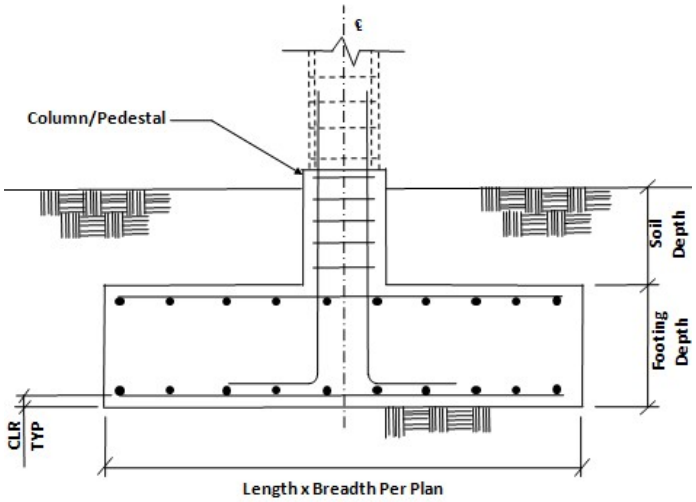
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

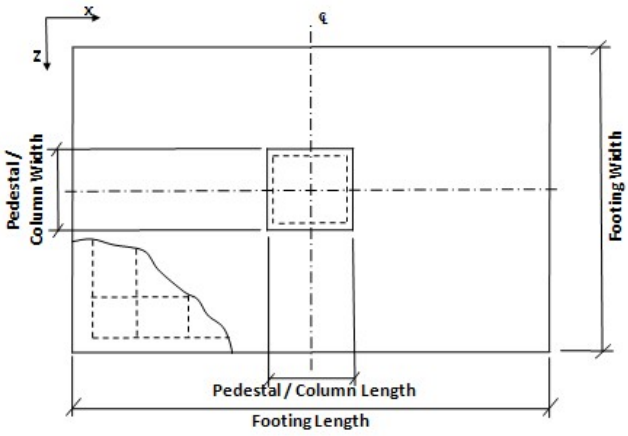
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 39



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1266.859	38.498	38.746	72.522	-82.074
102	1013.488	30.799	30.997	58.017	-65.659

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1900.289	57.748	58.119	108.782	-123.111
202	1216.185	36.959	37.196	69.621	-78.791
203	1520.231	46.198	46.495	87.026	-98.488
204	1140.174	34.649	34.872	65.269	-73.866

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

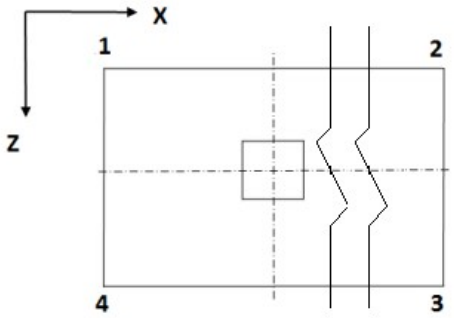
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.745 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.100	m	Governing Load Case :	# 101
Width (W_2) =	4.100	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.810	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**67.4787**	83.8134	98.4982	82.1634	0.000
101	67.4787	**83.8134**	98.4982	82.1634	0.000
101	67.4787	83.8134	**98.4982**	82.1634	0.000
101	67.4787	83.8134	98.4982	**82.1634**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**67.4787**	83.8134	98.4982	82.1634
101	67.4787	**83.8134**	98.4982	82.1634
101	67.4787	83.8134	**98.4982**	82.1634
101	67.4787	83.8134	98.4982	**82.1634**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 16.810 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	18.118	18.002	33.908	30.483

102	18.534	18.416	34.687	31.183

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 38.498 kN
Governing Restoring Force : 697.518 kN
Minimum Sliding Ratio for the Critical Load Case : 18.118
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 84.339 kNm
Governing Resisting Moment : 2859.771 kNm
Minimum Overturning Ratio for the Critical Load Case : 33.908

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 38.746 kN
Governing Restoring Force : 697.518 kN
Minimum Sliding Ratio for the Critical Load Case : 18.002
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -93.815 kNm
Governing Resisting Moment : 2859.771 kNm
Minimum Overturning Ratio for the Critical Load Case : 30.483

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 677.748 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1749.690007 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 611.814 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

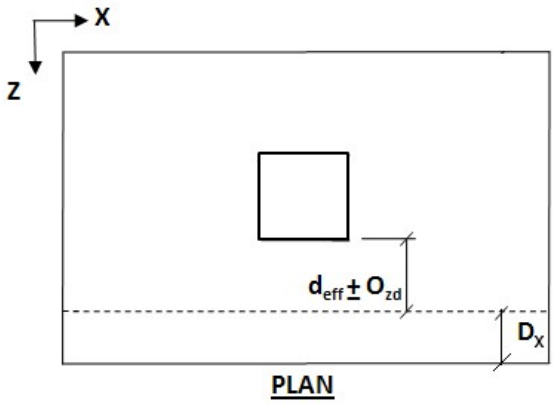
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1749.690007 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 689.555 kN

Shear Stress(T_v) = 372.088730 kN/m2

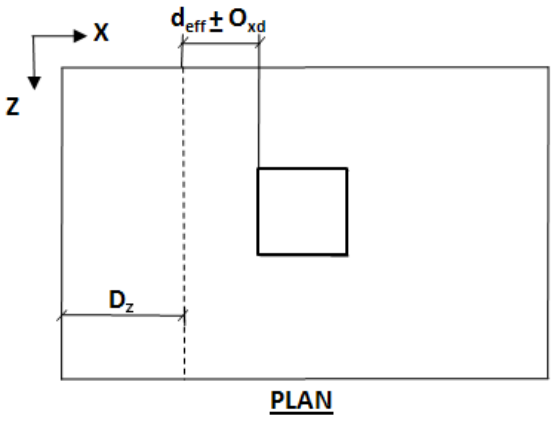
Percentage Of Steel(P_t) = 0.2645

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 373.780 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 646.161 kN

Shear Stress(T_v) = 348.672990 kN/m2

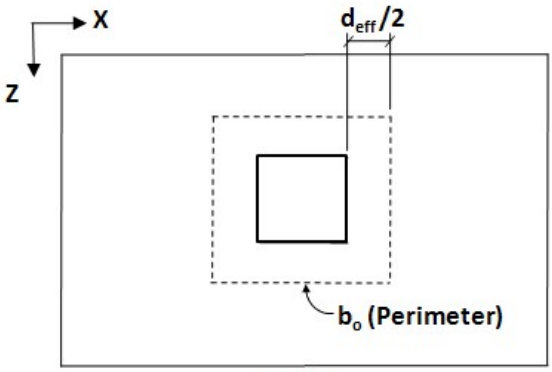
Percentage Of Steel(P_t) = 0.2945

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 391.532 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1798.966 kN

Shear Stress(T_v) = 1045.172 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

$K_s \times T_c$ = 1250.0000 kN/m2

$T_v <= K_s \times T_c$ hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

$$\text{Development Length}(l_d) = \frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}} = 1.289 \text{ m}$$

$$\text{Allowable Length}(l_{db}) = \left[\frac{(B - b)}{2} - cc\right] = 1.700 \text{ m}$$

$l_{db} >= l_d$ hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

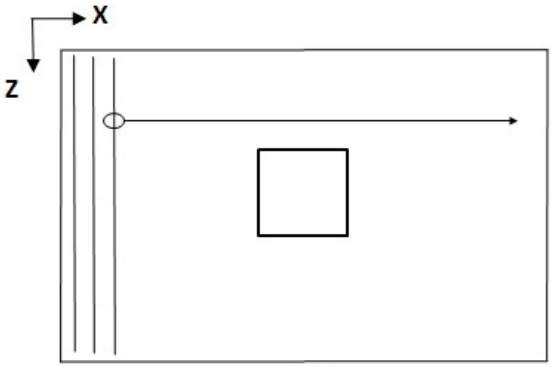
$$\text{Development Length}(l_d) = \frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}} = 1.289 \text{ m}$$

$$\text{Allowable Length}(l_{db}) = \left[\frac{(H - h)}{2} - cc\right] = 1.800 \text{ m}$$

$l_{db} >= l_d$ hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 5707.643 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5707.643 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 55.417 mm

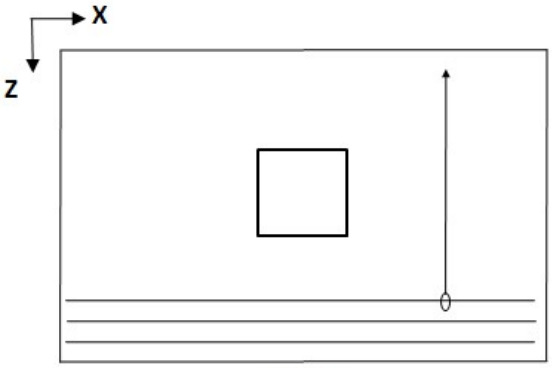
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 5115.069 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5115.069 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 61.385 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

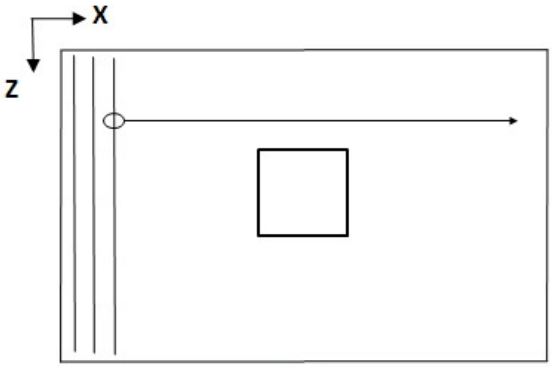
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 1746.600 mm2

Provided Area of Steel ($A_{st,Provided}$) = 2002.440 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 59.441 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 65.475 mm

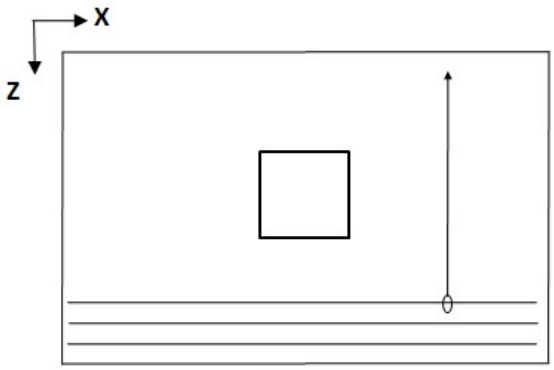
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 65 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 4477.102 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4477.102 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 53.189 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 70.000 mm

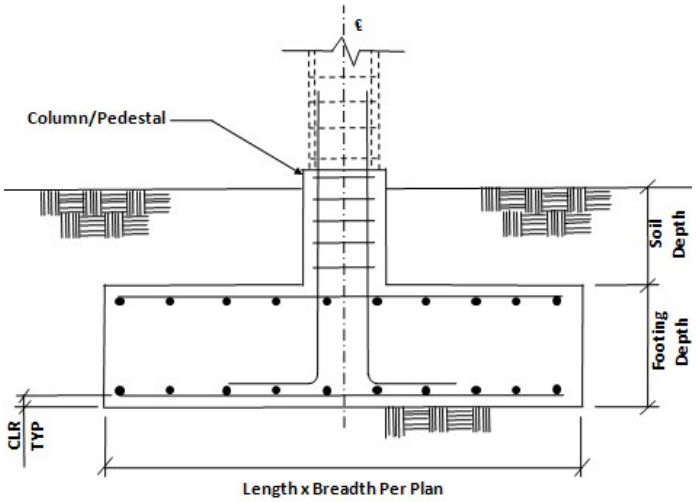
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

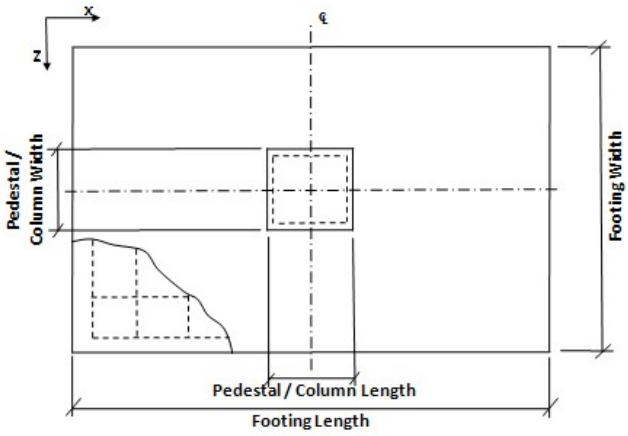
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 40



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1266.884	38.643	38.745	72.519	-82.230
102	1013.507	30.914	30.996	58.016	-65.784

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1900.326	57.964	58.117	108.779	-123.345
202	1216.209	37.097	37.195	69.619	-78.941
203	1520.261	46.371	46.493	87.023	-98.676
204	1140.196	34.779	34.870	65.268	-74.007

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

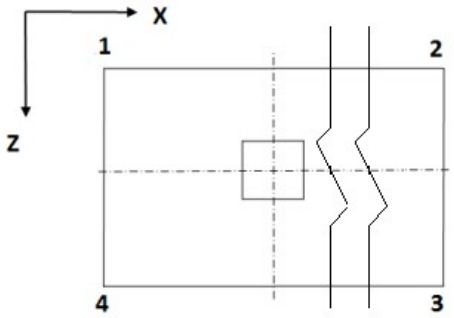
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.745 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.100	m	Governing Load Case :	# 101
Width (W_2) =	4.100	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.810	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**67.4629**	83.8326	98.5169	82.1472	0.000
101	67.4629	**83.8326**	98.5169	82.1472	0.000
101	67.4629	83.8326	**98.5169**	82.1472	0.000
101	67.4629	83.8326	98.5169	**82.1472**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**67.4629**	83.8326	98.5169	82.1472
101	67.4629	**83.8326**	98.5169	82.1472
101	67.4629	83.8326	**98.5169**	82.1472
101	67.4629	83.8326	98.5169	**82.1472**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 16.810 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	18.051	18.003	33.910	30.418

102	18.465	18.417	34.689	31.117

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 38.643 kN
Governing Restoring Force : 697.530 kN
Minimum Sliding Ratio for the Critical Load Case : 18.051
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 84.336 kNm
Governing Resisting Moment : 2859.821 kNm
Minimum Overturning Ratio for the Critical Load Case : 33.910

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 38.745 kN
Governing Restoring Force : 697.530 kN
Minimum Sliding Ratio for the Critical Load Case : 18.003
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -94.016 kNm
Governing Resisting Moment : 2859.821 kNm
Minimum Overturning Ratio for the Critical Load Case : 30.418

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 677.759 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1749.690007 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 611.919 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

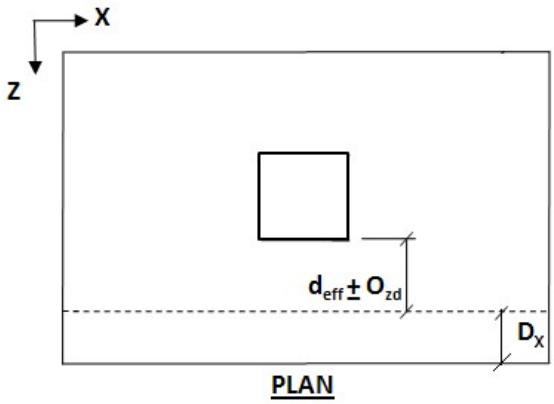
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1749.690007 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 689.566 kN

Shear Stress(T_v) = 372.094794 kN/m2

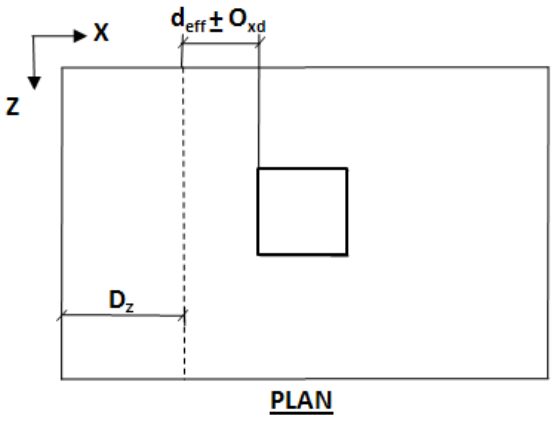
Percentage Of Steel(P_t) = 0.2645

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 373.809 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 646.268 kN

Shear Stress(T_v) = 348.730693 kN/m2

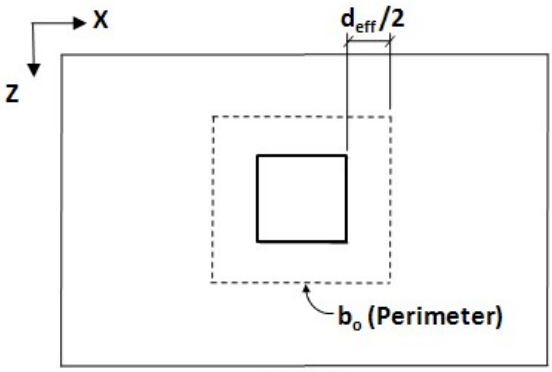
Percentage Of Steel(P_t) = 0.2945

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 391.535 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1799.001 kN

Shear Stress(T_v) = 1045.192 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

$K_s \times T_c$ = 1250.0000 kN/m2

$T_v <= K_s \times T_c$ hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

$$\text{Development Length}(l_d) = \frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}} = 1.289 \text{ m}$$

$$\text{Allowable Length}(l_{db}) = \left[\frac{(B - b)}{2} - cc\right] = 1.700 \text{ m}$$

$l_{db} >= l_d$ hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

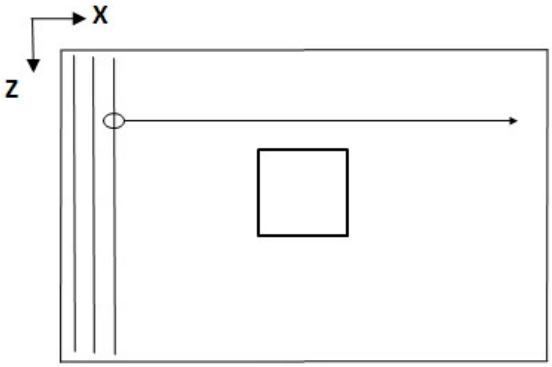
$$\text{Development Length}(l_d) = \frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}} = 1.289 \text{ m}$$

$$\text{Allowable Length}(l_{db}) = \left[\frac{(H - h)}{2} - cc\right] = 1.800 \text{ m}$$

$l_{db} >= l_d$ hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 5707.742 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5707.742 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 55.417 mm

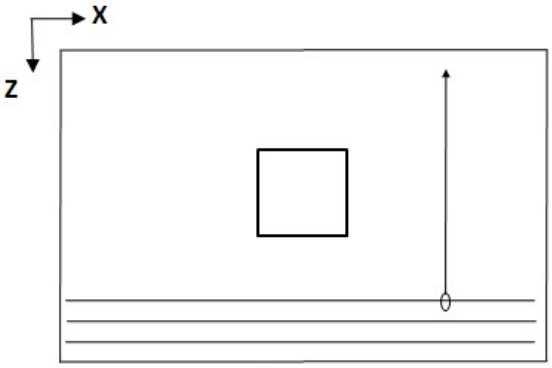
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 5116.008 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5116.008 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 61.385 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

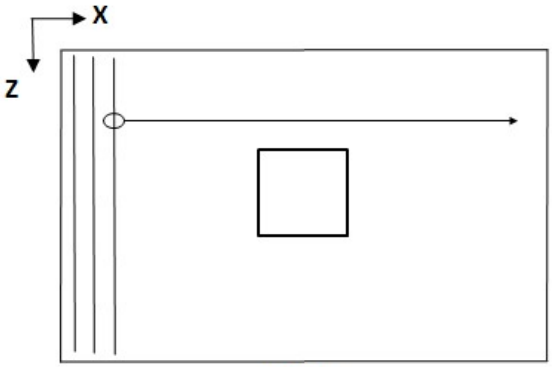
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 1746.600 mm2

Provided Area of Steel ($A_{st,Provided}$) = 2002.440 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 59.441 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 65.475 mm

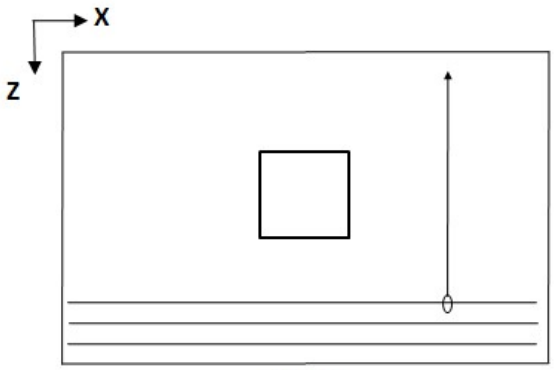
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 65 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 4477.925 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4477.925 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 53.189 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 70.000 mm

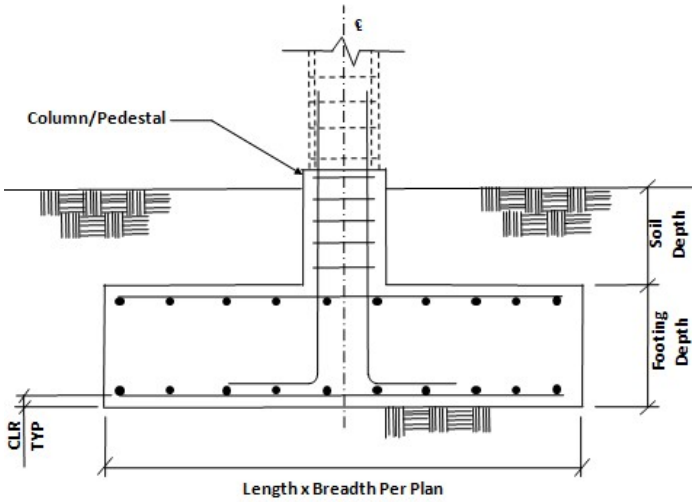
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

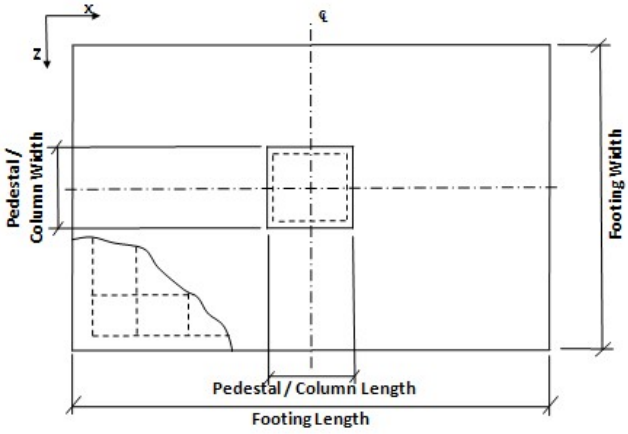
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 41



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1238.138	39.477	38.760	72.517	-83.090
102	990.511	31.582	31.008	58.014	-66.472

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1857.207	59.216	58.140	108.775	-124.635
202	1188.613	37.898	37.209	69.616	-79.767
203	1485.766	47.373	46.512	87.020	-99.708
204	1114.324	35.529	34.884	65.265	-74.781

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

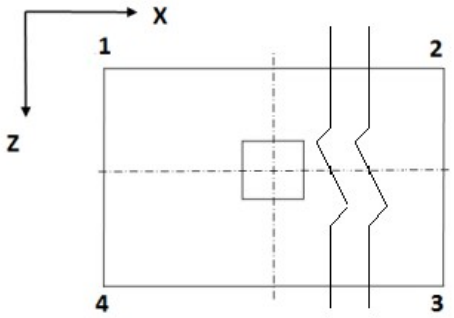
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.458 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.050	m	Governing Load Case :	# 101
Width (W_2) =	4.050	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.403	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**66.8997**	84.0845	99.3198	82.1350	0.000
101	66.8997	**84.0845**	99.3198	82.1350	0.000
101	66.8997	84.0845	**99.3198**	82.1350	0.000
101	66.8997	84.0845	99.3198	**82.1350**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**66.8997**	84.0845	99.3198	82.1350
101	66.8997	**84.0845**	99.3198	82.1350
101	66.8997	84.0845	**99.3198**	82.1350
101	66.8997	84.0845	99.3198	**82.1350**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 16.403 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	17.266	17.585	32.731	29.017

102	17.662	17.989	33.481	29.683

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 39.477 kN
Governing Restoring Force : 681.604 kN
Minimum Sliding Ratio for the Critical Load Case : 17.266
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 84.338 kNm
Governing Resisting Moment : 2760.445 kNm
Minimum Overturning Ratio for the Critical Load Case : 32.731

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 38.760 kN
Governing Restoring Force : 681.604 kN
Minimum Sliding Ratio for the Critical Load Case : 17.585
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -95.130 kNm
Governing Resisting Moment : 2760.445 kNm
Minimum Overturning Ratio for the Critical Load Case : 29.017

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 654.076 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 590.324 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

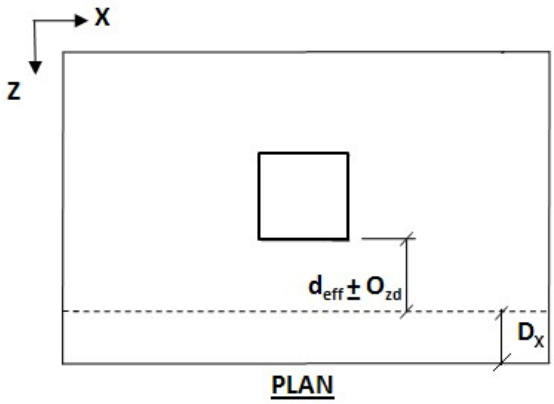
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 671.614 kN

Shear Stress(T_v) = 366.882117 kN/m2

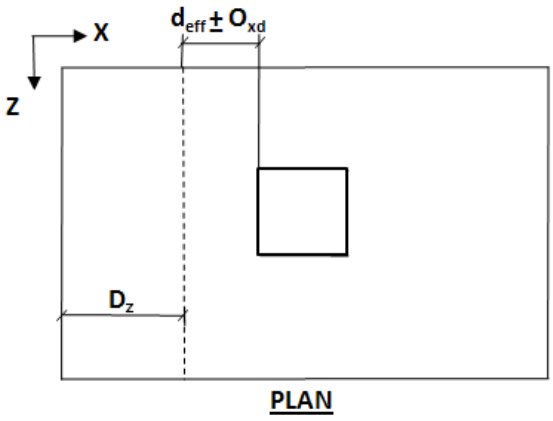
Percentage Of Steel(P_t) = 0.2581

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 369.815 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 629.322 kN

Shear Stress(T_v) = 343.779077 kN/m2

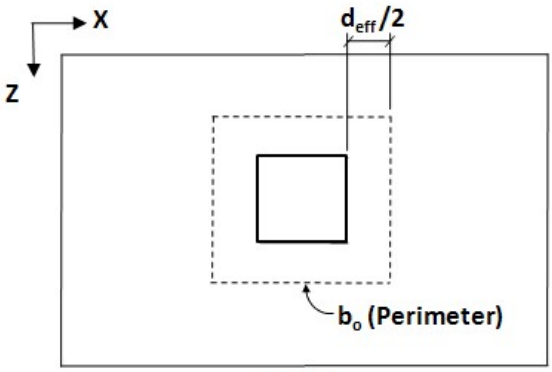
Percentage Of Steel(P_t) = 0.2874

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 387.425 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1755.722 kN

Shear Stress(T_v) = 1020.047 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

$K_s \times T_c$ = 1250.0000 kN/m2

$T_v <= K_s \times T_c$ hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

$$\text{Development Length}(l_d) = \frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}} = 1.289 \text{ m}$$

$$\text{Allowable Length}(l_{db}) = \left[\frac{(B - b)}{2} - cc\right] = 1.675 \text{ m}$$

$l_{db} >= l_d$ hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

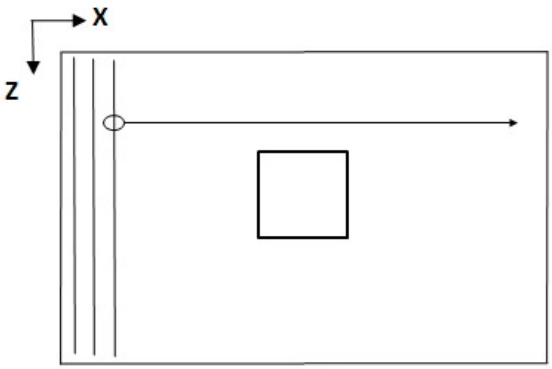
$$\text{Development Length}(l_d) = \frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}} = 1.289 \text{ m}$$

$$\text{Allowable Length}(l_{db}) = \left[\frac{(H - h)}{2} - cc\right] = 1.775 \text{ m}$$

$l_{db} >= l_d$ hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 5498.745 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5498.745 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 56.286 mm

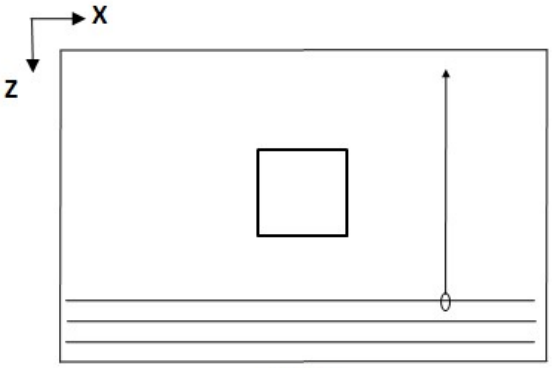
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4927.788 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4927.788 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 63.548 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

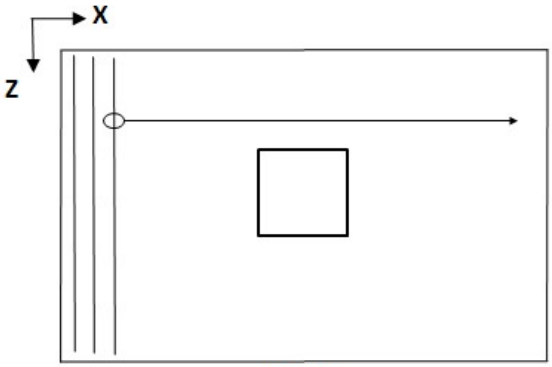
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 1725.300 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1978.020 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 57.140 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 64.656 mm

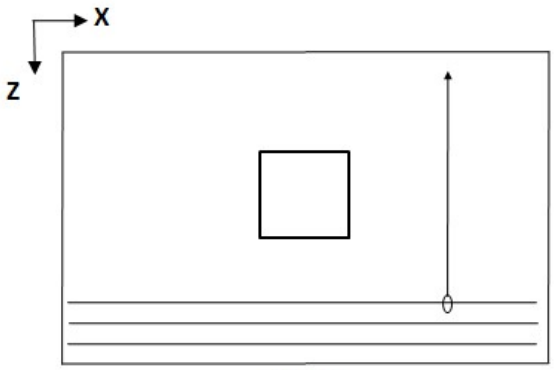
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 60 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin})	= 1978.020 mm2
Calculated Area of Steel (A_{st})	= 4313.027 mm2
Provided Area of Steel ($A_{st,Provided}$)	= 4313.027 mm2
A_{stmin} <= $A_{st,Provided}$	Steel area is accepted
Governing Moment	= 51.050 kNm

Selected bar Size (d_b)	= Ø10
Minimum spacing allowed (S_{min}) =	= 50.000 mm
Selected spacing (S)	= 72.963 mm

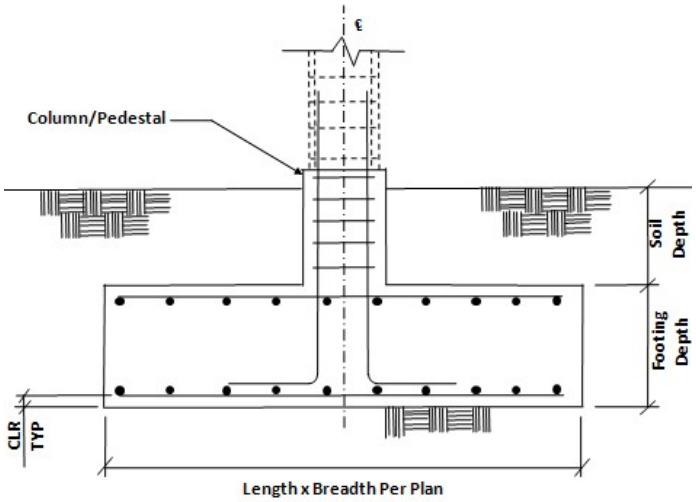
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

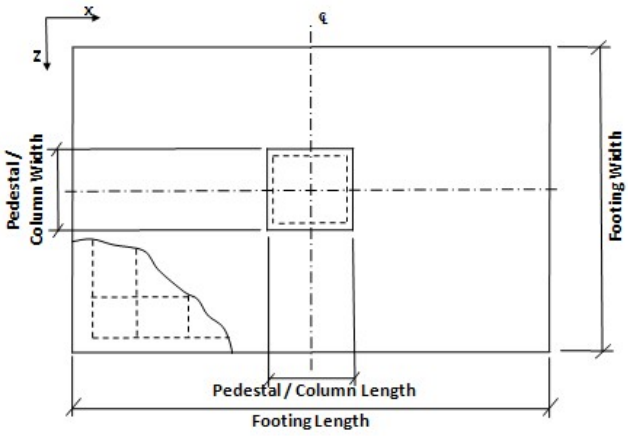
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 42



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1075.613	31.071	36.825	70.389	-78.596
102	860.491	24.857	29.460	56.311	-62.877

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1613.420	46.606	55.237	105.583	-117.895
202	1032.589	29.828	35.352	67.573	-75.453
203	1290.736	37.285	44.190	84.467	-94.316
204	968.052	27.964	33.142	63.350	-70.737

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

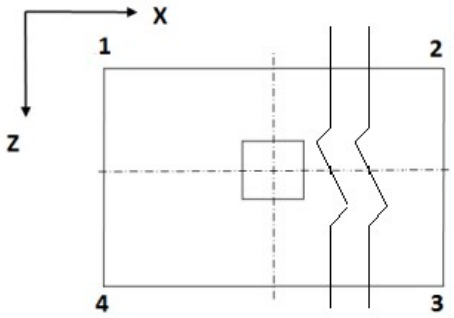
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 10.832 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	3.850	m	Governing Load Case :	# 101
Width (W_2) =	3.850	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	14.823	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**62.3493**	80.8697	98.0332	79.5128	0.000
101	62.3493	**80.8697**	98.0332	79.5128	0.000
101	62.3493	80.8697	**98.0332**	79.5128	0.000
101	62.3493	80.8697	98.0332	**79.5128**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**62.3493**	80.8697	98.0332	79.5128
101	62.3493	**80.8697**	98.0332	79.5128
101	62.3493	80.8697	**98.0332**	79.5128
101	62.3493	80.8697	98.0332	**79.5128**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 14.823 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	19.128	16.139	28.033	25.979

102	19.582	16.523	28.700	26.597

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 31.071 kN
Governing Restoring Force : 594.317 kN
Minimum Sliding Ratio for the Critical Load Case : 19.128
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 81.620 kNm
Governing Resisting Moment : 2288.080 kNm
Minimum Overturning Ratio for the Critical Load Case : 28.033

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 36.825 kN
Governing Restoring Force : 594.317 kN
Minimum Sliding Ratio for the Critical Load Case : 16.139
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -88.073 kNm
Governing Resisting Moment : 2288.080 kNm
Minimum Overturning Ratio for the Critical Load Case : 25.979

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #204

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 405.137 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 842.080760 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #204

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 362.444 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

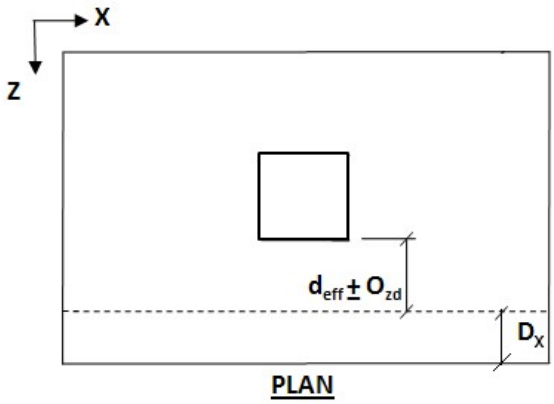
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 842.080760 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 575.705 kN

Shear Stress(T_v) = 330.826978 kN/m2

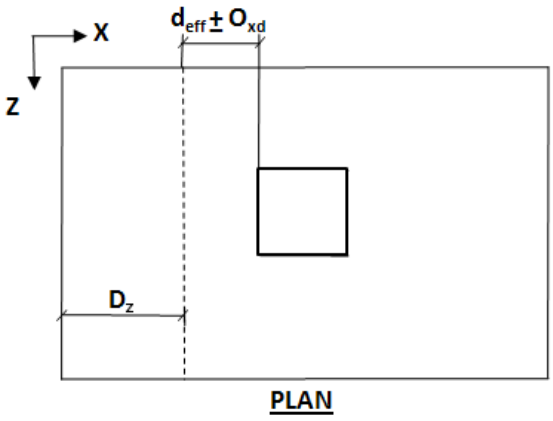
Percentage Of Steel(P_t) = 0.2208

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 345.489 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 535.186 kN

Shear Stress(T_v) = 307.543038 kN/m2

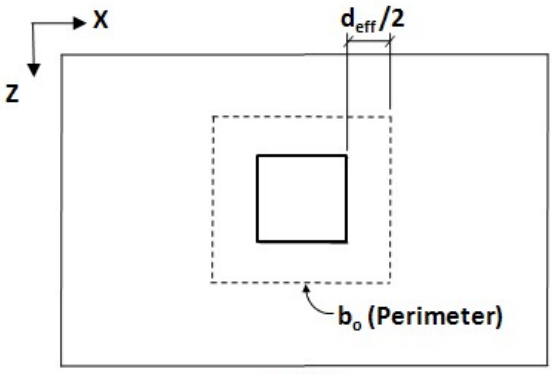
Percentage Of Steel(P_t) = 0.2480

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 363.463 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1515.858 kN

Shear Stress(T_v) = 880.690 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

$K_s \times T_c$ = 1250.0000 kN/m2

$T_v <= K_s \times T_c$ hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.575 m

$l_{db} >= l_d$ hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

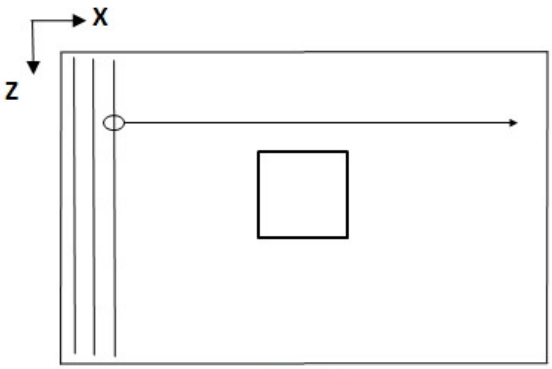
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.675 m

$l_{db} >= l_d$ hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #204

Minimum Area of Steel (A_{stmin}) = 1418.340 mm2

Calculated Area of Steel (A_{st}) = 4856.428 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4856.428 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 61.311 mm

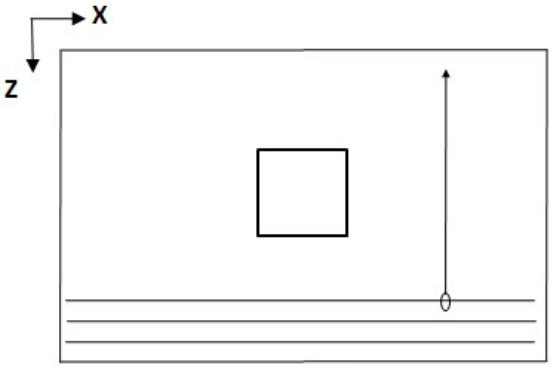
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #204

Minimum Area of Steel (A_{stmin}) = 1418.340 mm2

Calculated Area of Steel (A_{st}) = 4300.005 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4300.005 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 69.259 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

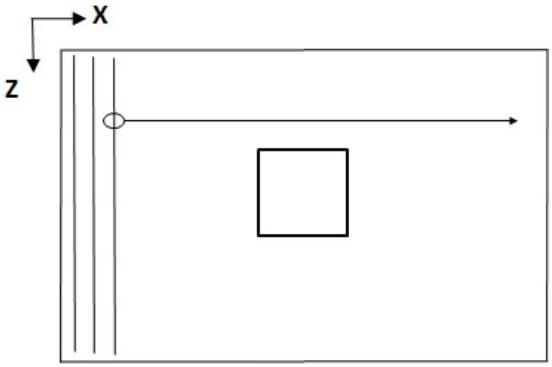
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 65.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1418.340 mm2

Calculated Area of Steel (A_{st}) = 1409.100 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1418.340 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 48.529 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 76.408 mm

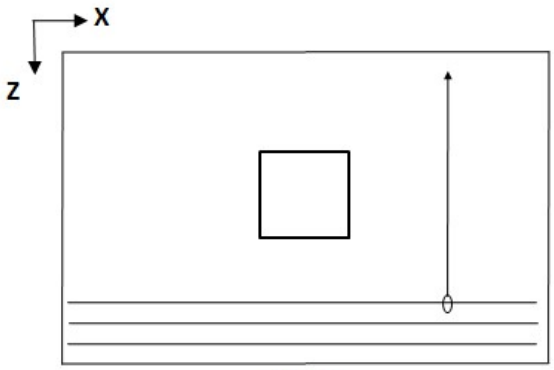
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 75 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1418.340 mm2

Calculated Area of Steel (A_{st}) = 4300.005 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4300.005 mm2

A_{stmin}<= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 43.065 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 69.259 mm

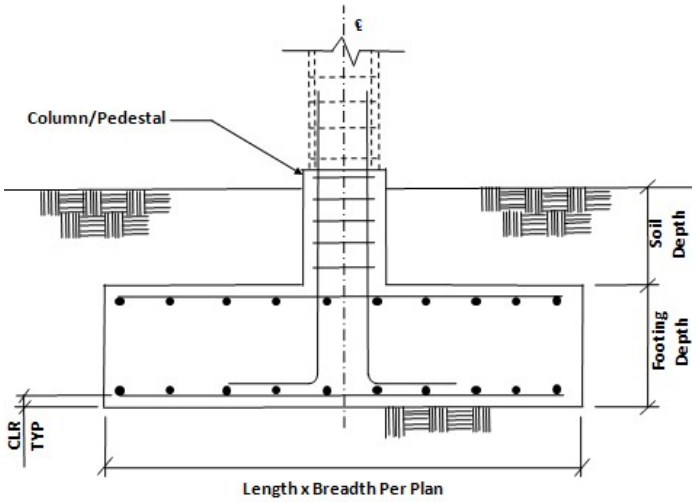
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

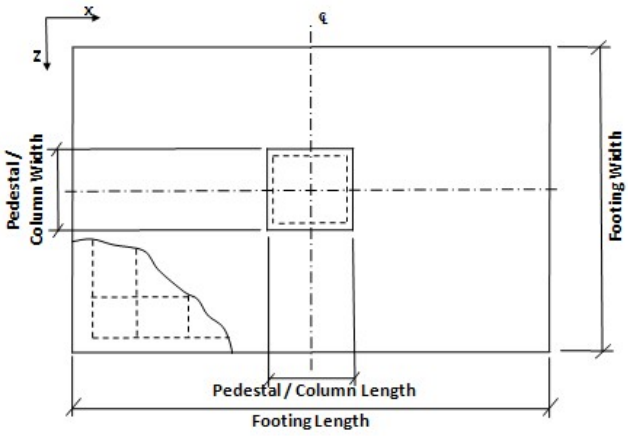
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 65 mm o.c.

Isolated Footing 73



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	785.148	28.648	36.361	69.904	-67.283
102	628.119	22.918	29.088	55.923	-53.826

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1177.722	42.972	54.541	104.856	-100.924
202	753.742	27.502	34.906	67.108	-64.592
203	942.178	34.377	43.633	83.885	-80.740
204	706.633	25.783	32.725	62.913	-60.555

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

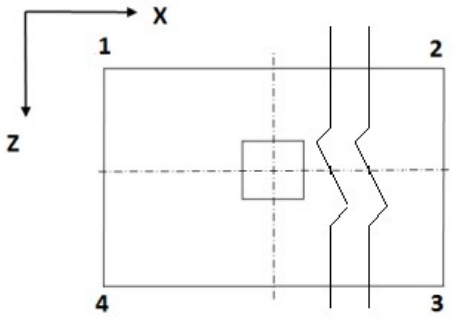
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 7.928 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	3.400	m	Governing Load Case :	# 101
Width (W_2) =	3.400	m	Governing Load Case :	# 101
Depth (D_2) =	0.406	m	Governing Load Case :	# 101
Area (A_2) =	11.560	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**51.5748**	74.7852	99.5140	76.3036	0.000
101	51.5748	**74.7852**	99.5140	76.3036	0.000
101	51.5748	74.7852	**99.5140**	76.3036	0.000
101	51.5748	74.7852	99.5140	**76.3036**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**51.5748**	74.7852	99.5140	76.3036
101	51.5748	**74.7852**	99.5140	76.3036
101	51.5748	74.7852	**99.5140**	76.3036
101	51.5748	74.7852	99.5140	**76.3036**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 11.560 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	15.242	12.009	18.329	19.529

102	15.627	12.312	18.792	20.021

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 28.648 kN
Governing Restoring Force : 436.647 kN
Minimum Sliding Ratio for the Critical Load Case : 15.242
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 80.994 kNm
Governing Resisting Moment : 1484.571 kNm
Minimum Overturning Ratio for the Critical Load Case : 18.329

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 36.361 kN
Governing Restoring Force : 436.647 kN
Minimum Sliding Ratio for the Critical Load Case : 12.009
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -76.020 kNm
Governing Resisting Moment : 1484.571 kNm
Minimum Overturning Ratio for the Critical Load Case : 19.529

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 351.831 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 743.655736 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 305.257 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

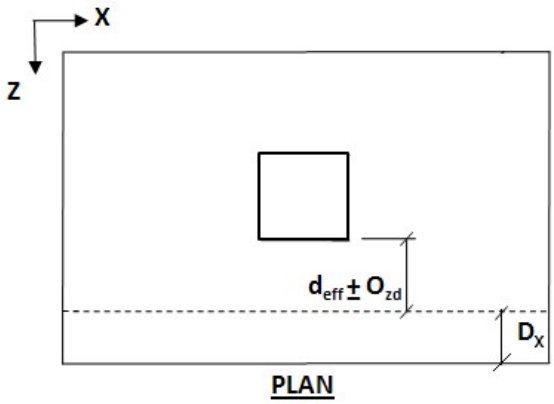
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 743.655736 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.352 m

Shear Force(S) = 445.603 kN

Shear Stress(T_v) = 372.328687 kN/m2

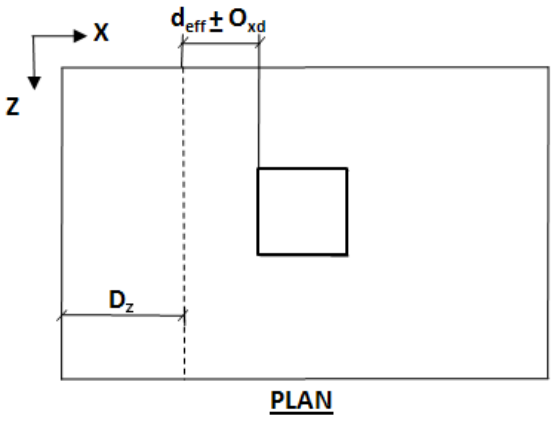
Percentage Of Steel(P_t) = 0.2623

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 372.432 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.352 m

Shear Force(S) = 405.924 kN

Shear Stress(T_v) = 339.174434 kN/m2

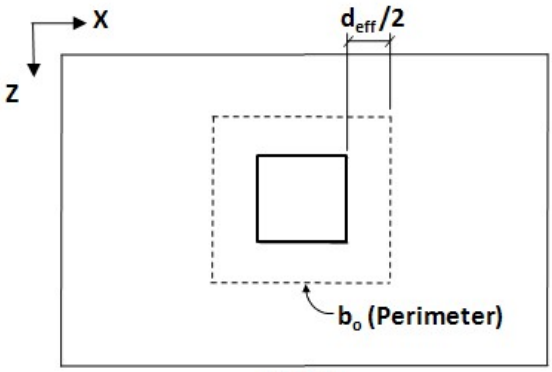
Percentage Of Steel(P_t) = 0.3045

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 397.188 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1104.787 kN

Shear Stress(T_v) = 920.950 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

$K_s \times T_c$ = 1250.0000 kN/m2

$T_v <= K_s \times T_c$ hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.350 m

$l_{db} >= l_d$ hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

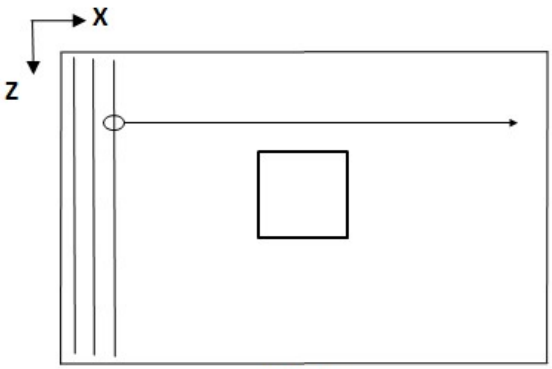
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.450 m

$l_{db} >= l_d$ hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 4210.476 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4210.476 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 62.075 mm

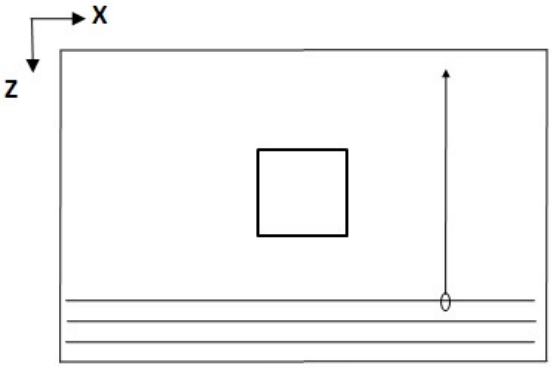
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 3607.204 mm2

Provided Area of Steel ($A_{st,Provided}$) = 3607.204 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 73.111 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

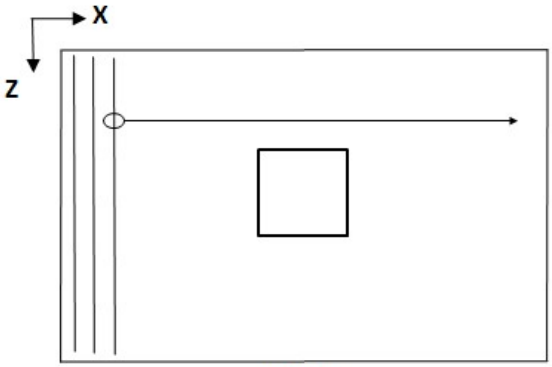
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 1244.400 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1252.560 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 32.406 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 74.864 mm

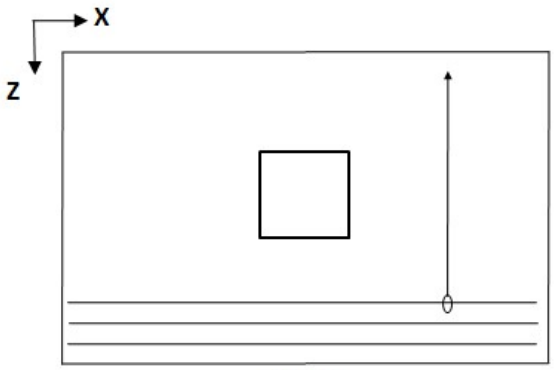
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 70 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 2652.659 mm2

Provided Area of Steel ($A_{st,Provided}$) = 2652.659 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 28.229 kNm

Selected bar Size (d_b) = Ø8

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 63.308 mm

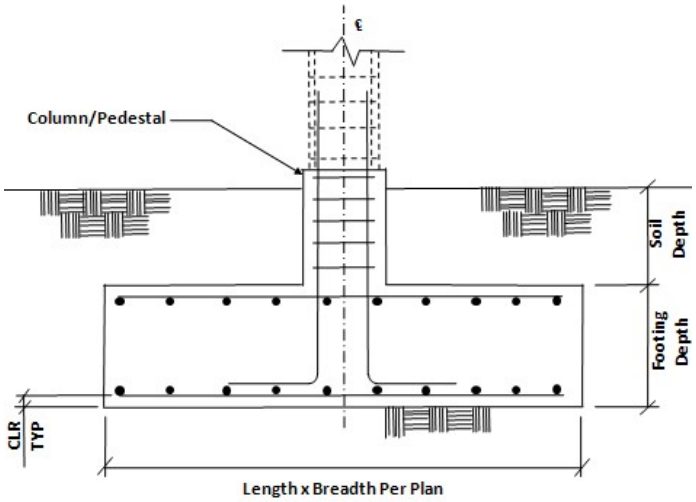
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

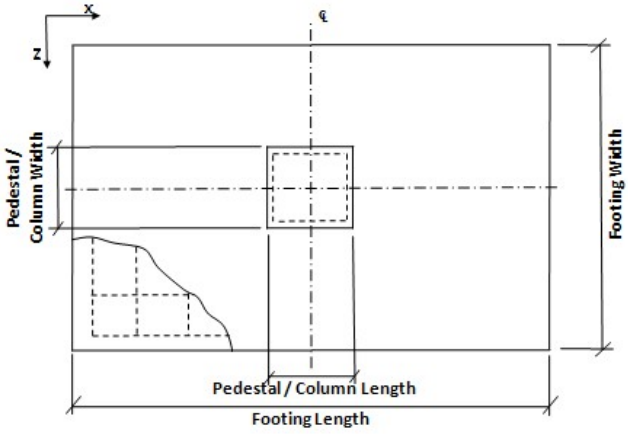
Based on spacing reinforcement increment; provided reinforcement is

Ø8 @ 60 mm o.c.

Isolated Footing 74



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1238.162	37.987	38.278	71.993	-81.521
102	990.530	30.390	30.622	57.595	-65.217

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1857.243	56.981	57.417	107.990	-122.281
202	1188.636	36.468	36.747	69.114	-78.260
203	1485.795	45.585	45.934	86.392	-97.825
204	1114.346	34.189	34.450	64.794	-73.369

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

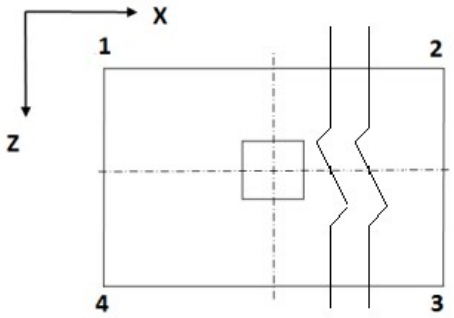
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.458 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.050	m	Governing Load Case :	# 101
Width (W_2) =	4.050	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.403	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**67.1445**	83.9637	99.0779	82.2586	0.000
101	67.1445	**83.9637**	99.0779	82.2586	0.000
101	67.1445	83.9637	**99.0779**	82.2586	0.000
101	67.1445	83.9637	99.0779	**82.2586**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**67.1445**	83.9637	99.0779	82.2586
101	67.1445	**83.9637**	99.0779	82.2586
101	67.1445	83.9637	**99.0779**	82.2586
101	67.1445	83.9637	99.0779	**82.2586**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 16.403 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	17.943	17.807	32.993	29.649

102	18.355	18.215	33.750	30.329

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 37.987 kN
Governing Restoring Force : 681.616 kN
Minimum Sliding Ratio for the Critical Load Case : 17.943
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 83.668 kNm
Governing Resisting Moment : 2760.493 kNm
Minimum Overturning Ratio for the Critical Load Case : 32.993

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 38.278 kN
Governing Restoring Force : 681.616 kN
Minimum Sliding Ratio for the Critical Load Case : 17.807
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -93.107 kNm
Governing Resisting Moment : 2760.493 kNm
Minimum Overturning Ratio for the Critical Load Case : 29.649

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 653.745 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 589.388 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

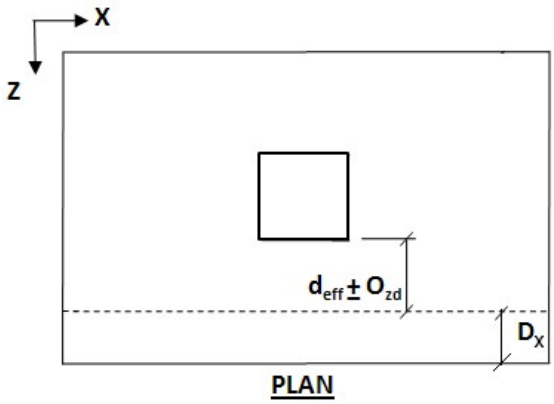
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 671.293 kN

Shear Stress(T_v) = 366.706329 kN/m2

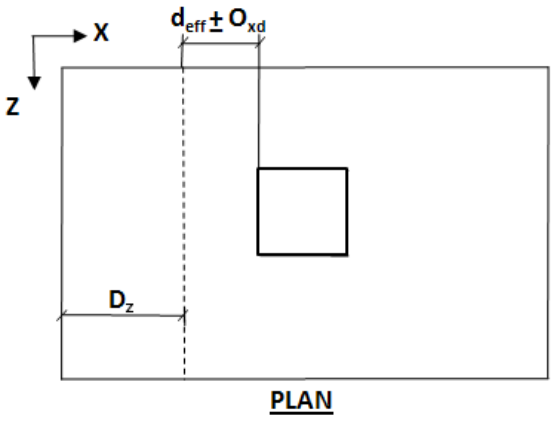
Percentage Of Steel(P_t) = 0.2576

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 369.548 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 628.364 kN

Shear Stress(T_v) = 343.255766 kN/m2

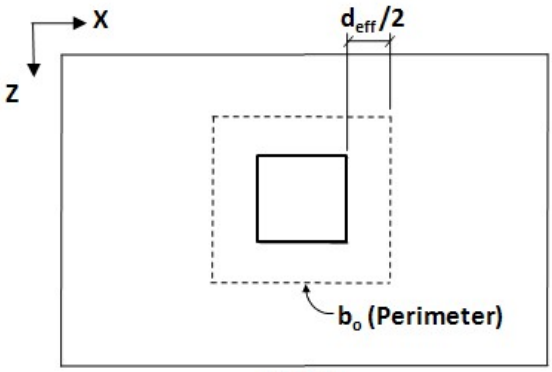
Percentage Of Steel(P_t) = 0.2872

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 387.336 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1755.755 kN

Shear Stress(T_v) = 1020.067 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.675 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

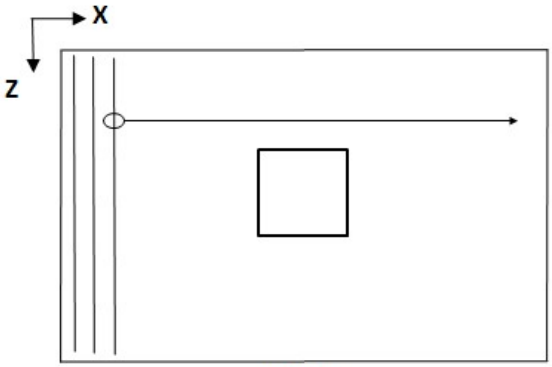
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.775 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 5495.754 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5495.754 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 57.101 mm

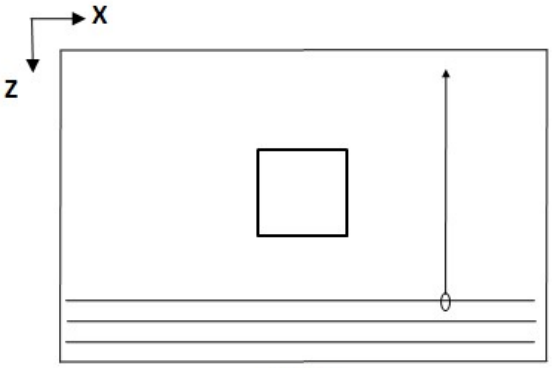
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4919.464 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4919.464 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 63.548 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

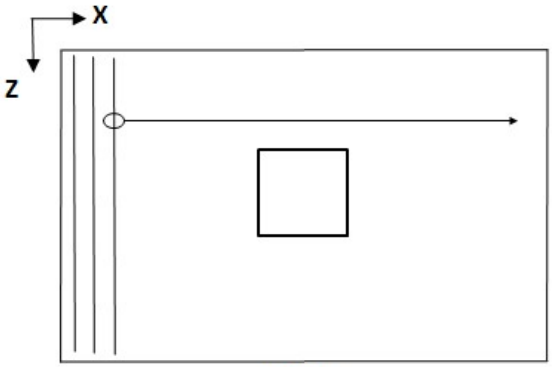
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 1725.300 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1978.020 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 57.140 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 64.656 mm

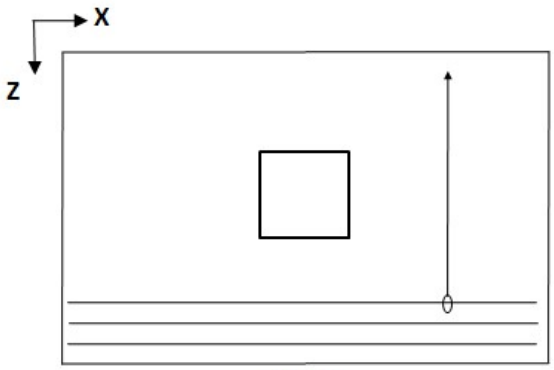
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 60 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4305.731 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4305.731 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 51.050 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 72.963 mm

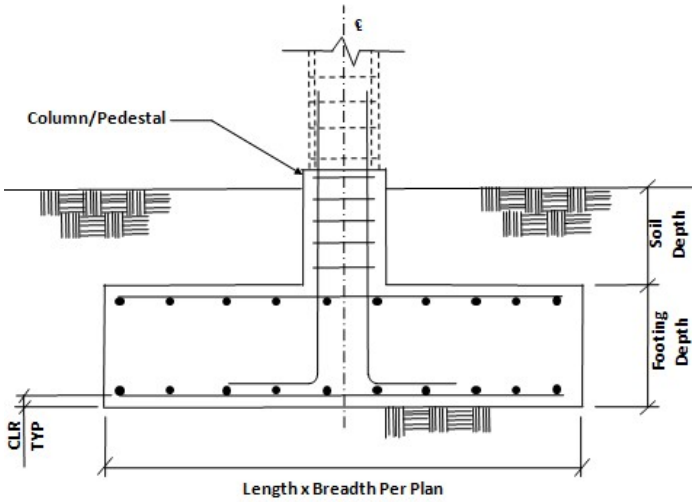
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

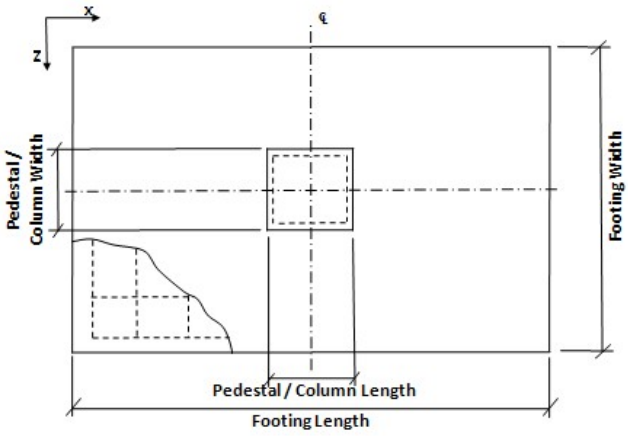
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 75



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1275.476	38.493	38.275	72.006	-82.082
102	1020.380	30.794	30.620	57.604	-65.666

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1913.213	57.739	57.412	108.008	-123.123
202	1224.457	36.953	36.744	69.125	-78.799
203	1530.571	46.191	45.930	86.407	-98.498
204	1147.928	34.644	34.447	64.805	-73.874

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

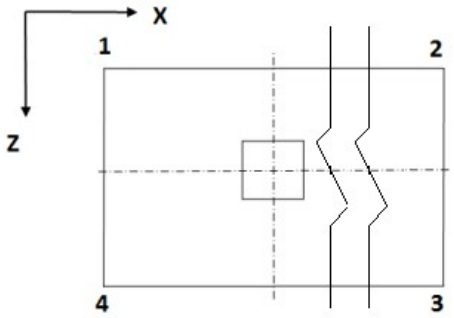
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.831 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.100	m	Governing Load Case :	# 101
Width (W_2) =	4.100	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.810	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**68.0481**	84.3840	98.9539	82.6180	0.000
101	68.0481	**84.3840**	98.9539	82.6180	0.000
101	68.0481	84.3840	**98.9539**	82.6180	0.000
101	68.0481	84.3840	98.9539	**82.6180**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**68.0481**	84.3840	98.9539	82.6180
101	68.0481	**84.3840**	98.9539	82.6180
101	68.0481	84.3840	**98.9539**	82.6180
101	68.0481	84.3840	98.9539	**82.6180**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 16.810 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	18.233	18.336	34.386	30.669

102	18.649	18.755	35.171	31.369

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 38.493 kN
Governing Restoring Force : 701.826 kN
Minimum Sliding Ratio for the Critical Load Case : 18.233
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 83.679 kNm
Governing Resisting Moment : 2877.433 kNm
Minimum Overturning Ratio for the Critical Load Case : 34.386

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 38.275 kN
Governing Restoring Force : 701.826 kN
Minimum Sliding Ratio for the Critical Load Case : 18.336
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -93.822 kNm
Governing Resisting Moment : 2877.433 kNm
Minimum Overturning Ratio for the Critical Load Case : 30.669

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 681.725 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1749.690007 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 615.679 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

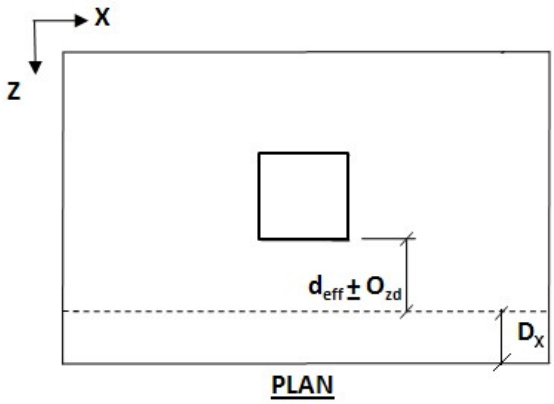
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1749.690007 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 693.636 kN

Shear Stress(T_v) = 374.291019 kN/m2

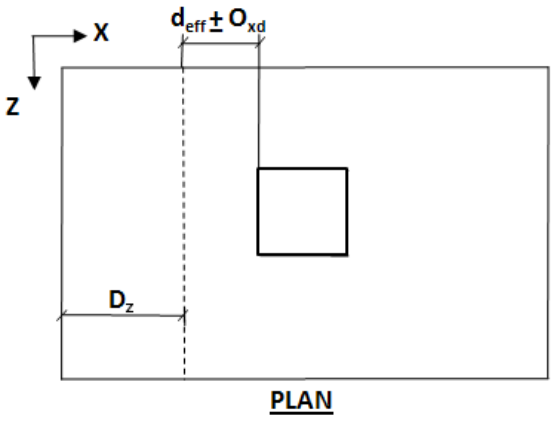
Percentage Of Steel(P_t) = 0.2662

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 374.850 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 650.256 kN

Shear Stress(T_v) = 350.882527 kN/m2

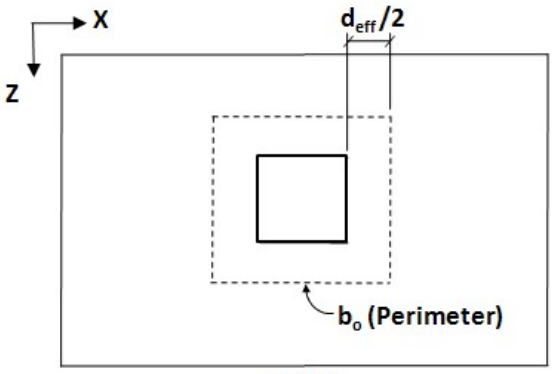
Percentage Of Steel(P_t) = 0.2964

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 392.570 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1811.201 kN

Shear Stress(T_v) = 1052.280 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.700 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

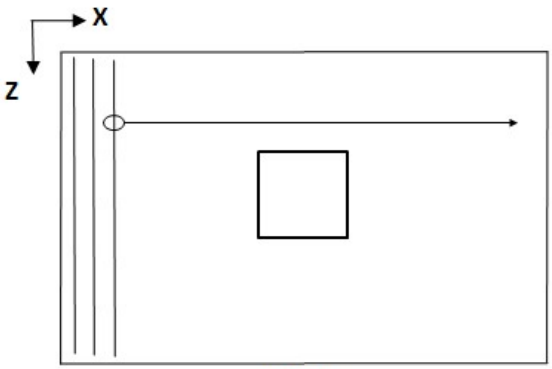
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.800 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 5743.683 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5743.683 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 54.658 mm

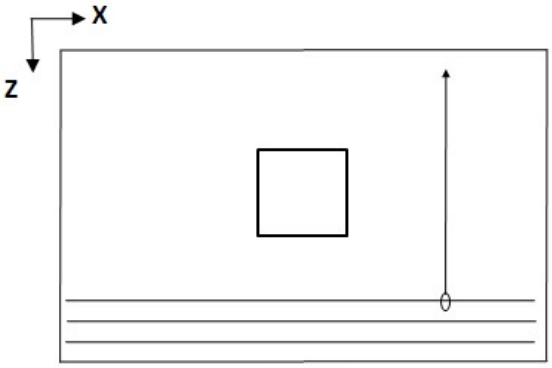
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 50.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 5149.549 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5149.549 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 61.385 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

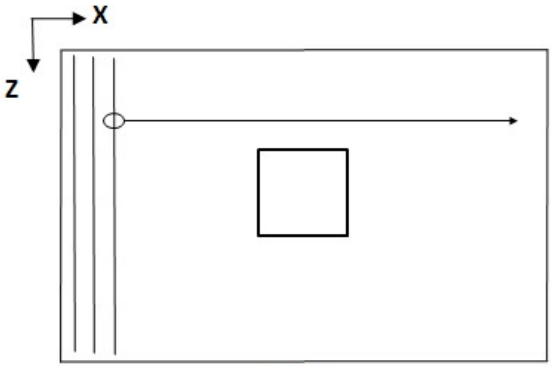
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 1746.600 mm2

Provided Area of Steel ($A_{st,Provided}$) = 2002.440 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 59.441 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 65.475 mm

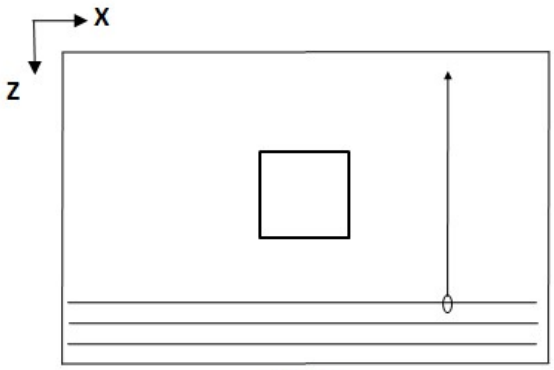
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 65 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 4507.324 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4507.324 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 53.189 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 70.000 mm

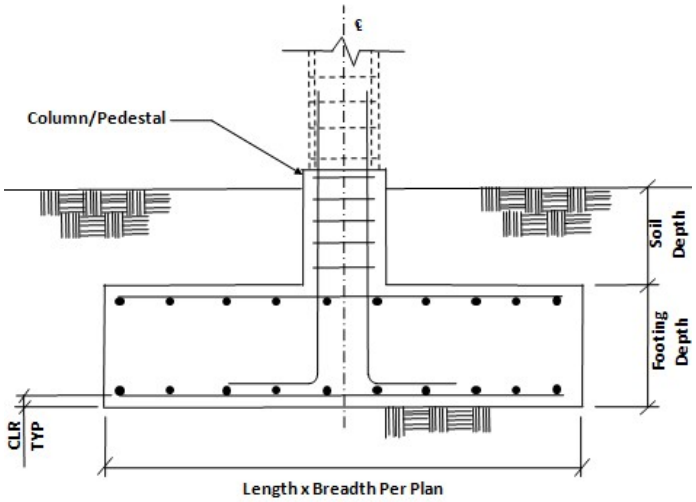
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

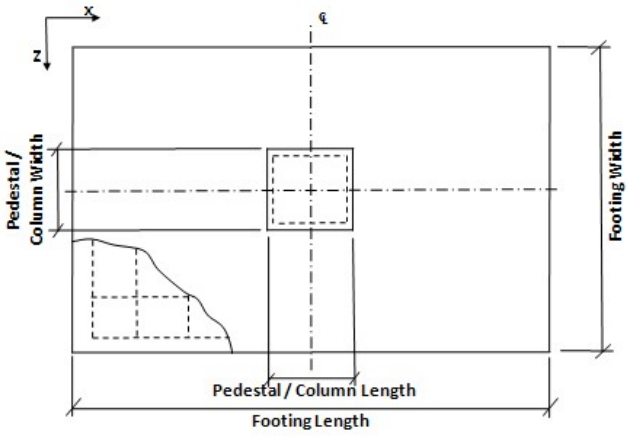
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 76



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1275.482	38.659	38.275	72.006	-82.269
102	1020.386	30.927	30.620	57.604	-65.815

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1913.223	57.989	57.412	108.008	-123.403
202	1224.463	37.113	36.744	69.125	-78.978
203	1530.579	46.391	45.930	86.407	-98.723
204	1147.934	34.793	34.447	64.805	-74.042

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

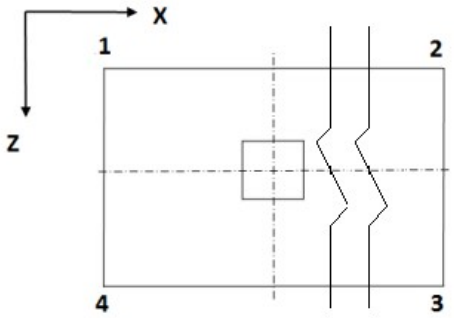
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.831 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.100	m	Governing Load Case :	# 101
Width (W_2) =	4.100	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.810	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**68.0279**	84.4051	98.9750	82.5977	0.000
101	68.0279	**84.4051**	98.9750	82.5977	0.000
101	68.0279	84.4051	**98.9750**	82.5977	0.000
101	68.0279	84.4051	98.9750	**82.5977**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**68.0279**	84.4051	98.9750	82.5977
101	68.0279	**84.4051**	98.9750	82.5977
101	68.0279	84.4051	**98.9750**	82.5977
101	68.0279	84.4051	98.9750	**82.5977**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 16.810 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	18.154	18.337	34.387	30.592

102	18.569	18.755	35.172	31.290

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 38.659 kN
Governing Restoring Force : 701.829 kN
Minimum Sliding Ratio for the Critical Load Case : 18.154
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 83.679 kNm
Governing Resisting Moment : 2877.448 kNm
Minimum Overturning Ratio for the Critical Load Case : 34.387

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 38.275 kN
Governing Restoring Force : 701.829 kN
Minimum Sliding Ratio for the Critical Load Case : 18.337
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -94.060 kNm
Governing Resisting Moment : 2877.448 kNm
Minimum Overturning Ratio for the Critical Load Case : 30.592

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 681.728 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1749.690007 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 615.793 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

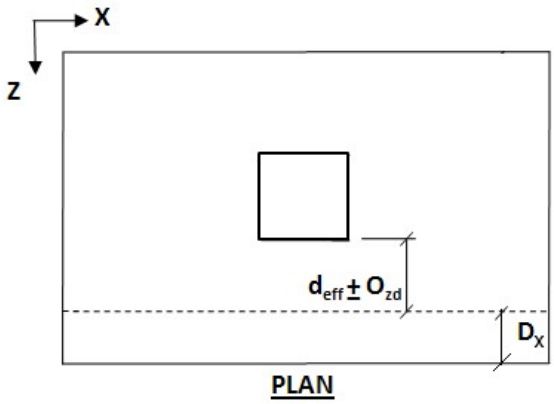
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1749.690007 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 693.640 kN

Shear Stress(T_v) = 374.292905 kN/m2

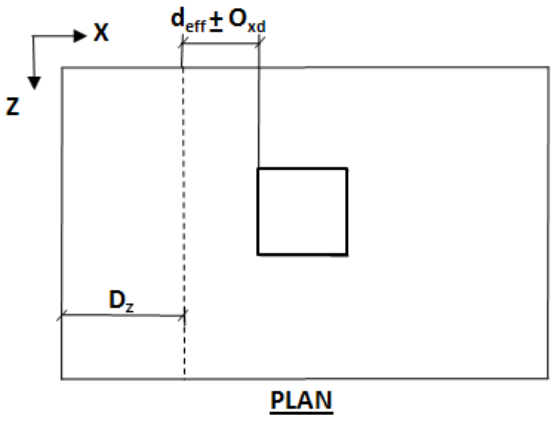
Percentage Of Steel(P_t) = 0.2663

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 374.882 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 650.372 kN

Shear Stress(T_v) = 350.945141 kN/m2

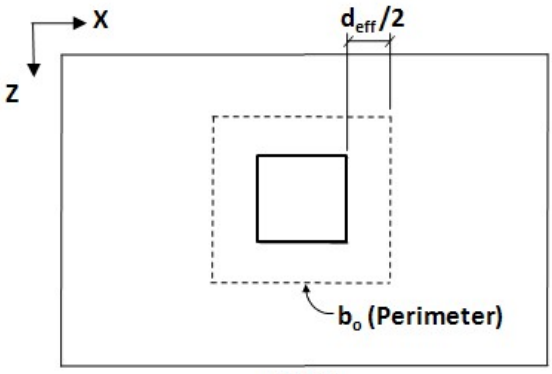
Percentage Of Steel(P_t) = 0.2964

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 392.571 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1811.211 kN

Shear Stress(T_v) = 1052.286 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.700 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

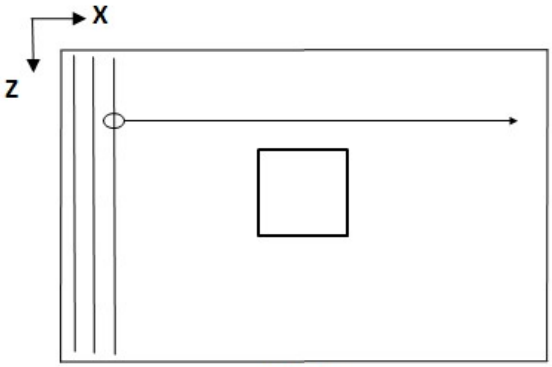
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.800 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 5743.714 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5743.714 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 54.658 mm

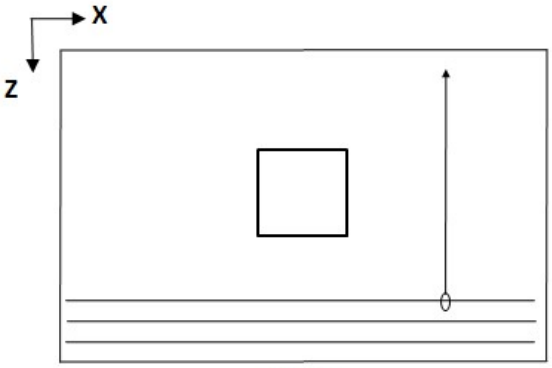
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 50.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 5150.572 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5150.572 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 61.385 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

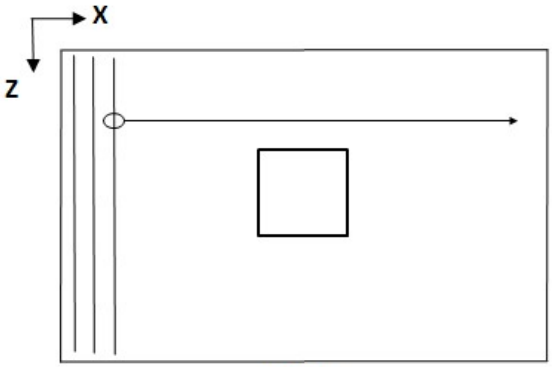
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 1746.600 mm2

Provided Area of Steel ($A_{st,Provided}$) = 2002.440 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 59.441 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 65.475 mm

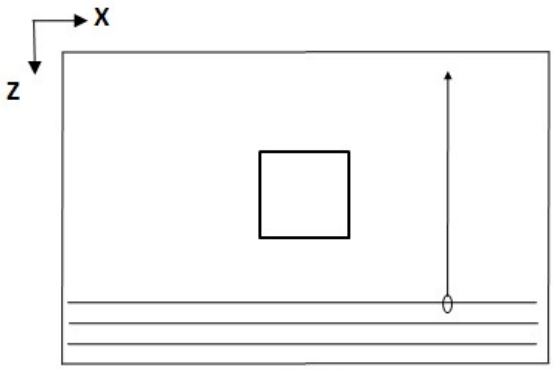
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 65 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 2002.440 mm2

Calculated Area of Steel (A_{st}) = 4508.221 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4508.221 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 53.189 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 70.000 mm

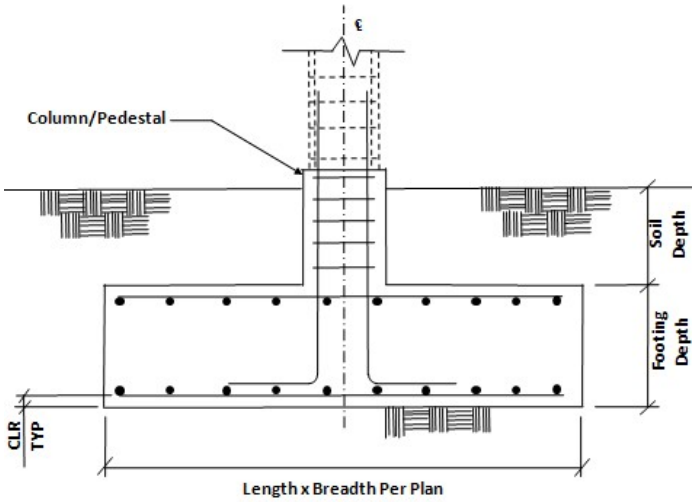
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

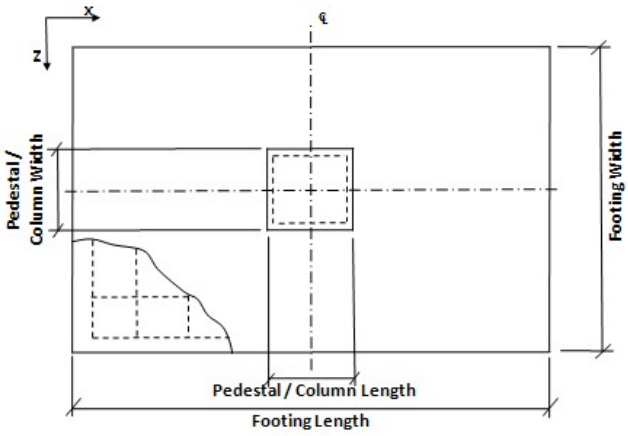
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 77



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1245.202	39.533	38.278	71.993	-83.176
102	996.162	31.626	30.622	57.595	-66.541

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1867.803	59.300	57.417	107.990	-124.764
202	1195.394	37.952	36.747	69.114	-79.849
203	1494.243	47.440	45.934	86.392	-99.812
204	1120.682	35.580	34.450	64.794	-74.859

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

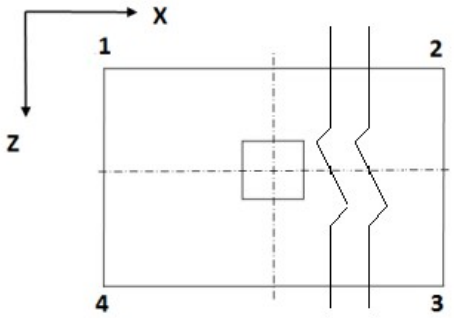
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.528 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.050	m	Governing Load Case :	# 101
Width (W_2) =	4.050	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.403	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**67.3816**	84.5850	99.6992	82.4958	0.000
101	67.3816	**84.5850**	99.6992	82.4958	0.000
101	67.3816	84.5850	**99.6992**	82.4958	0.000
101	67.3816	84.5850	99.6992	**82.4958**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**67.3816**	84.5850	99.6992	82.4958
101	67.3816	**84.5850**	99.6992	82.4958
101	67.3816	84.5850	**99.6992**	82.4958
101	67.3816	84.5850	99.6992	**82.4958**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 16.403 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	17.331	17.899	33.164	29.136

102	17.726	18.307	33.921	29.801

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 39.533 kN
Governing Restoring Force : 685.136 kN
Minimum Sliding Ratio for the Critical Load Case : 17.331
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 83.668 kNm
Governing Resisting Moment : 2774.749 kNm
Minimum Overturning Ratio for the Critical Load Case : 33.164

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 38.278 kN
Governing Restoring Force : 685.136 kN
Minimum Sliding Ratio for the Critical Load Case : 17.899
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -95.234 kNm
Governing Resisting Moment : 2774.749 kNm
Minimum Overturning Ratio for the Critical Load Case : 29.136

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 657.218 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 593.486 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

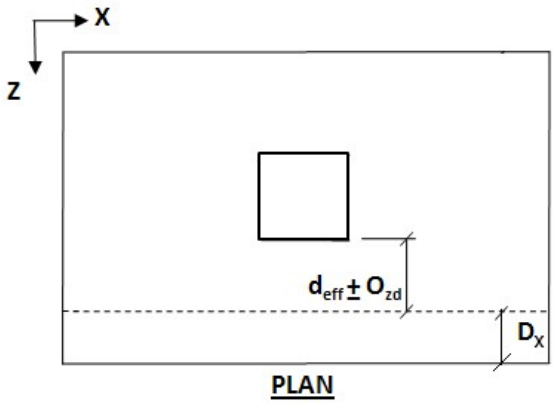
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 674.873 kN

Shear Stress(T_v) = 368.661956 kN/m2

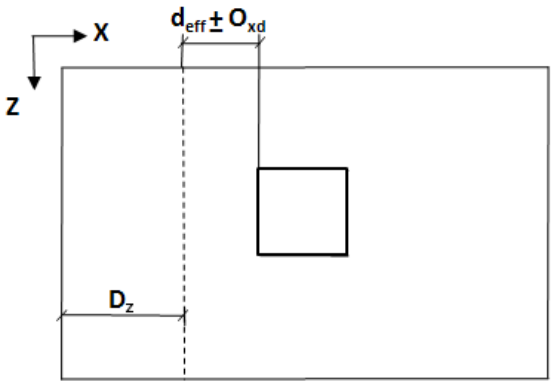
Percentage Of Steel(P_t) = 0.2595

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 370.713 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 632.702 kN

Shear Stress(T_v) = 345.625385 kN/m2

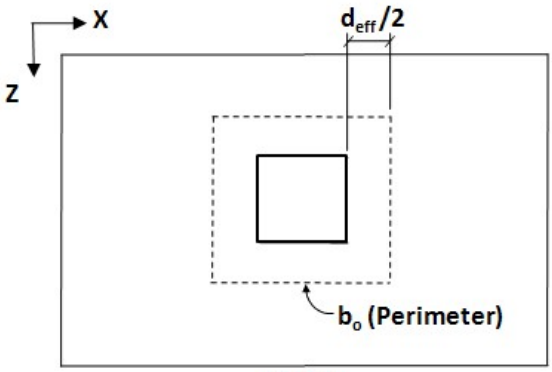
Percentage Of Steel(P_t) = 0.2888

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 388.267 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1765.738 kN

Shear Stress(T_v) = 1025.867 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

$K_s \times T_c$ = 1250.0000 kN/m2

$T_v <= K_s \times T_c$ hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.675 m

$l_{db} >= l_d$ hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

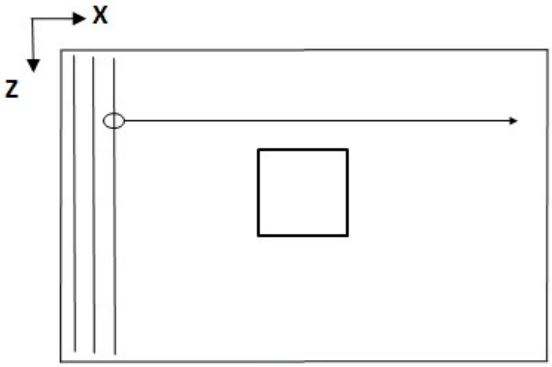
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.775 m

$l_{db} >= l_d$ hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 5527.112 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5527.112 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 56.286 mm

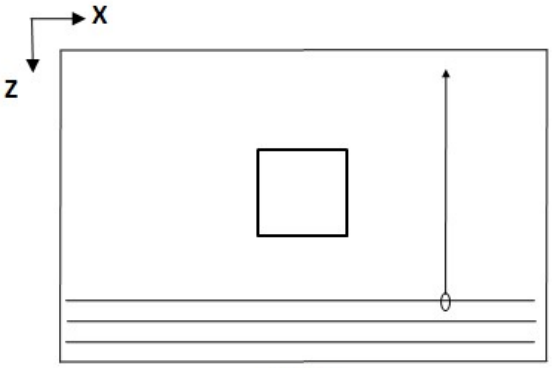
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4955.906 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4955.906 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 62.540 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

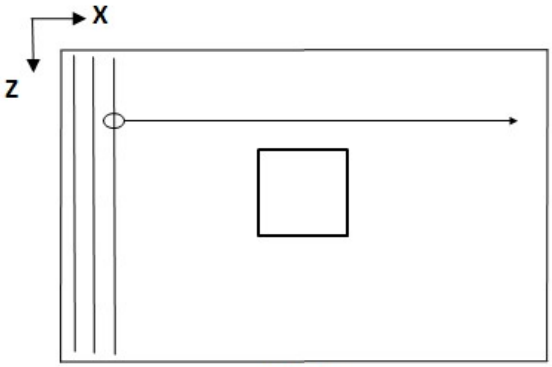
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 1725.300 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1978.020 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 57.140 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 64.656 mm

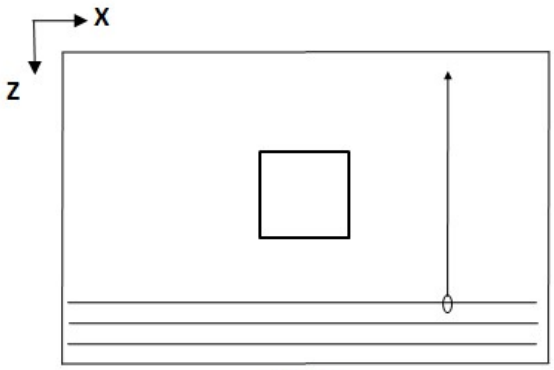
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 60 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin})	= 1978.020 mm2
Calculated Area of Steel (A_{st})	= 4337.671 mm2
Provided Area of Steel ($A_{st,Provided}$)	= 4337.671 mm2
A_{stmin} <= $A_{st,Provided}$	Steel area is accepted
Governing Moment	= 51.050 kNm

Selected bar Size (d_b)	= Ø10
Minimum spacing allowed (S_{min}) =	= 50.000 mm
Selected spacing (S)	= 71.636 mm

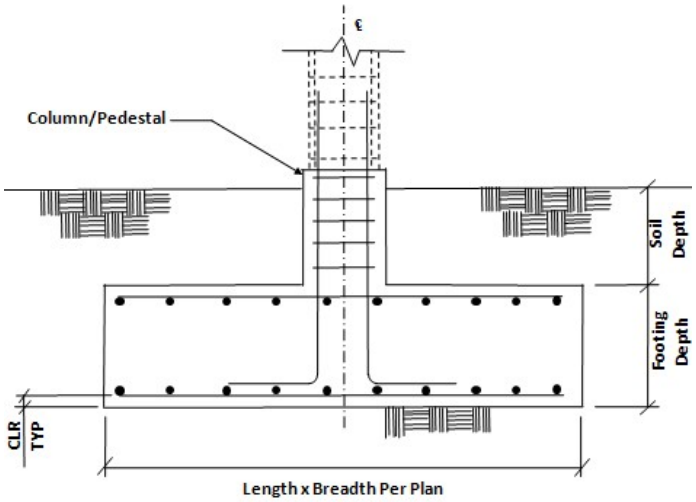
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

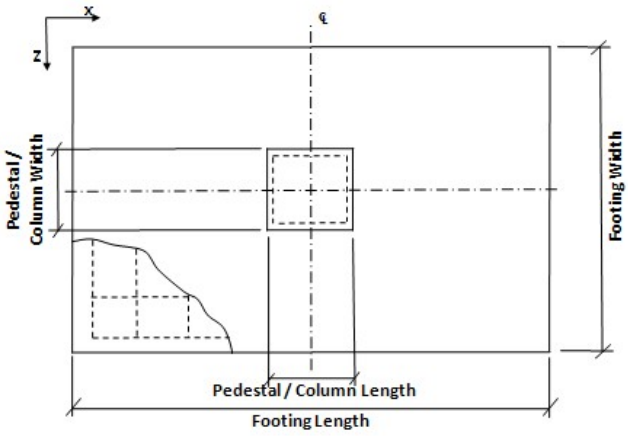
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 78



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1078.239	31.061	36.361	69.904	-78.624
102	862.591	24.849	29.088	55.923	-62.899

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1617.358	46.592	54.541	104.856	-117.936
202	1035.109	29.819	34.906	67.108	-75.479
203	1293.887	37.273	43.633	83.885	-94.349
204	970.415	27.955	32.725	62.913	-70.761

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

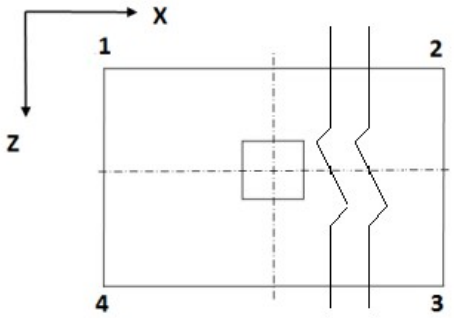
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 10.859 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	3.850	m	Governing Load Case :	# 101
Width (W_2) =	3.850	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	14.823	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**62.5898**	81.1153	98.1470	79.6215	0.000
101	62.5898	**81.1153**	98.1470	79.6215	0.000
101	62.5898	81.1153	**98.1470**	79.6215	0.000
101	62.5898	81.1153	98.1470	**79.6215**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**62.5898**	81.1153	98.1470	79.6215
101	62.5898	**81.1153**	98.1470	79.6215
101	62.5898	81.1153	**98.1470**	79.6215
101	62.5898	81.1153	98.1470	**79.6215**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 14.823 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	19.176	16.381	28.313	26.030

102	19.631	16.770	28.984	26.647

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 31.061 kN
Governing Restoring Force : 595.630 kN
Minimum Sliding Ratio for the Critical Load Case : 19.176
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 80.994 kNm
Governing Resisting Moment : 2293.134 kNm
Minimum Overturning Ratio for the Critical Load Case : 28.313

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 36.361 kN
Governing Restoring Force : 595.630 kN
Minimum Sliding Ratio for the Critical Load Case : 16.381
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -88.097 kNm
Governing Resisting Moment : 2293.134 kNm
Minimum Overturning Ratio for the Critical Load Case : 26.030

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #204

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 405.812 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 842.080760 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #204

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 363.262 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

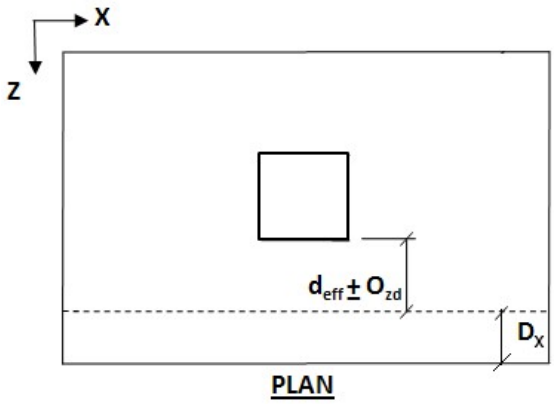
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 842.080760 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 576.683 kN

Shear Stress(T_v) = 331.388989 kN/m2

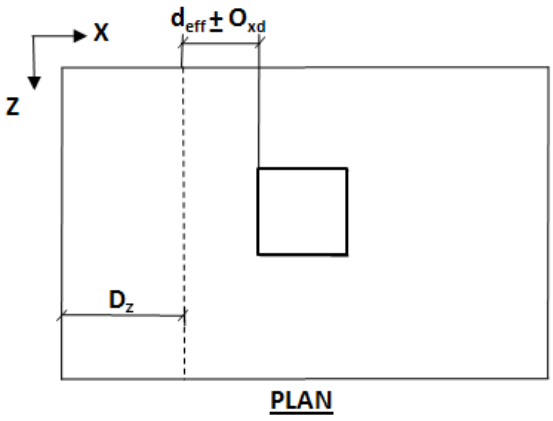
Percentage Of Steel(P_t) = 0.2213

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 345.845 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 536.398 kN

Shear Stress(T_v) = 308.239506 kN/m2

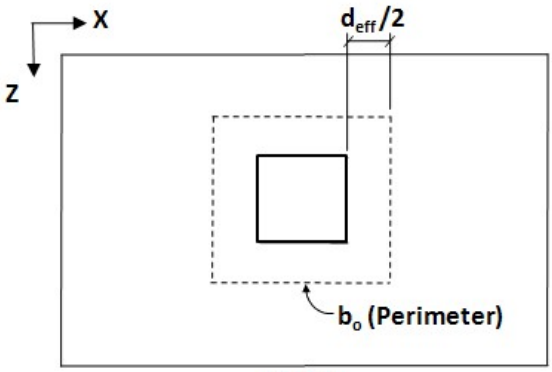
Percentage Of Steel(P_t) = 0.2484

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 363.738 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1519.558 kN

Shear Stress(T_v) = 882.840 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.575 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

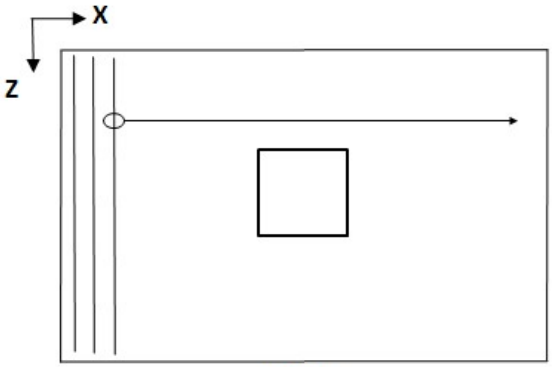
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.675 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #204

Minimum Area of Steel (A_{stmin}) = 1418.340 mm2

Calculated Area of Steel (A_{st}) = 4865.324 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4865.324 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 61.311 mm

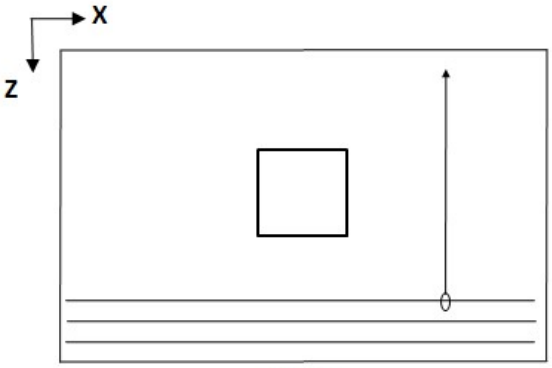
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #204

Minimum Area of Steel (A_{stmin}) = 1418.340 mm2

Calculated Area of Steel (A_{st}) = 4310.560 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4310.560 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 69.259 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

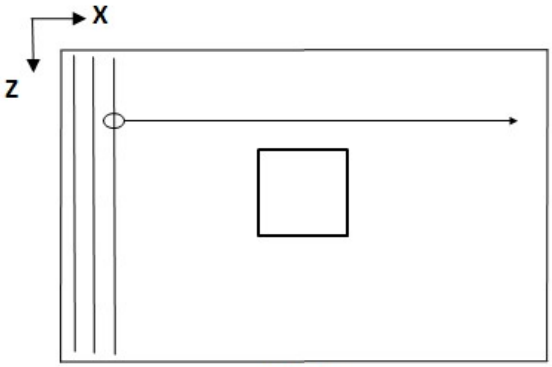
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 65.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1418.340 mm2

Calculated Area of Steel (A_{st}) = 1409.100 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1418.340 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 48.529 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 76.408 mm

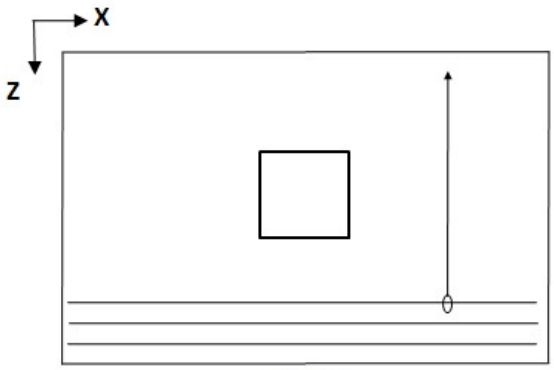
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 75 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1418.340 mm2

Calculated Area of Steel (A_{st}) = 4310.560 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4310.560 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 43.065 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 69.259 mm

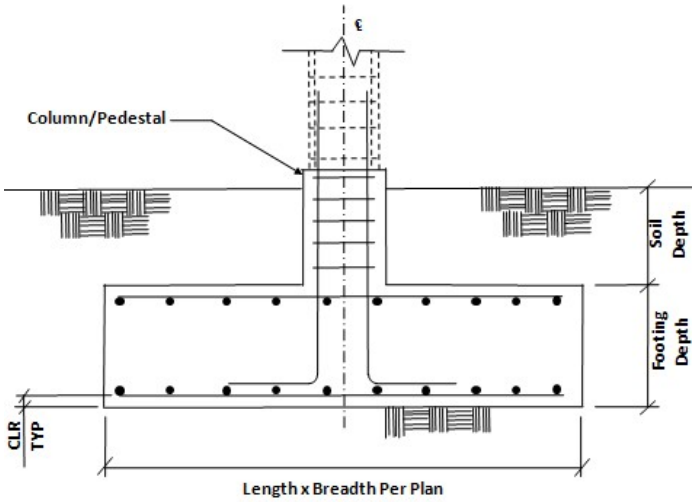
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

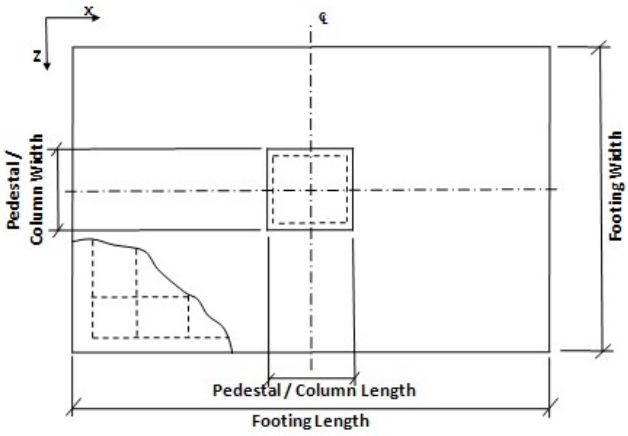
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 65 mm o.c.

Isolated Footing 109



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	767.350	28.720	37.085	70.642	-67.310
102	613.880	22.976	29.668	56.514	-53.848

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1151.025	43.080	55.628	105.963	-100.966
202	736.656	27.571	35.602	67.816	-64.618
203	920.820	34.464	44.503	84.771	-80.773
204	690.615	25.848	33.377	63.578	-60.579

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

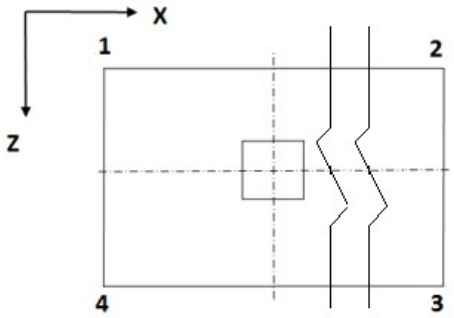
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 7.750 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	3.400	m	Governing Load Case :	# 101
Width (W_2) =	3.400	m	Governing Load Case :	# 101
Depth (D_2) =	0.406	m	Governing Load Case :	# 101
Area (A_2) =	11.560	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**49.8811**	73.1066	98.1283	74.9028	0.000
101	49.8811	**73.1066**	98.1283	74.9028	0.000
101	49.8811	73.1066	**98.1283**	74.9028	0.000
101	49.8811	73.1066	98.1283	**74.9028**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**49.8811**	73.1066	98.1283	74.9028
101	49.8811	**73.1066**	98.1283	74.9028
101	49.8811	73.1066	**98.1283**	74.9028
101	49.8811	73.1066	98.1283	**74.9028**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 11.560 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	14.894	11.534	17.746	19.118

102	15.277	11.831	18.203	19.611

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 28.720 kN
Governing Restoring Force : 427.747 kN
Minimum Sliding Ratio for the Critical Load Case : 14.894
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 81.953 kNm
Governing Resisting Moment : 1454.314 kNm
Minimum Overturning Ratio for the Critical Load Case : 17.746

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 37.085 kN
Governing Restoring Force : 427.747 kN
Minimum Sliding Ratio for the Critical Load Case : 11.534
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -76.070 kNm
Governing Resisting Moment : 1454.314 kNm
Minimum Overturning Ratio for the Critical Load Case : 19.118

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 345.239 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 743.655736 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 299.123 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

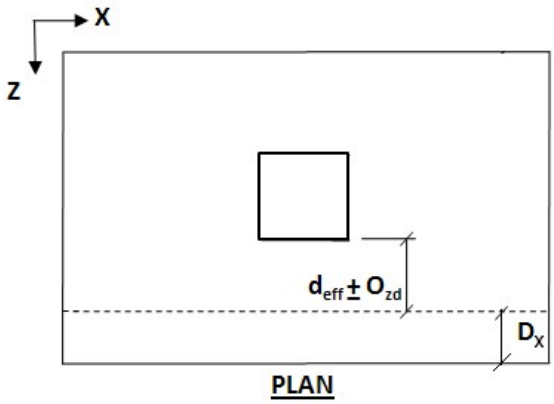
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 743.655736 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.352 m

Shear Force(S) = 437.157 kN

Shear Stress(T_v) = 365.271137 kN/m2

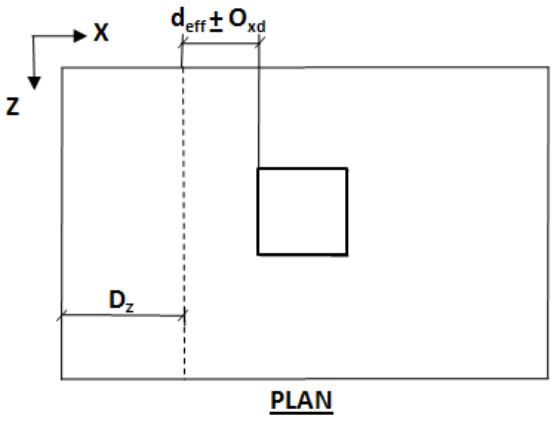
Percentage Of Steel(P_t) = 0.2568

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 369.014 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.352 m

Shear Force(S) = 397.723 kN

Shear Stress(T_v) = 332.321813 kN/m2

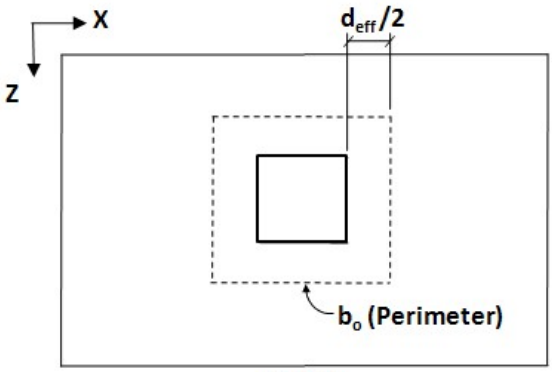
Percentage Of Steel(P_t) = 0.2985

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 393.803 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1079.742 kN

Shear Stress(T_v) = 900.073 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

$K_s \times T_c$ = 1250.0000 kN/m2

$T_v <= K_s \times T_c$ hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

$$\text{Development Length}(l_d) = \frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}} = 1.289 \text{ m}$$

$$\text{Allowable Length}(l_{db}) = \left[\frac{(B - b)}{2} - cc\right] = 1.350 \text{ m}$$

$l_{db} >= l_d$ hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

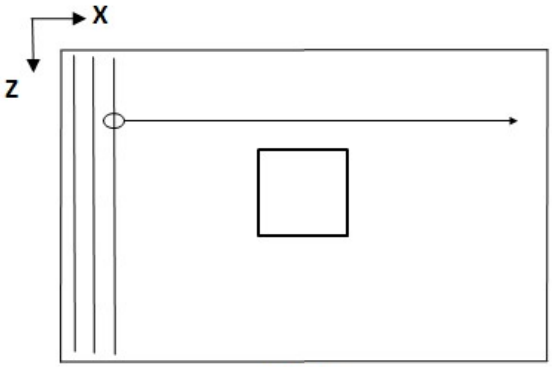
$$\text{Development Length}(l_d) = \frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}} = 1.289 \text{ m}$$

$$\text{Allowable Length}(l_{db}) = \left[\frac{(H - h)}{2} - cc\right] = 1.450 \text{ m}$$

$l_{db} >= l_d$ hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 4124.069 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4124.069 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 63.269 mm

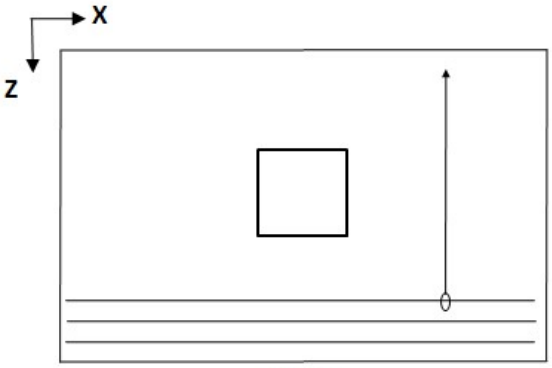
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 3528.967 mm2

Provided Area of Steel ($A_{st,Provided}$) = 3528.967 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 74.773 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

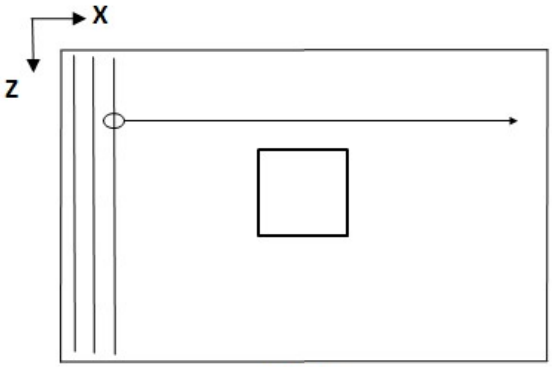
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 1244.400 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1252.560 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 32.406 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 74.864 mm

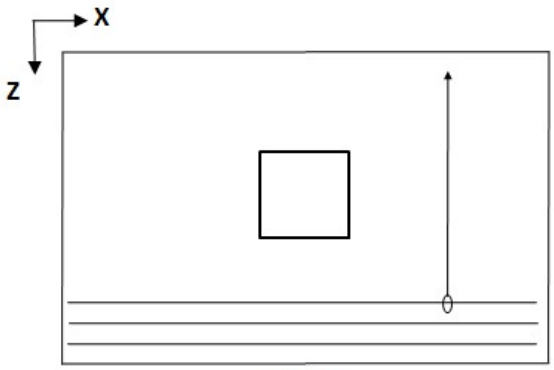
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 70 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1252.560 mm2

Calculated Area of Steel (A_{st}) = 2596.369 mm2

Provided Area of Steel ($A_{st,Provided}$) = 2596.369 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 28.229 kNm

Selected bar Size (d_b) = Ø8

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 64.549 mm

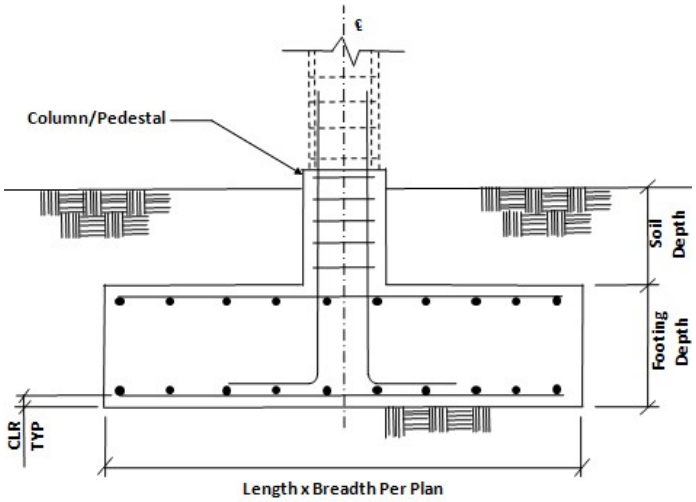
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

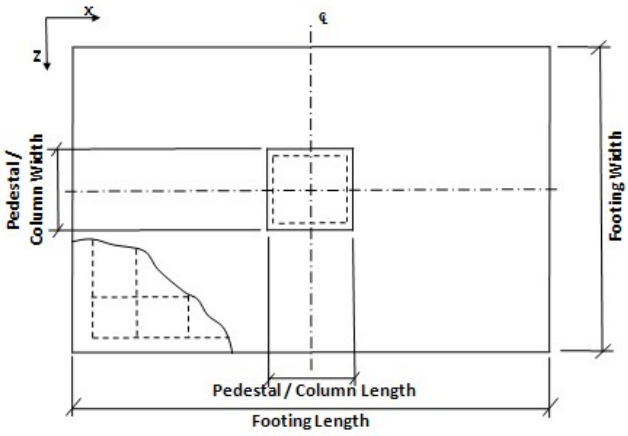
Based on spacing reinforcement increment; provided reinforcement is

Ø8 @ 60 mm o.c.

Isolated Footing 110



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1214.051	37.967	39.199	72.966	-81.463
102	971.241	30.374	31.359	58.373	-65.171

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1821.076	56.950	58.798	109.449	-122.195
202	1165.489	36.448	37.631	70.047	-78.205
203	1456.861	45.560	47.038	87.559	-97.756
204	1092.646	34.170	35.279	65.669	-73.317

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

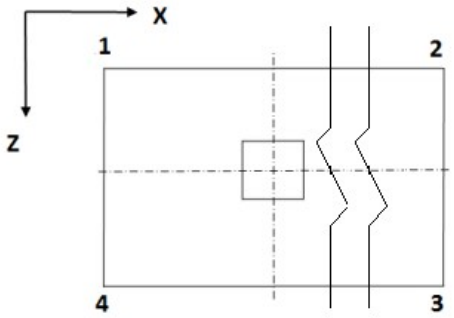
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.217 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.050	m	Governing Load Case :	# 101
Width (W_2) =	4.050	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.403	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**65.5671**	82.3748	97.7153	80.9076	0.000
101	65.5671	**82.3748**	97.7153	80.9076	0.000
101	65.5671	82.3748	**97.7153**	80.9076	0.000
101	65.5671	82.3748	97.7153	**80.9076**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**65.5671**	82.3748	97.7153	80.9076
101	65.5671	**82.3748**	97.7153	80.9076
101	65.5671	82.3748	**97.7153**	80.9076
101	65.5671	82.3748	97.7153	**80.9076**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A

Plan area of footing = 16.403 sq.m

Area not in contact with soil = 0.000 sq.m

% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	17.635	17.081	31.932	29.144

102	18.047	17.480	32.677	29.825

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 37.967 kN
Governing Restoring Force : 669.560 kN
Minimum Sliding Ratio for the Critical Load Case : 17.635
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 84.921 kNm
Governing Resisting Moment : 2711.668 kNm
Minimum Overturning Ratio for the Critical Load Case : 31.932

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 39.199 kN
Governing Restoring Force : 669.560 kN
Minimum Sliding Ratio for the Critical Load Case : 17.081
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -93.043 kNm
Governing Resisting Moment : 2711.668 kNm
Minimum Overturning Ratio for the Critical Load Case : 29.144

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 642.489 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 578.729 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

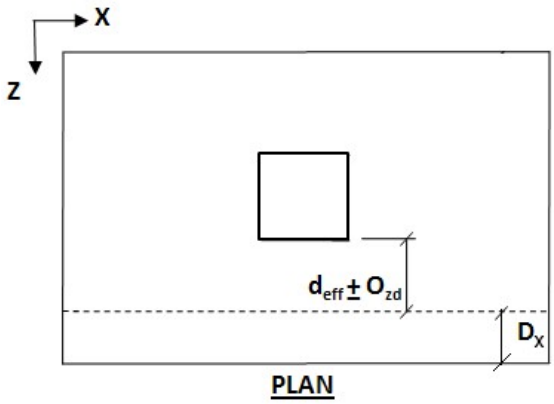
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 659.656 kN

Shear Stress(T_v) = 360.349386 kN/m2

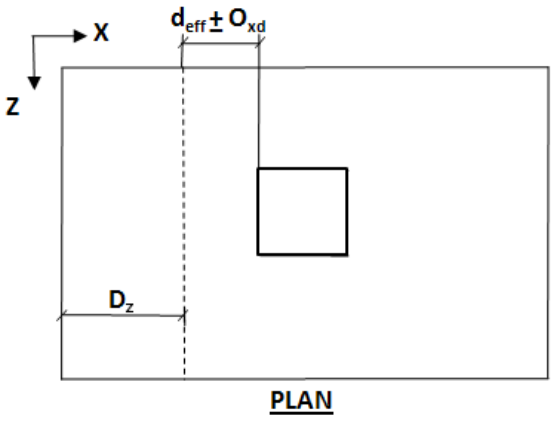
Percentage Of Steel(P_t) = 0.2528

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 366.496 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 616.965 kN

Shear Stress(T_v) = 337.029075 kN/m2

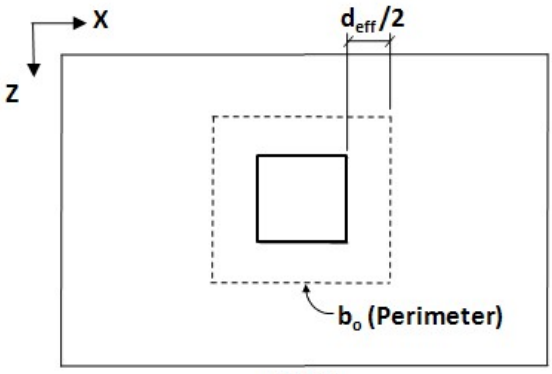
Percentage Of Steel(P_t) = 0.2820

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 384.300 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1721.565 kN

Shear Stress(T_v) = 1000.203 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.675 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

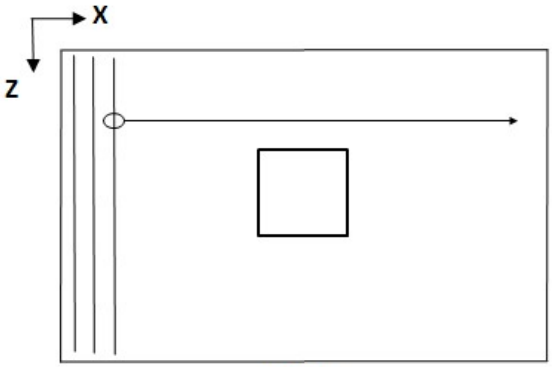
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.775 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 5394.326 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5394.326 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 57.941 mm

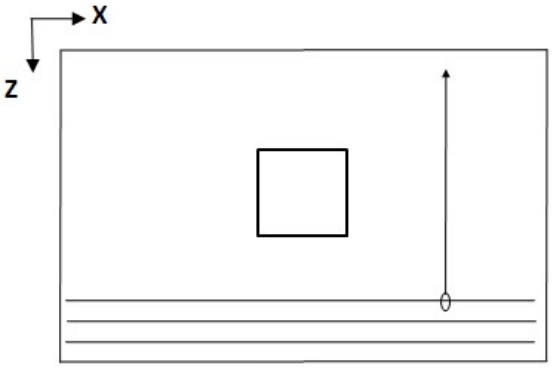
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4824.861 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4824.861 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 64.590 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

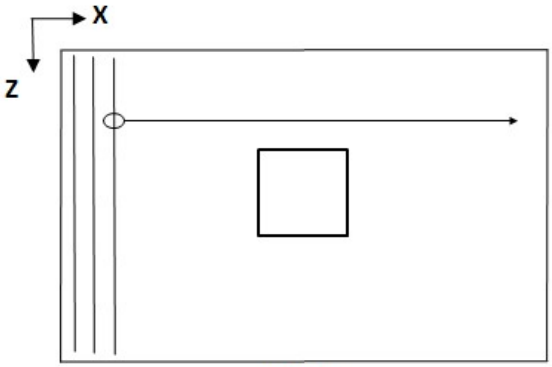
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 1725.300 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1978.020 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 57.140 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 64.656 mm

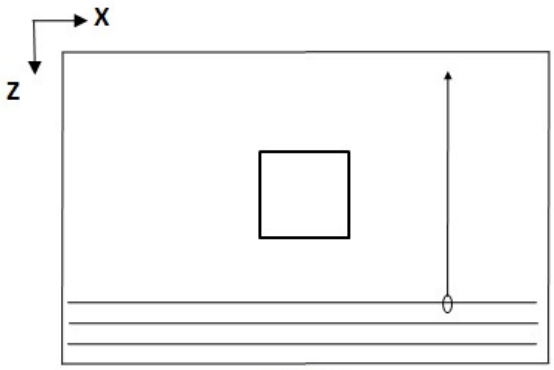
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 60 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4222.818 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4222.818 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 51.050 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 74.340 mm

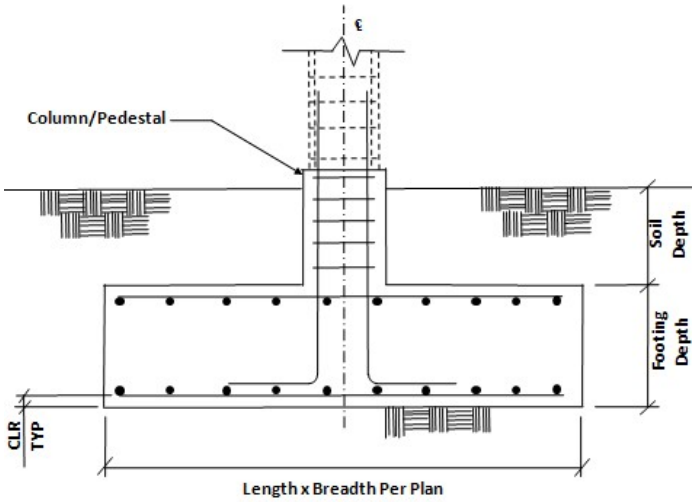
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

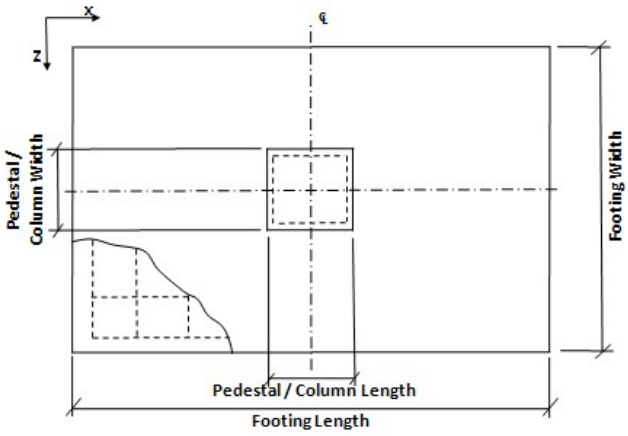
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 111



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1249.806	38.481	39.223	73.009	-82.046
102	999.845	30.785	31.378	58.407	-65.637

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1874.709	57.722	58.834	109.513	-123.069
202	1199.814	36.942	37.654	70.089	-78.764
203	1499.767	46.177	47.067	87.611	-98.455
204	1124.825	34.633	35.300	65.708	-73.841

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

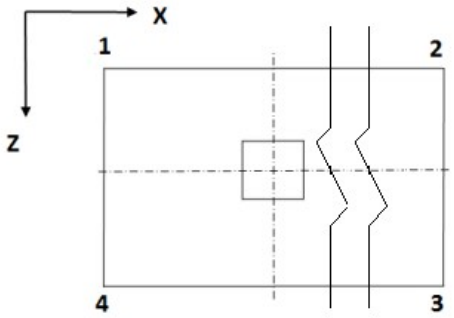
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.574 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.050	m	Governing Load Case :	# 101
Width (W_2) =	4.050	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.403	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**67.6756**	84.6169	99.9665	83.0253	0.000
101	67.6756	**84.6169**	99.9665	83.0253	0.000
101	67.6756	84.6169	**99.9665**	83.0253	0.000
101	67.6756	84.6169	99.9665	**83.0253**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**67.6756**	84.6169	99.9665	83.0253
101	67.6756	**84.6169**	99.9665	83.0253
101	67.6756	84.6169	**99.9665**	83.0253
101	67.6756	84.6169	99.9665	**83.0253**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 16.403 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	17.864	17.527	32.765	29.687

102	18.271	17.925	33.510	30.362

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 38.481 kN
Governing Restoring Force : 687.438 kN
Minimum Sliding Ratio for the Critical Load Case : 17.864
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 84.972 kNm
Governing Resisting Moment : 2784.071 kNm
Minimum Overturning Ratio for the Critical Load Case : 32.765

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 39.223 kN
Governing Restoring Force : 687.438 kN
Minimum Sliding Ratio for the Critical Load Case : 17.527
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -93.782 kNm
Governing Resisting Moment : 2784.071 kNm
Minimum Overturning Ratio for the Critical Load Case : 29.687

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 660.157 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 594.837 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

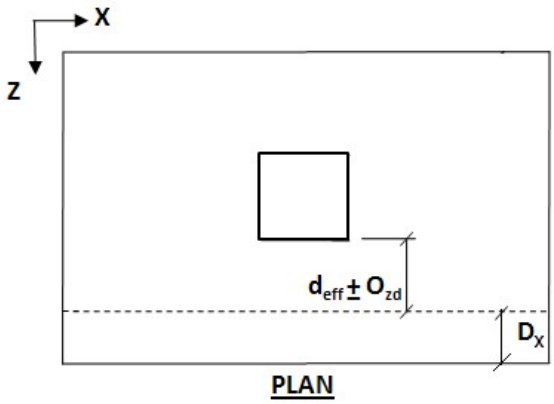
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 677.863 kN

Shear Stress(T_v) = 370.295493 kN/m2

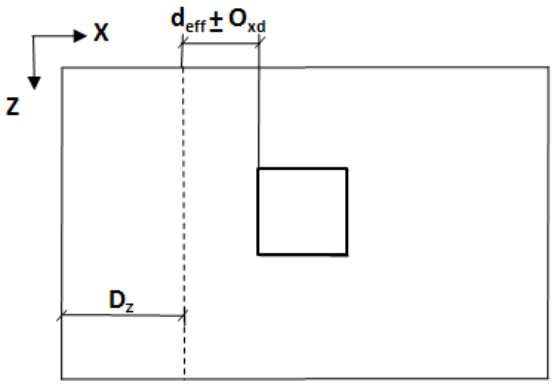
Percentage Of Steel(P_t) = 0.2601

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 371.096 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 634.177 kN

Shear Stress(T_v) = 346.431436 kN/m2

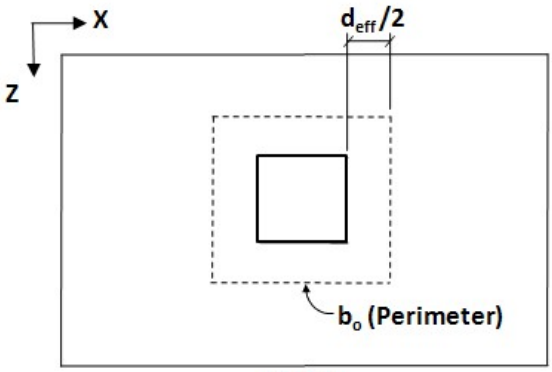
Percentage Of Steel(P_t) = 0.2902

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 389.052 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1772.267 kN

Shear Stress(T_v) = 1029.660 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

$K_s \times T_c$ = 1250.0000 kN/m2

$T_v <= K_s \times T_c$ hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.675 m

$l_{db} >= l_d$ hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

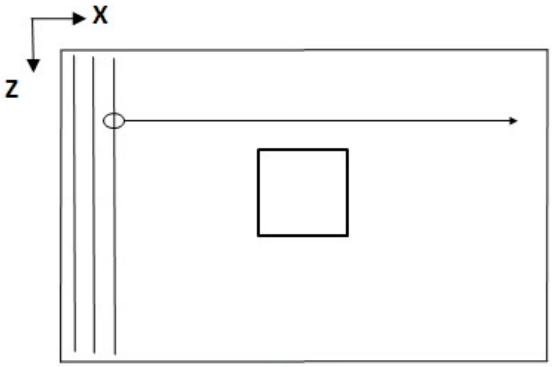
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.775 m

$l_{db} >= l_d$ hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 5553.661 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5553.661 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 56.286 mm

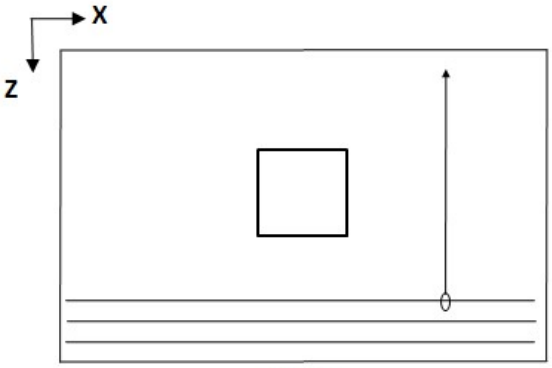
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4967.919 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4967.919 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 62.540 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

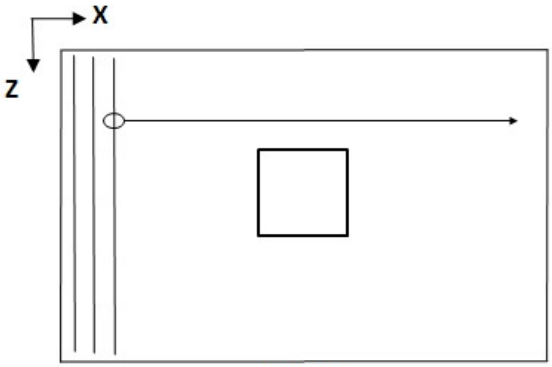
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 1725.300 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1978.020 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 57.140 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 64.656 mm

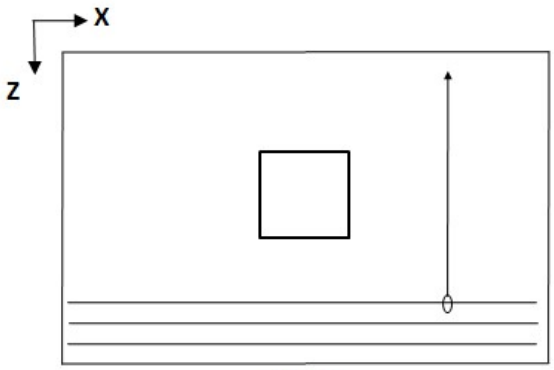
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 60 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4348.201 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4348.201 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 51.050 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 71.636 mm

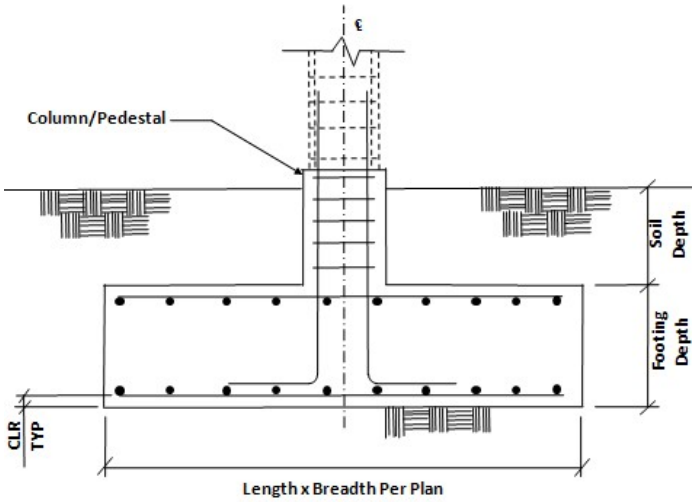
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

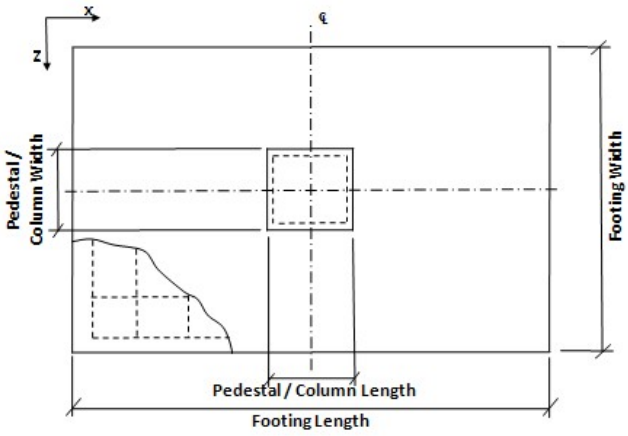
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 112



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1249.831	38.660	39.224	73.011	-82.258
102	999.865	30.928	31.380	58.409	-65.806

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1874.746	57.990	58.837	109.517	-123.387
202	1199.837	37.114	37.655	70.091	-78.968
203	1499.797	46.392	47.069	87.613	-98.710
204	1124.848	34.794	35.302	65.710	-74.032

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

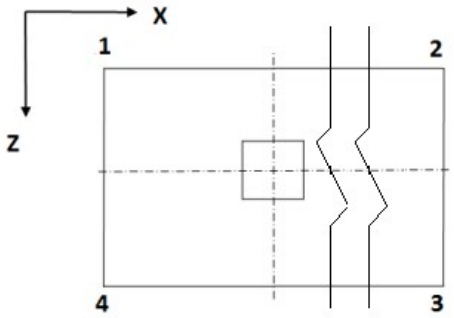
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.575 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.050	m	Governing Load Case :	# 101
Width (W_2) =	4.050	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.403	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**67.6528**	84.6422	99.9924	83.0029	0.000
101	67.6528	**84.6422**	99.9924	83.0029	0.000
101	67.6528	84.6422	**99.9924**	83.0029	0.000
101	67.6528	84.6422	99.9924	**83.0029**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**67.6528**	84.6422	99.9924	83.0029
101	67.6528	**84.6422**	99.9924	83.0029
101	67.6528	84.6422	**99.9924**	83.0029
101	67.6528	84.6422	99.9924	**83.0029**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A

Plan area of footing = 16.403 sq.m

Area not in contact with soil = 0.000 sq.m

% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	17.782	17.526	32.764	29.603

102	18.186	17.925	33.509	30.276

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 38.660 kN
Governing Restoring Force : 687.450 kN
Minimum Sliding Ratio for the Critical Load Case : 17.782
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 84.974 kNm
Governing Resisting Moment : 2784.121 kNm
Minimum Overturning Ratio for the Critical Load Case : 32.764

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 39.224 kN
Governing Restoring Force : 687.450 kN
Minimum Sliding Ratio for the Critical Load Case : 17.526
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -94.049 kNm
Governing Resisting Moment : 2784.121 kNm
Minimum Overturning Ratio for the Critical Load Case : 29.603

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 660.170 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 594.972 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

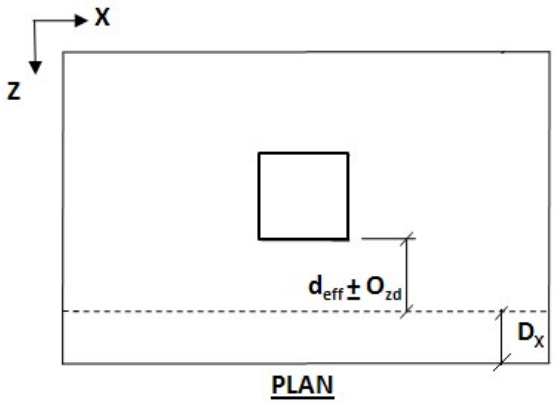
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 677.877 kN

Shear Stress(T_v) = 370.303058 kN/m2

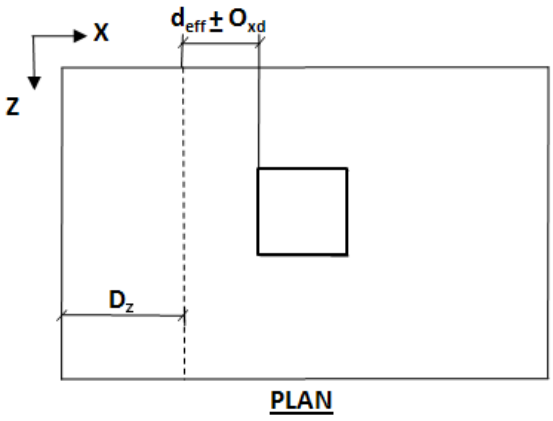
Percentage Of Steel(P_t) = 0.2602

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 371.135 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 634.317 kN

Shear Stress(T_v) = 346.507614 kN/m2

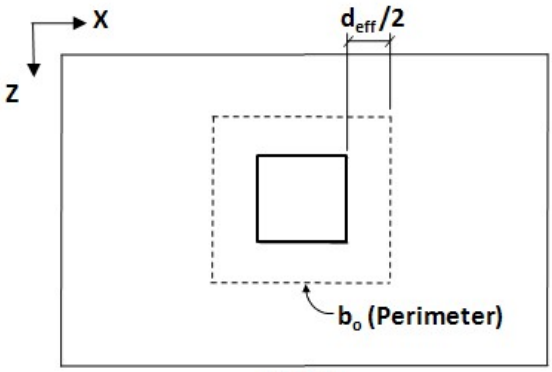
Percentage Of Steel(P_t) = 0.2902

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 389.056 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1772.302 kN

Shear Stress(T_v) = 1029.680 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

$K_s \times T_c$ = 1250.0000 kN/m2

$T_v <= K_s \times T_c$ hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.675 m

$l_{db} >= l_d$ hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

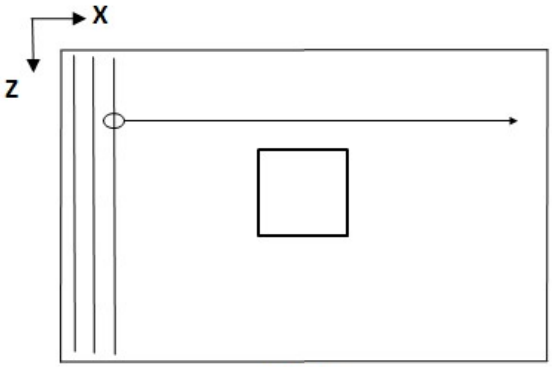
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.775 m

$l_{db} >= l_d$ hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 5553.783 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5553.783 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 56.286 mm

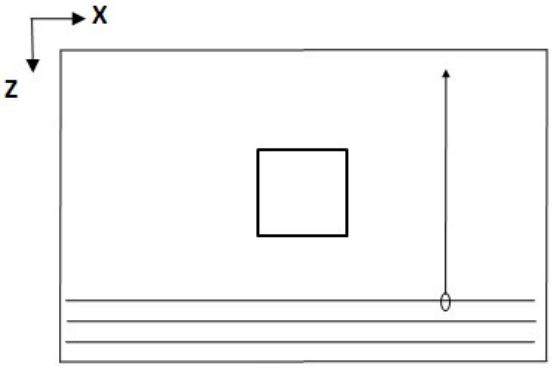
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4969.127 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4969.127 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 62.540 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

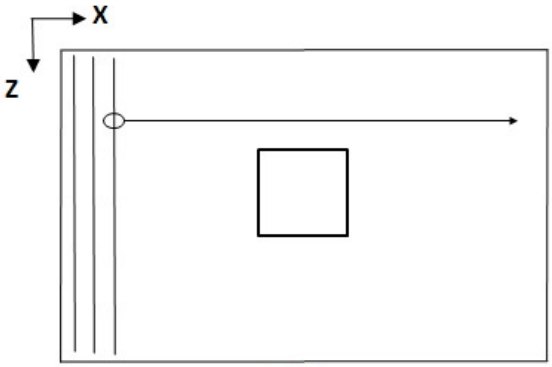
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 1725.300 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1978.020 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 57.140 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 64.656 mm

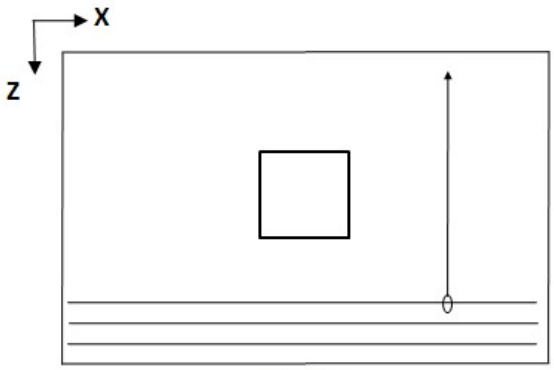
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 60 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4349.259 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4349.259 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 51.050 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 71.636 mm

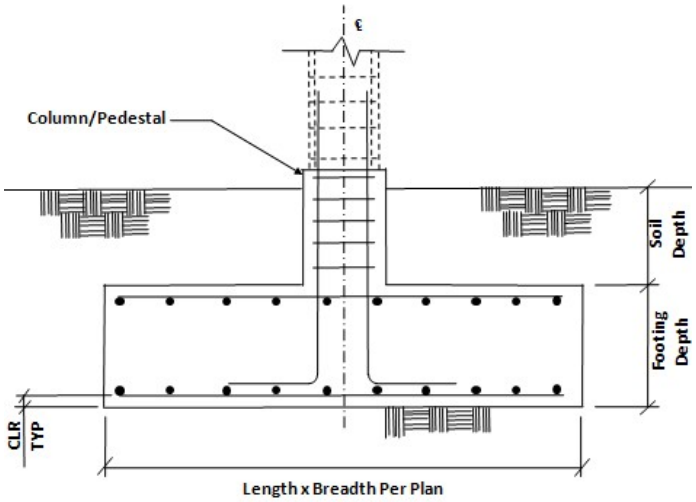
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

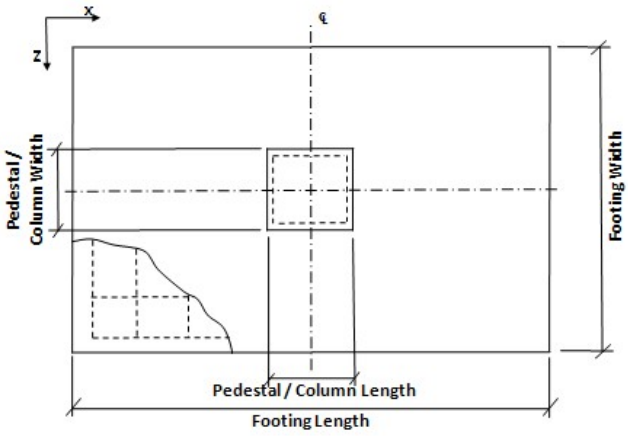
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 113



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1221.320	39.502	39.194	72.965	-83.138
102	977.056	31.601	31.355	58.372	-66.511

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1831.979	59.253	58.791	109.448	-124.707
202	1172.467	37.922	37.626	70.047	-79.813
203	1465.584	47.402	47.033	87.558	-99.766
204	1099.188	35.552	35.274	65.669	-74.824

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

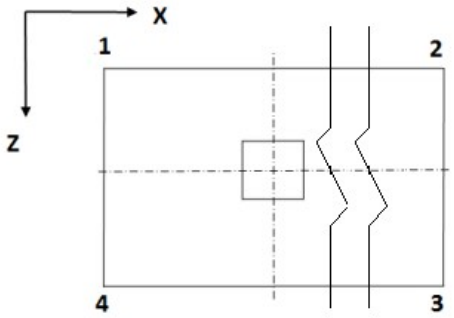
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.289 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.050	m	Governing Load Case :	# 101
Width (W_2) =	4.050	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.403	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**65.8169**	83.0117	98.3519	81.1570	0.000
101	65.8169	**83.0117**	98.3519	81.1570	0.000
101	65.8169	83.0117	**98.3519**	81.1570	0.000
101	65.8169	83.0117	98.3519	**81.1570**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**65.8169**	83.0117	98.3519	81.1570
101	65.8169	**83.0117**	98.3519	81.1570
101	65.8169	83.0117	**98.3519**	81.1570
101	65.8169	83.0117	98.3519	**81.1570**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 16.403 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	17.042	17.176	32.106	28.643

102	17.438	17.575	32.851	29.308

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 39.502 kN
Governing Restoring Force : 673.194 kN
Minimum Sliding Ratio for the Critical Load Case : 17.042
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 84.919 kNm
Governing Resisting Moment : 2726.387 kNm
Minimum Overturning Ratio for the Critical Load Case : 32.106

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 39.194 kN
Governing Restoring Force : 673.194 kN
Minimum Sliding Ratio for the Critical Load Case : 17.176
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -95.186 kNm
Governing Resisting Moment : 2726.387 kNm
Minimum Overturning Ratio for the Critical Load Case : 28.643

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 646.074 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 582.936 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

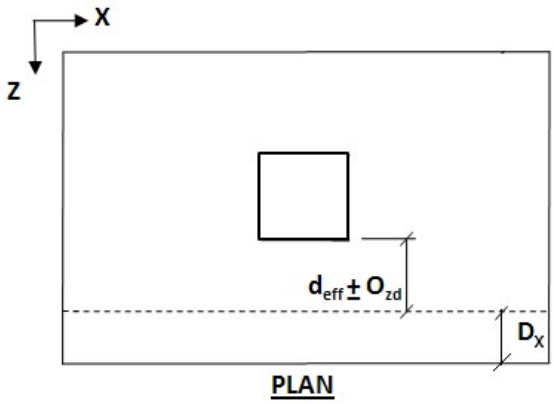
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 663.351 kN

Shear Stress(T_v) = 362.368054 kN/m2

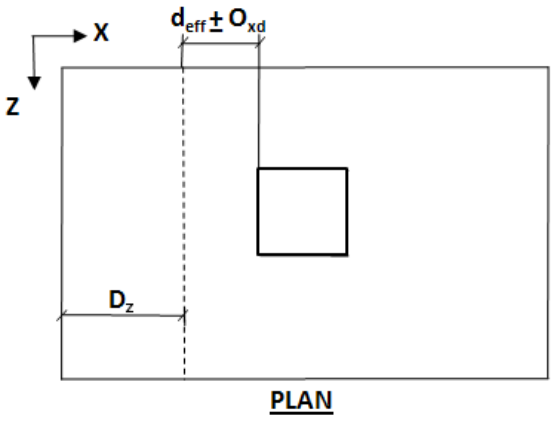
Percentage Of Steel(P_t) = 0.2547

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 367.705 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 621.419 kN

Shear Stress(T_v) = 339.461884 kN/m2

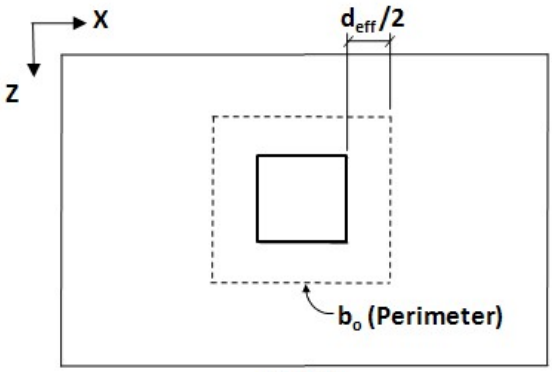
Percentage Of Steel(P_t) = 0.2837

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 385.270 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1731.872 kN

Shear Stress(T_v) = 1006.191 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

$K_s \times T_c$ = 1250.0000 kN/m2

$T_v <= K_s \times T_c$ hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.675 m

$l_{db} >= l_d$ hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

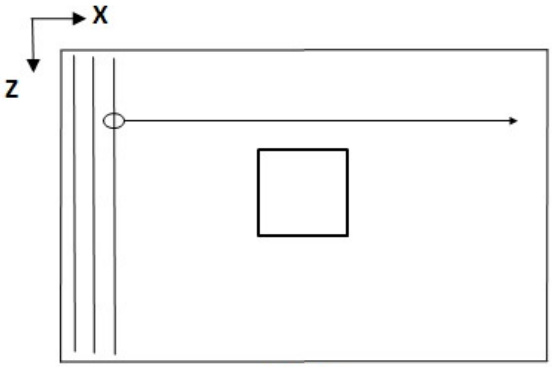
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.775 m

$l_{db} >= l_d$ hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 5426.607 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5426.607 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 57.101 mm

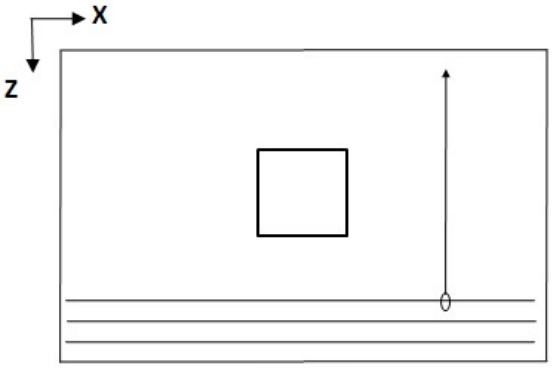
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4862.174 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4862.174 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 64.590 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

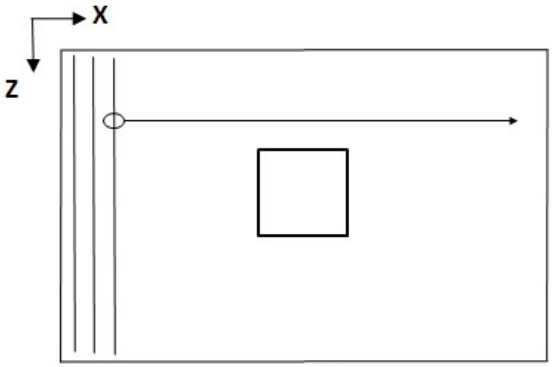
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 1725.300 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1978.020 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 57.140 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 64.656 mm

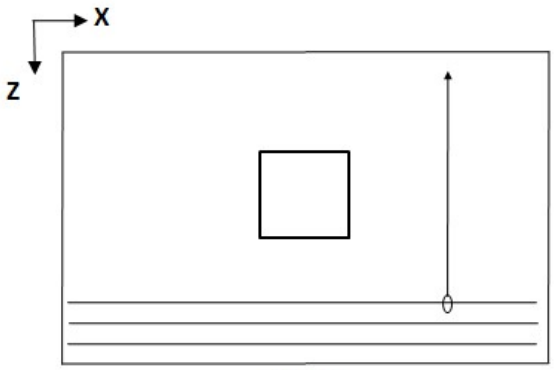
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 60 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4255.520 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4255.520 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 51.050 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 72.963 mm

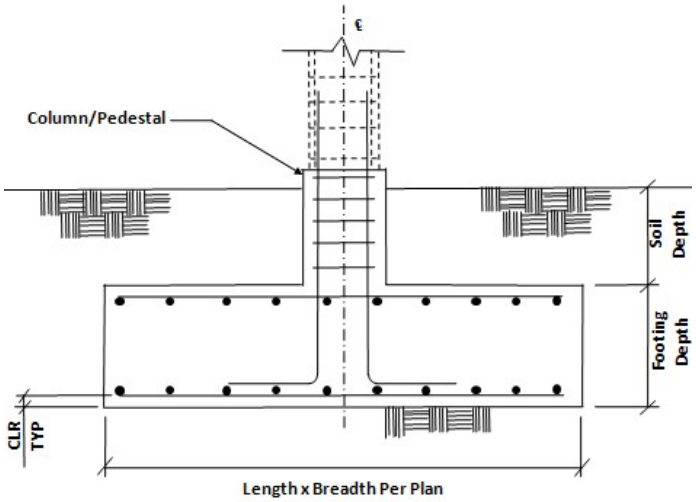
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

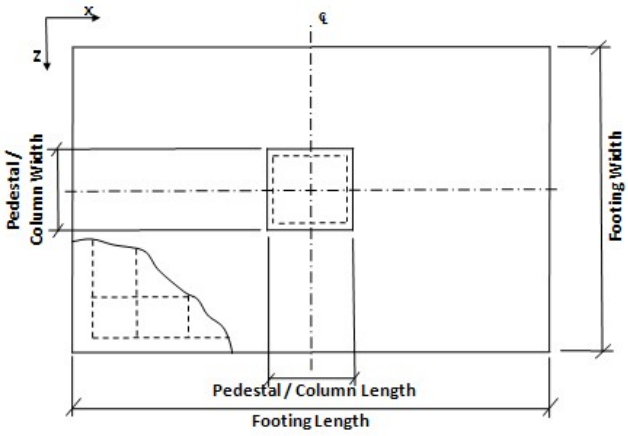
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 114



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1061.574	30.922	37.053	70.641	-78.479
102	849.259	24.737	29.642	56.513	-62.783

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1592.360	46.383	55.579	105.962	-117.719
202	1019.111	29.685	35.571	67.816	-75.340
203	1273.888	37.106	44.463	84.770	-94.175
204	955.416	27.830	33.347	63.577	-70.631

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

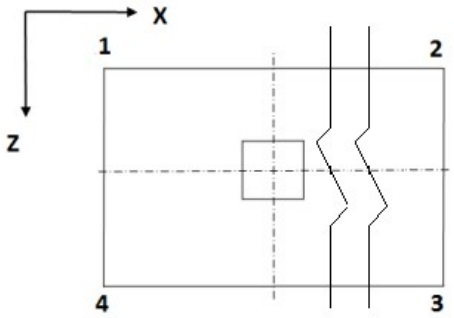
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 10.692 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	3.800	m	Governing Load Case :	# 101
Width (W_2) =	3.800	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	14.440	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**62.5682**	81.7937	99.7141	80.4886	0.000
101	62.5682	**81.7937**	99.7141	80.4886	0.000
101	62.5682	81.7937	**99.7141**	80.4886	0.000
101	62.5682	81.7937	99.7141	**80.4886**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**62.5682**	81.7937	99.7141	80.4886
101	62.5682	**81.7937**	99.7141	80.4886
101	62.5682	81.7937	**99.7141**	80.4886
101	62.5682	81.7937	99.7141	**80.4886**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 14.440 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	18.946	15.811	27.167	25.323

102	19.391	16.182	27.805	25.918

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 30.922 kN
Governing Restoring Force : 585.839 kN
Minimum Sliding Ratio for the Critical Load Case : 18.946
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 81.942 kNm
Governing Resisting Moment : 2226.148 kNm
Minimum Overturning Ratio for the Critical Load Case : 27.167

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 37.053 kN
Governing Restoring Force : 585.839 kN
Minimum Sliding Ratio for the Critical Load Case : 15.811
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -87.910 kNm
Governing Resisting Moment : 2226.148 kNm
Minimum Overturning Ratio for the Critical Load Case : 25.323

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #204

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 394.396 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 831.144646 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #204

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 352.111 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

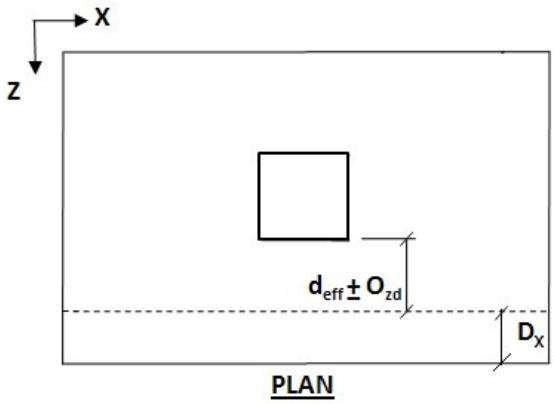
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 831.144646 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 565.770 kN

Shear Stress(T_v) = 329.395927 kN/m2

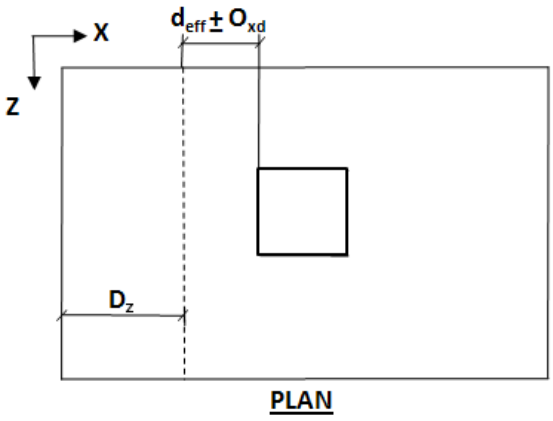
Percentage Of Steel(P_t) = 0.2172

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 343.000 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 524.959 kN

Shear Stress(T_v) = 305.635388 kN/m2

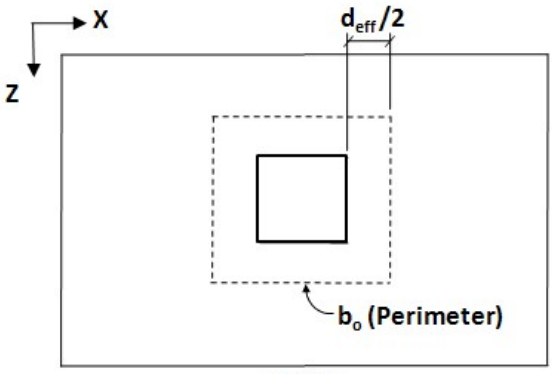
Percentage Of Steel(P_t) = 0.2444

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 361.189 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1493.521 kN

Shear Stress(T_v) = 867.713 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.550 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

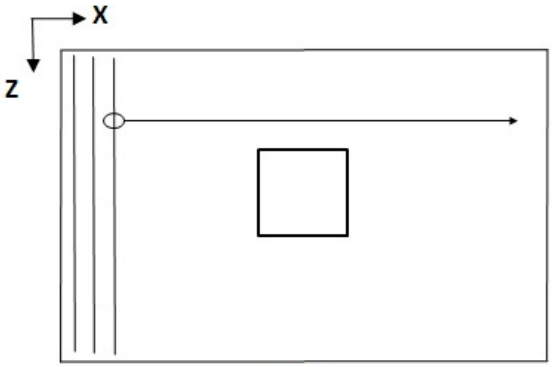
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.650 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #204

Minimum Area of Steel (A_{stmin}) = 1399.920 mm2

Calculated Area of Steel (A_{st}) = 4721.235 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4721.235 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 61.500 mm

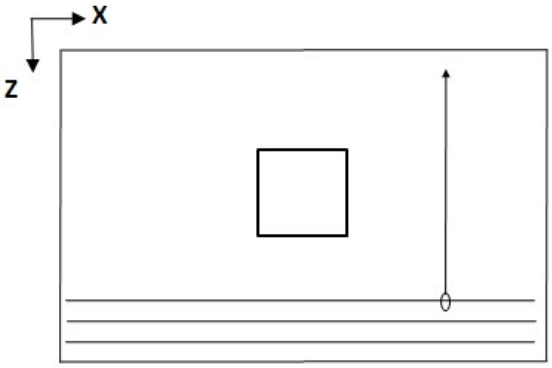
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #204

Minimum Area of Steel (A_{stmin}) = 1399.920 mm2

Calculated Area of Steel (A_{st}) = 4171.772 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4171.772 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 69.623 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

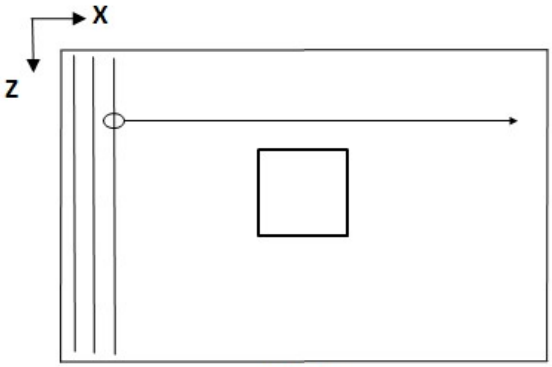
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 65.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1399.920 mm2

Calculated Area of Steel (A_{st}) = 1390.800 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1399.920 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 46.520 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 75.388 mm

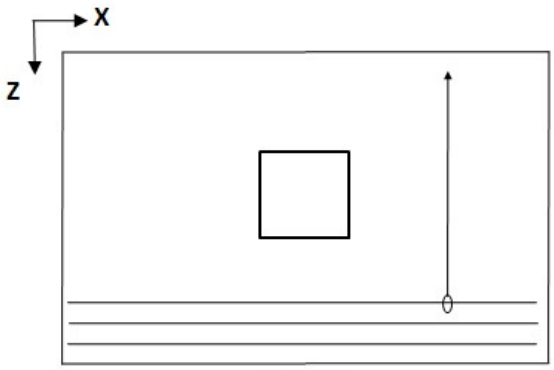
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 75 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1399.920 mm2

Calculated Area of Steel (A_{st}) = 4171.772 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4171.772 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 41.208 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 69.623 mm

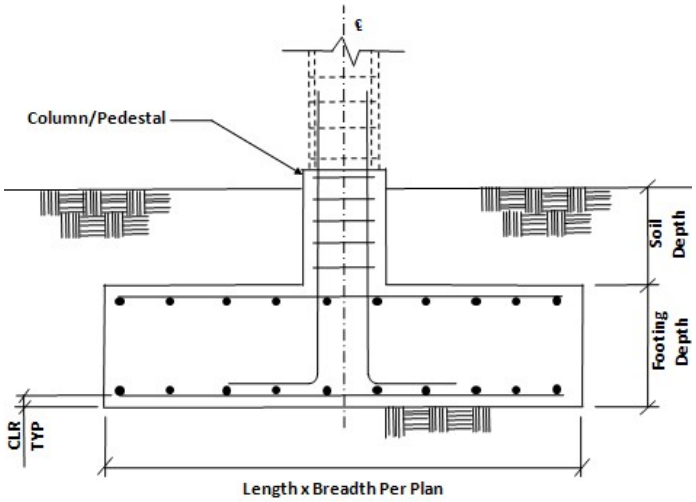
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

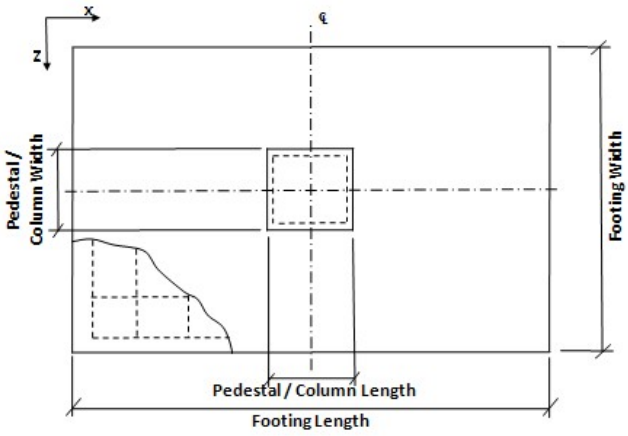
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 65 mm o.c.

Isolated Footing 145



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	830.799	24.305	27.113	62.071	-64.642
102	664.639	19.444	21.690	49.657	-51.714

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1246.198	36.458	40.669	93.106	-96.963
202	797.567	23.333	26.028	59.588	-62.056
203	996.958	29.167	32.535	74.485	-77.571
204	747.719	21.875	24.401	55.864	-58.178

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

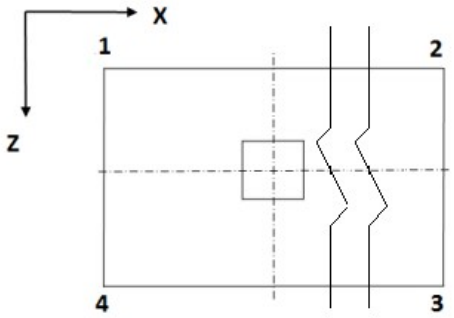
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 8.384 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	3.450	m	Governing Load Case :	# 101
Width (W_2) =	3.450	m	Governing Load Case :	# 101
Depth (D_2) =	0.456	m	Governing Load Case :	# 101
Area (A_2) =	11.903	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**56.6189**	77.6759	98.2318	77.1748	0.000
101	56.6189	**77.6759**	98.2318	77.1748	0.000
101	56.6189	77.6759	**98.2318**	77.1748	0.000
101	56.6189	77.6759	98.2318	**77.1748**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**56.6189**	77.6759	98.2318	77.1748
101	56.6189	**77.6759**	98.2318	77.1748
101	56.6189	77.6759	**98.2318**	77.1748
101	56.6189	77.6759	98.2318	**77.1748**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 11.903 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	18.958	16.995	22.599	22.062

102	19.425	17.413	23.156	22.605

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 24.305 kN
Governing Restoring Force : 460.778 kN
Minimum Sliding Ratio for the Critical Load Case : 18.958
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 70.340 kNm
Governing Resisting Moment : 1589.654 kNm
Minimum Overturning Ratio for the Critical Load Case : 22.599

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 27.113 kN
Governing Restoring Force : 460.778 kN
Minimum Sliding Ratio for the Critical Load Case : 16.995
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -72.055 kNm
Governing Resisting Moment : 1589.654 kNm
Minimum Overturning Ratio for the Critical Load Case : 22.062

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.302 m

Governing moment (M_u) = 370.936 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1083.739521 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.302 m

Governing moment (M_u) = 325.490 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

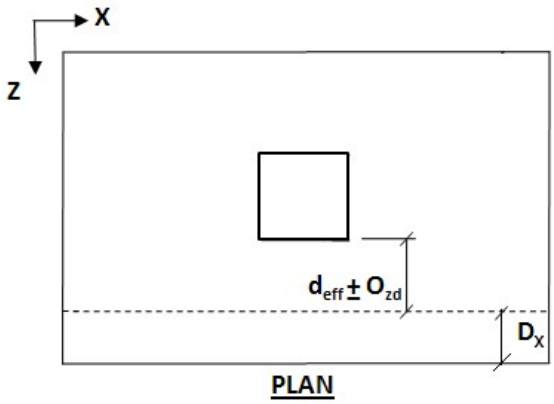
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1083.739521 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.402 m

Shear Force(S) = 445.934 kN

Shear Stress(T_v) = 321.533023 kN/m2

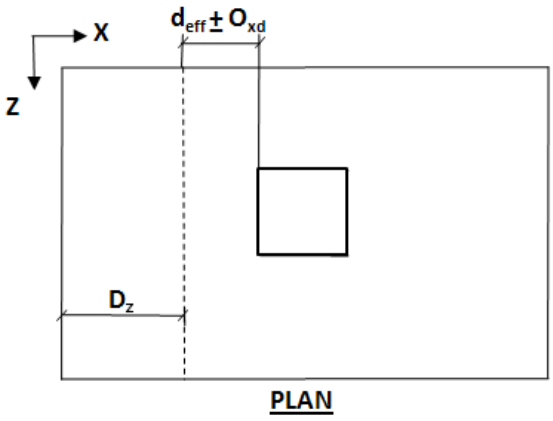
Percentage Of Steel(P_t) = 0.2094

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 337.530 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.402 m

Shear Force(S) = 408.735 kN

Shear Stress(T_v) = 294.711467 kN/m2

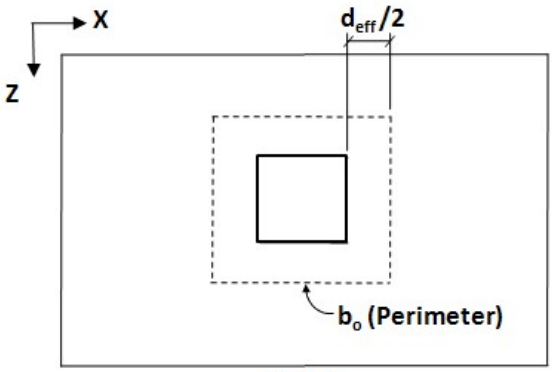
Percentage Of Steel(P_t) = 0.2399

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 358.256 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1162.060 kN

Shear Stress(T_v) = 801.191 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.375 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

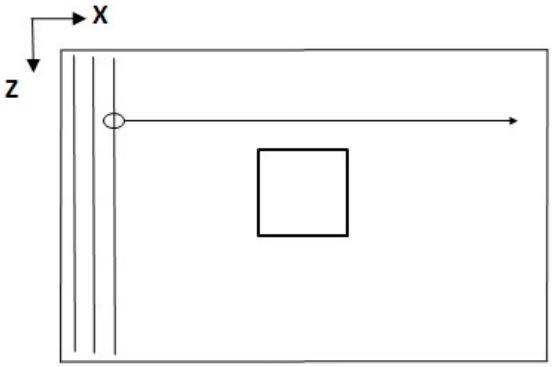
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.475 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1477.980 mm2

Calculated Area of Steel (A_{st}) = 3609.574 mm2

Provided Area of Steel ($A_{st,Provided}$) = 3609.574 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 74.222 mm

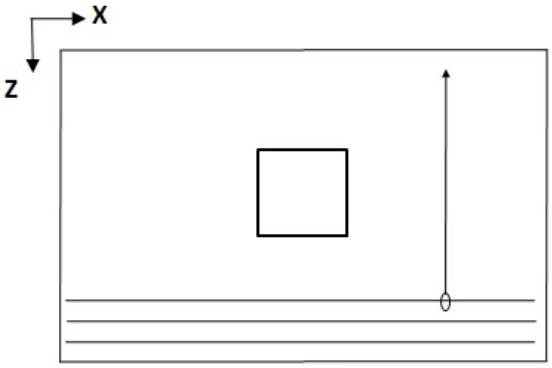
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1473.840 mm2

Calculated Area of Steel (A_{st}) = 3142.521 mm2

Provided Area of Steel ($A_{st,Provided}$) = 3142.521 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø8

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 53.903 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

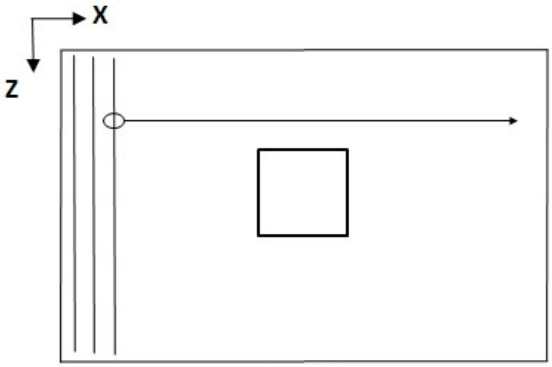
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø8 @ 50.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1477.980 mm2

Calculated Area of Steel (A_{st}) = 1262.700 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1477.980 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 33.987 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 76.000 mm

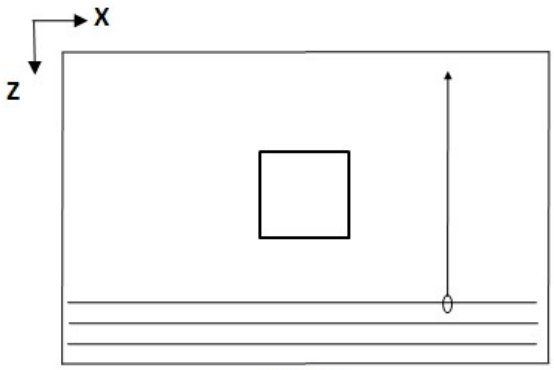
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 75 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1473.840 mm2

Calculated Area of Steel (A_{st}) = 2836.759 mm2

Provided Area of Steel ($A_{st,Provided}$) = 2836.759 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 29.676 kNm

Selected bar Size (d_b) = Ø8

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 59.679 mm

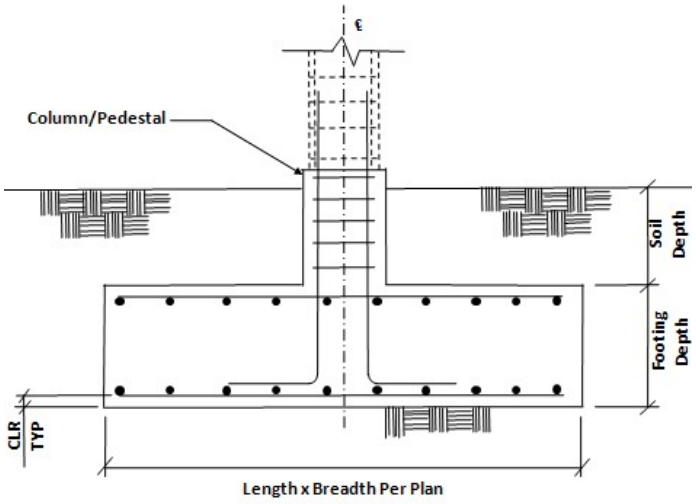
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

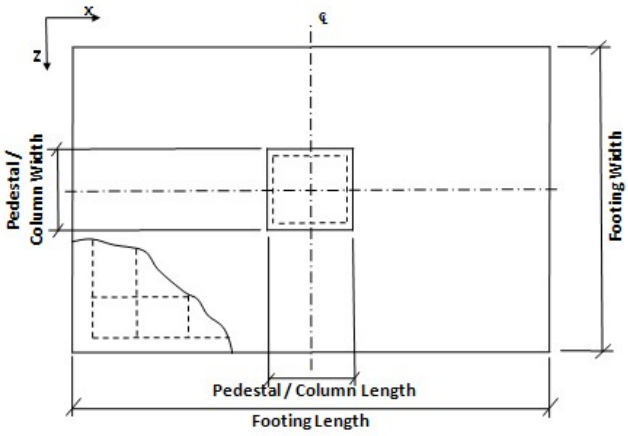
Based on spacing reinforcement increment; provided reinforcement is

Ø8 @ 55 mm o.c.

Isolated Footing 146



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1225.452	35.561	27.862	65.216	-78.822
102	980.361	28.449	22.290	52.173	-63.057

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1838.177	53.341	41.793	97.824	-118.233
202	1176.434	34.138	26.748	62.607	-75.669
203	1470.542	42.673	33.435	78.259	-94.586
204	1102.906	32.005	25.076	58.694	-70.940

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

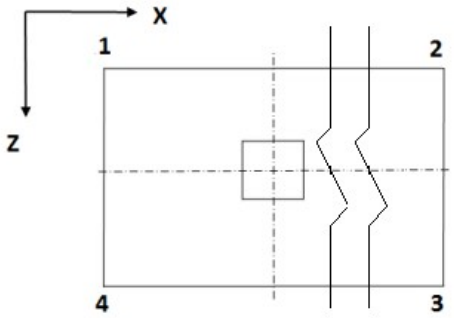
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.331 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.000	m	Governing Load Case :	# 101
Width (W_2) =	4.000	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.000	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**68.8984**	85.7115	99.5330	82.7200	0.000
101	68.8984	**85.7115**	99.5330	82.7200	0.000
101	68.8984	85.7115	**99.5330**	82.7200	0.000
101	68.8984	85.7115	99.5330	**82.7200**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**68.8984**	85.7115	99.5330	82.7200
101	68.8984	**85.7115**	99.5330	82.7200
101	68.8984	85.7115	**99.5330**	82.7200
101	68.8984	85.7115	99.5330	**82.7200**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 16.000 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	18.946	24.181	36.558	30.054

102	19.375	24.728	37.386	30.734

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 35.561 kN
Governing Restoring Force : 673.726 kN
Minimum Sliding Ratio for the Critical Load Case : 18.946
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 73.714 kNm
Governing Resisting Moment : 2694.854 kNm
Minimum Overturning Ratio for the Critical Load Case : 36.558

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 27.862 kN
Governing Restoring Force : 673.726 kN
Minimum Sliding Ratio for the Critical Load Case : 24.181
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -89.668 kNm
Governing Resisting Moment : 2694.854 kNm
Minimum Overturning Ratio for the Critical Load Case : 30.054

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 633.198 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1707.014641 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 573.043 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

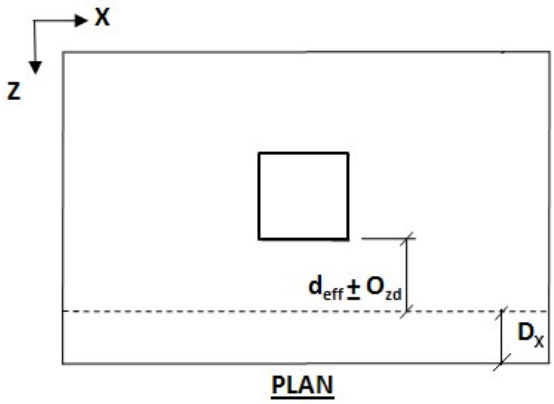
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1707.014641 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 656.524 kN

Shear Stress(T_v) = 363.121523 kN/m2

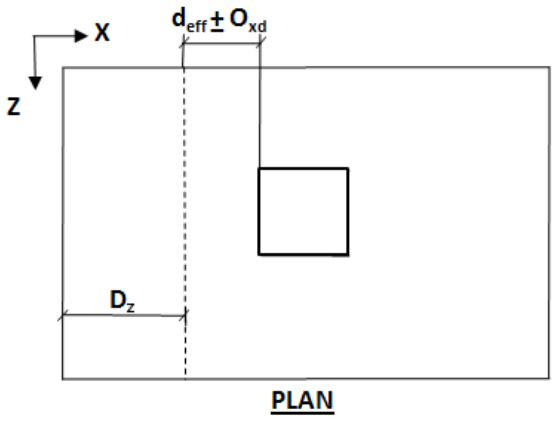
Percentage Of Steel(P_t) = 0.2534

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 366.921 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 616.819 kN

Shear Stress(T_v) = 341.161209 kN/m2

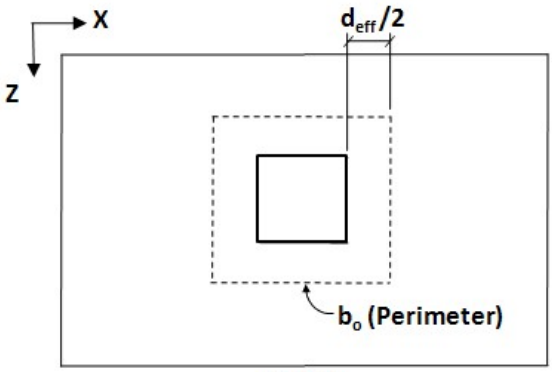
Percentage Of Steel(P_t) = 0.2814

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 383.927 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1735.205 kN

Shear Stress(T_v) = 1008.127 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.650 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

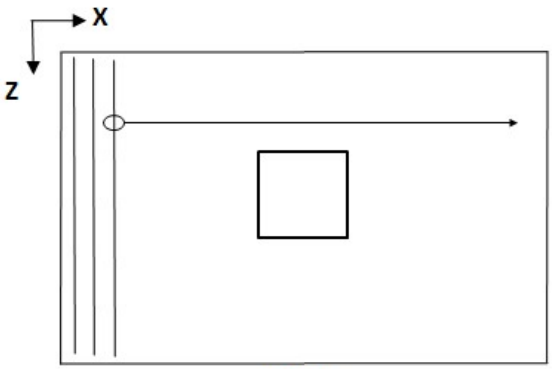
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.750 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1953.600 mm2

Calculated Area of Steel (A_{st}) = 5315.504 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5315.504 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 58.060 mm

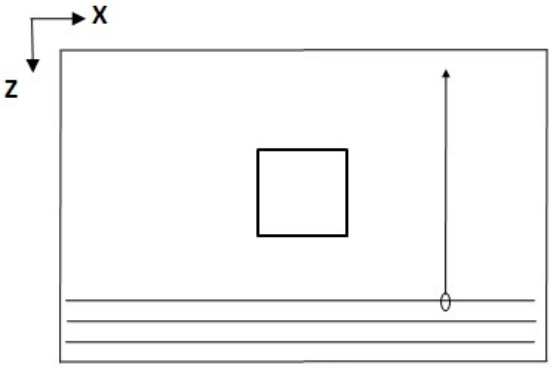
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1953.600 mm2

Calculated Area of Steel (A_{st}) = 4778.225 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4778.225 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 64.833 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

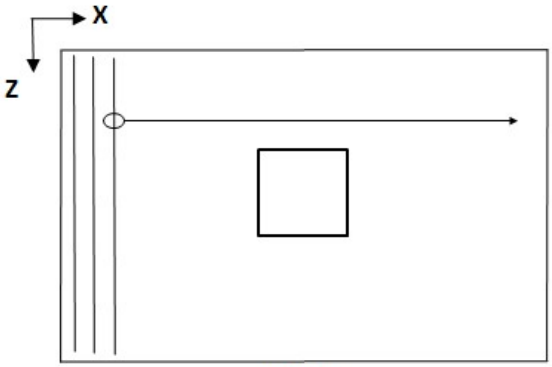
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1953.600 mm2

Calculated Area of Steel (A_{st}) = 1704.000 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1953.600 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 54.899 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 64.900 mm

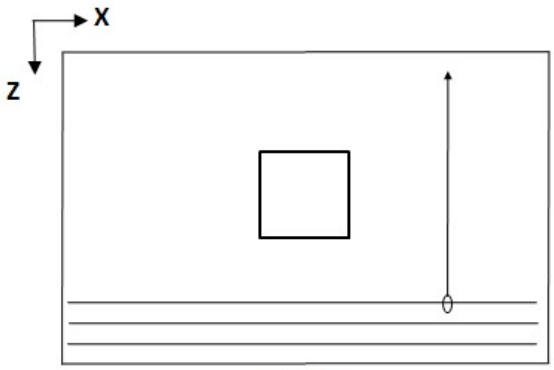
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 60 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1953.600 mm2

Calculated Area of Steel (A_{st}) = 4182.016 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4182.016 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 48.969 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 73.396 mm

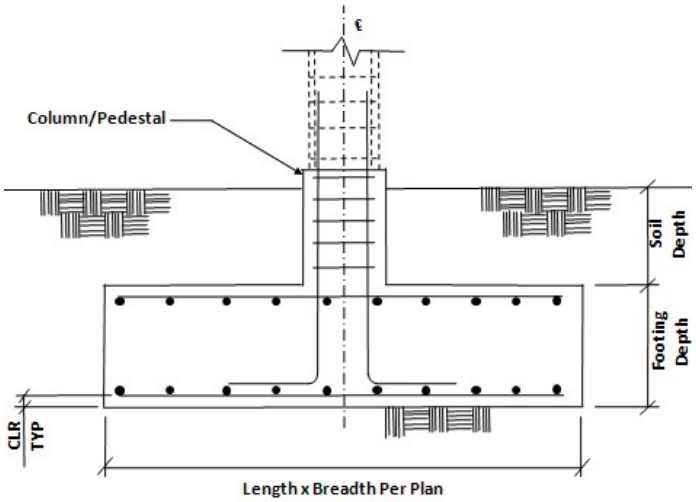
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

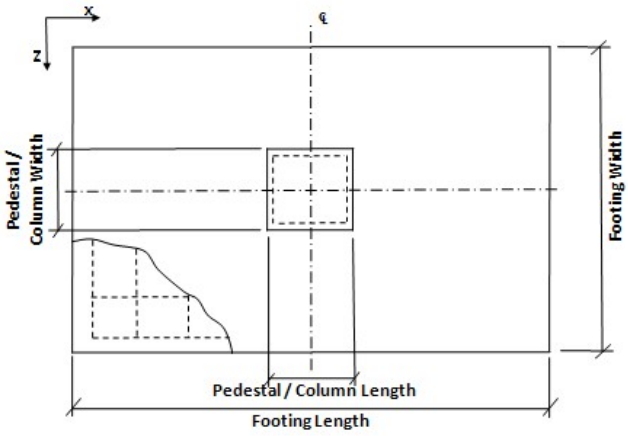
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 147



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1251.754	35.976	27.957	65.340	-79.359
102	1001.403	28.781	22.365	52.272	-63.487

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1877.631	53.964	41.935	98.010	-119.038
202	1201.684	34.537	26.838	62.726	-76.184
203	1502.105	43.171	33.548	78.408	-95.230
204	1126.579	32.378	25.161	58.806	-71.423

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

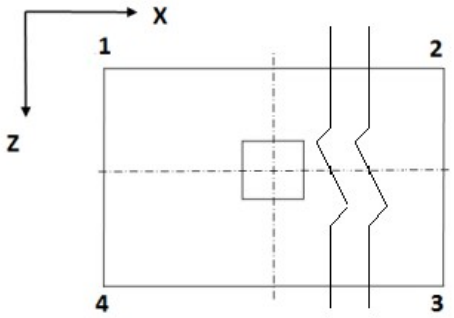
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.594 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.050	m	Governing Load Case :	# 101
Width (W_2) =	4.050	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.403	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**69.1091**	85.4270	98.7705	82.4527	0.000
101	69.1091	**85.4270**	98.7705	82.4527	0.000
101	69.1091	85.4270	**98.7705**	82.4527	0.000
101	69.1091	85.4270	98.7705	**82.4527**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**69.1091**	85.4270	98.7705	82.4527
101	69.1091	**85.4270**	98.7705	82.4527
101	69.1091	85.4270	**98.7705**	82.4527
101	69.1091	85.4270	98.7705	**82.4527**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A

Plan area of footing = 16.403 sq.m

Area not in contact with soil = 0.000 sq.m

% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	19.135	24.624	37.744	30.864

102	19.570	25.184	38.601	31.565

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 35.976 kN
Governing Restoring Force : 688.412 kN
Minimum Sliding Ratio for the Critical Load Case : 19.135
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 73.867 kNm
Governing Resisting Moment : 2788.016 kNm
Minimum Overturning Ratio for the Critical Load Case : 37.744

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 27.957 kN
Governing Restoring Force : 688.412 kN
Minimum Sliding Ratio for the Critical Load Case : 24.624
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -90.331 kNm
Governing Resisting Moment : 2788.016 kNm
Minimum Overturning Ratio for the Critical Load Case : 30.864

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 655.436 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 594.080 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

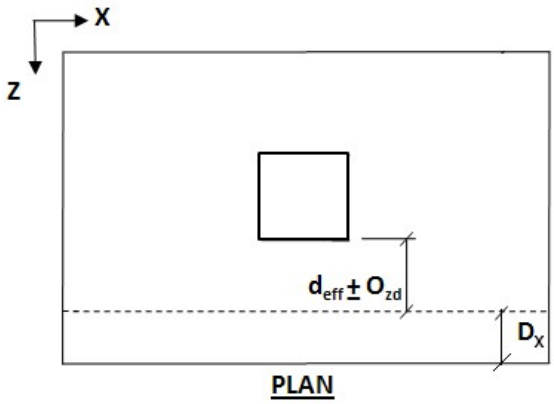
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 673.324 kN

Shear Stress(T_v) = 367.815853 kN/m2

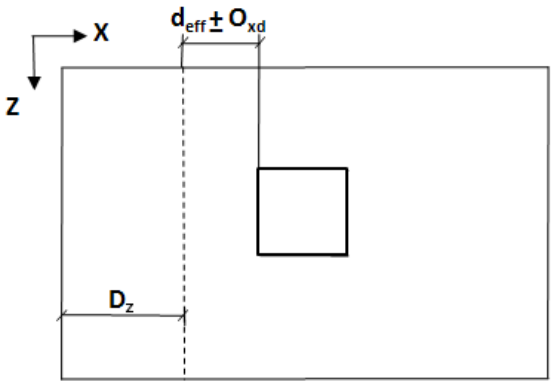
Percentage Of Steel(P_t) = 0.2598

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 370.882 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 633.443 kN

Shear Stress(T_v) = 346.030269 kN/m2

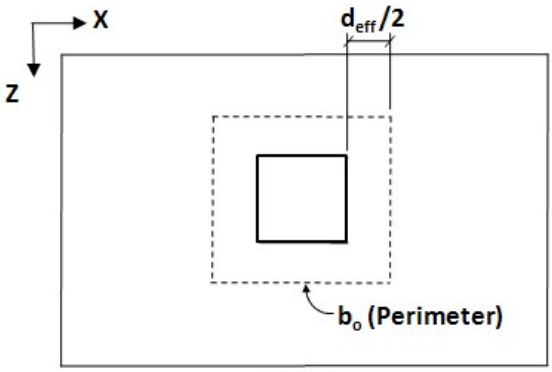
Percentage Of Steel(P_t) = 0.2880

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 387.790 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1775.029 kN

Shear Stress(T_v) = 1031.265 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

$K_s \times T_c$ = 1250.0000 kN/m2

$T_v <= K_s \times T_c$ hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.675 m

$l_{db} >= l_d$ hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

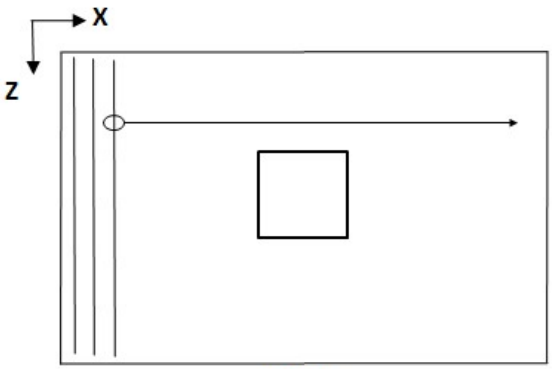
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.775 m

$l_{db} >= l_d$ hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 5511.017 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5511.017 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 56.286 mm

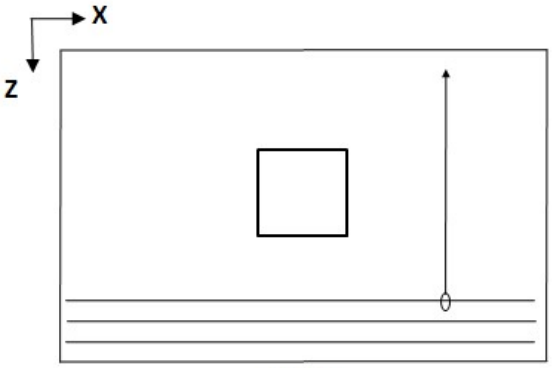
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4961.190 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4961.190 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 62.540 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

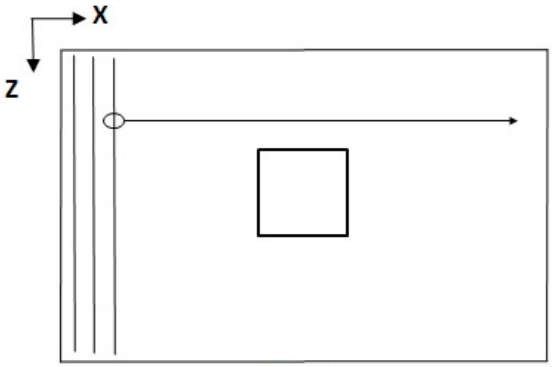
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 1725.300 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1978.020 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 57.140 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 64.656 mm

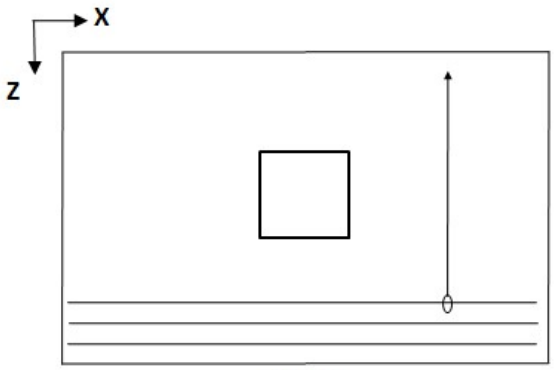
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 60 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin})	= 1978.020 mm2
Calculated Area of Steel (A_{st})	= 4342.302 mm2
Provided Area of Steel ($A_{st,Provided}$)	= 4342.302 mm2
A_{stmin} <= $A_{st,Provided}$	Steel area is accepted
Governing Moment	= 51.050 kNm

Selected bar Size (d_b)	= Ø10
Minimum spacing allowed (S_{min}) =	= 50.000 mm
Selected spacing (S)	= 71.636 mm

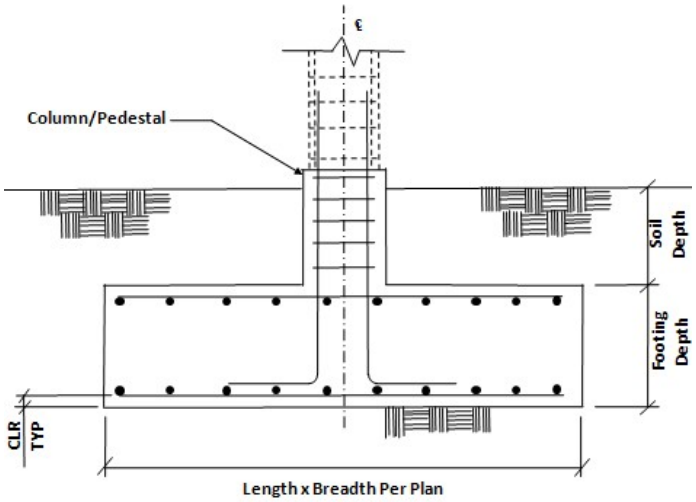
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

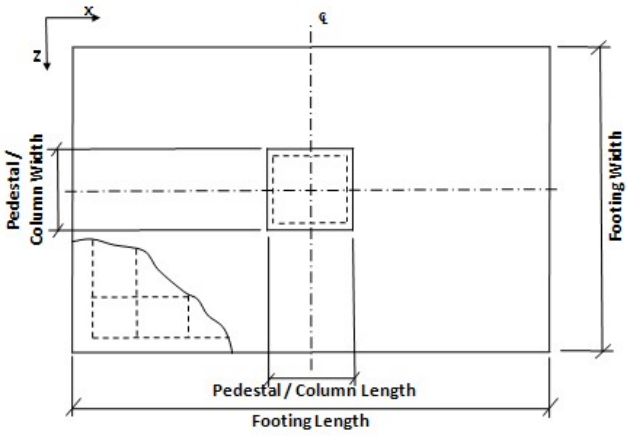
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 148



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1251.661	36.171	27.965	65.351	-79.618
102	1001.329	28.936	22.372	52.281	-63.694

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1877.492	54.256	41.948	98.026	-119.427
202	1201.595	34.724	26.847	62.737	-76.433
203	1501.994	43.405	33.558	78.421	-95.541
204	1126.495	32.553	25.169	58.816	-71.656

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

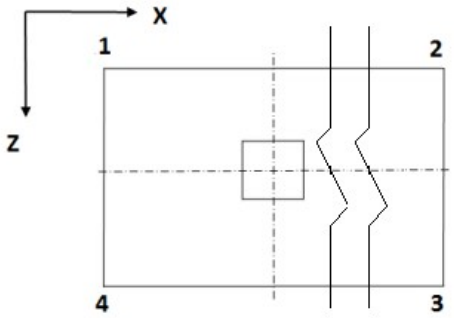
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.593 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.050	m	Governing Load Case :	# 101
Width (W_2) =	4.050	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.403	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**69.0735**	85.4488	98.7949	82.4195	0.000
101	69.0735	**85.4488**	98.7949	82.4195	0.000
101	69.0735	85.4488	**98.7949**	82.4195	0.000
101	69.0735	85.4488	98.7949	**82.4195**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**69.0735**	85.4488	98.7949	82.4195
101	69.0735	**85.4488**	98.7949	82.4195
101	69.0735	85.4488	**98.7949**	82.4195
101	69.0735	85.4488	98.7949	**82.4195**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 16.403 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	19.031	24.615	37.734	30.754

102	19.463	25.174	38.591	31.452

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 36.171 kN
Governing Restoring Force : 688.365 kN
Minimum Sliding Ratio for the Critical Load Case : 19.031
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 73.880 kNm
Governing Resisting Moment : 2787.828 kNm
Minimum Overturning Ratio for the Critical Load Case : 37.734

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 27.965 kN
Governing Restoring Force : 688.365 kN
Minimum Sliding Ratio for the Critical Load Case : 24.615
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -90.650 kNm
Governing Resisting Moment : 2787.828 kNm
Minimum Overturning Ratio for the Critical Load Case : 30.754

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 655.397 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 594.188 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

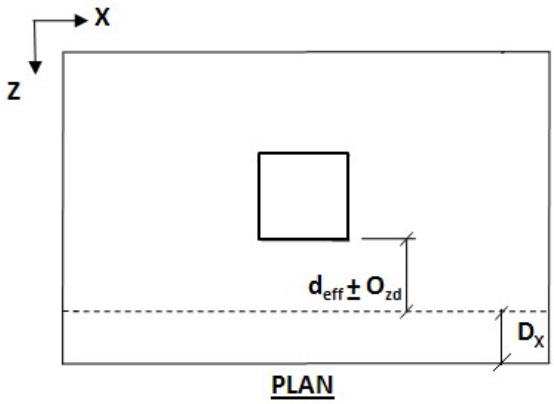
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 673.283 kN

Shear Stress(T_v) = 367.793696 kN/m2

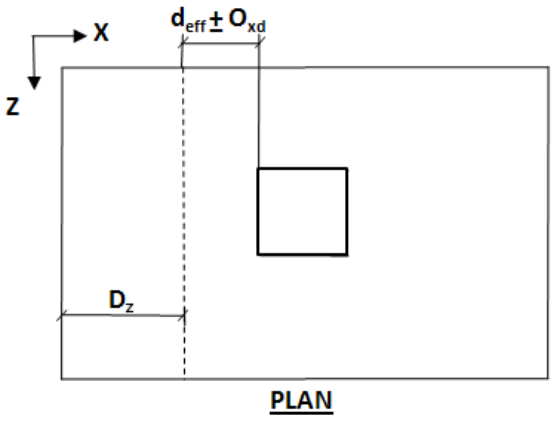
Percentage Of Steel(P_t) = 0.2598

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 370.912 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 633.552 kN

Shear Stress(T_v) = 346.089639 kN/m2

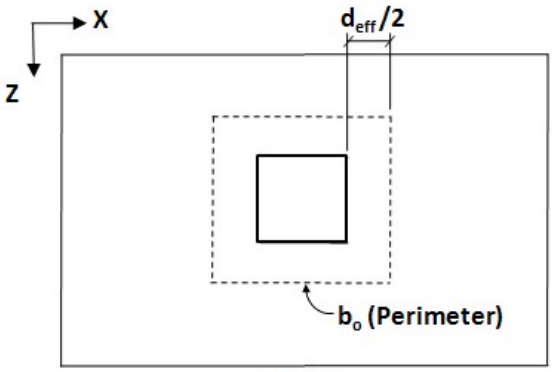
Percentage Of Steel(P_t) = 0.2880

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 387.779 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1774.898 kN

Shear Stress(T_v) = 1031.188 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.675 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

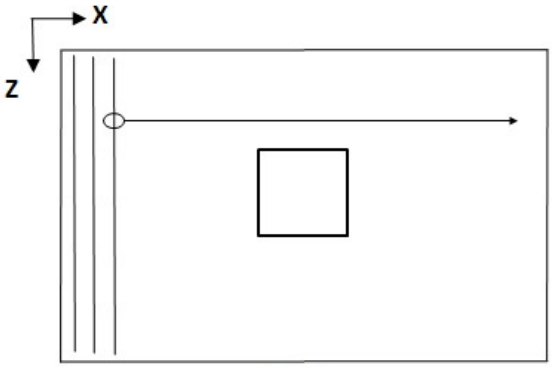
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.775 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 5510.665 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5510.665 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 56.286 mm

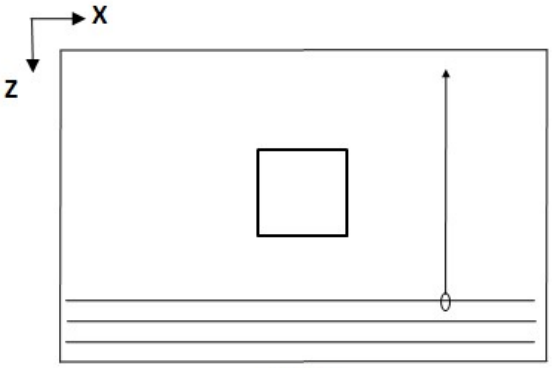
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4962.151 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4962.151 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 62.540 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

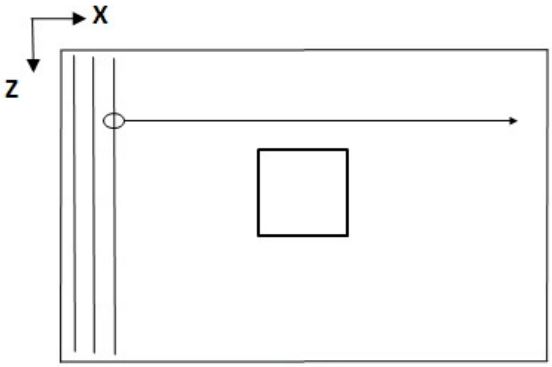
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 1725.300 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1978.020 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 57.140 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 64.656 mm

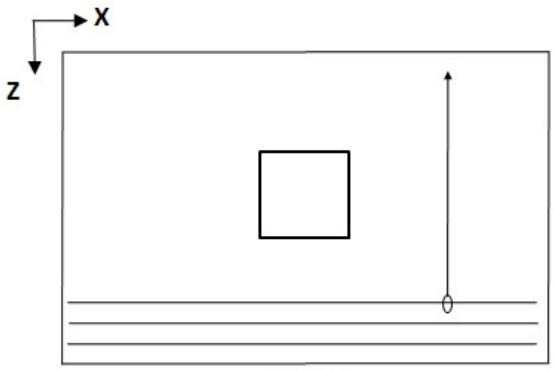
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 60 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4343.145 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4343.145 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 51.050 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 71.636 mm

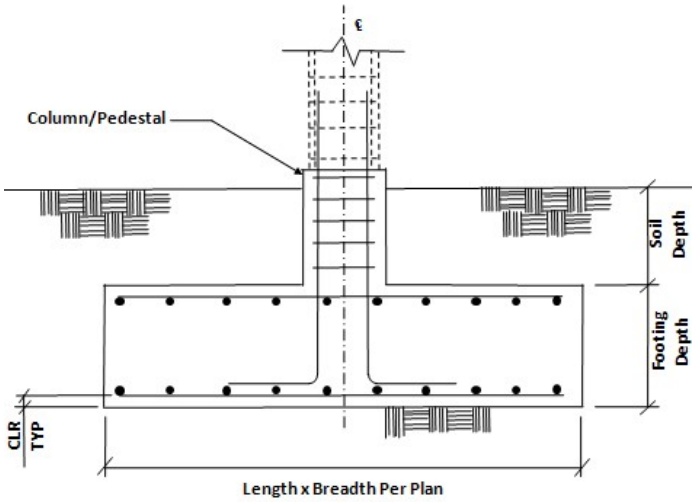
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

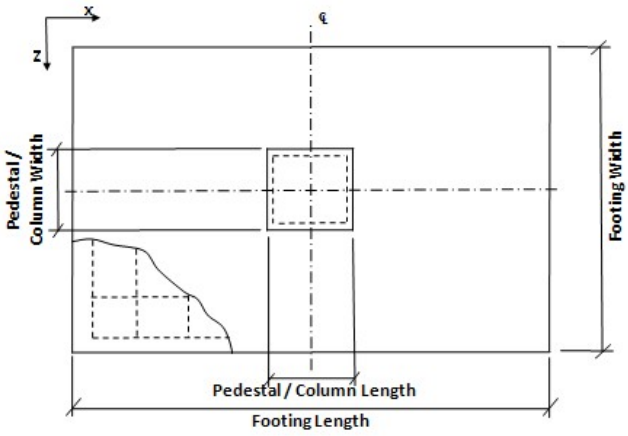
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 149



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1230.739	36.775	27.864	65.232	-80.268
102	984.591	29.420	22.292	52.186	-64.215

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1846.108	55.163	41.797	97.848	-120.403
202	1181.509	35.304	26.750	62.623	-77.058
203	1476.886	44.130	33.437	78.278	-96.322
204	1107.665	33.098	25.078	58.709	-72.242

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

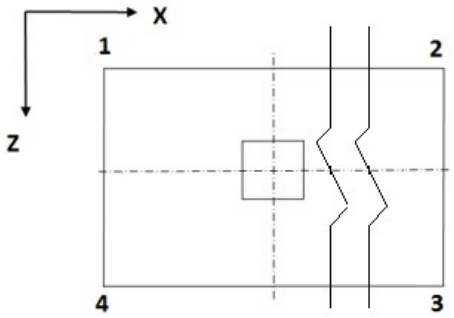
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 12.384 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	4.050	m	Governing Load Case :	# 101
Width (W_2) =	4.050	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	16.403	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**67.7360**	84.2622	97.5812	81.0550	0.000
101	67.7360	**84.2622**	97.5812	81.0550	0.000
101	67.7360	84.2622	**97.5812**	81.0550	0.000
101	67.7360	84.2622	97.5812	**81.0550**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**67.7360**	84.2622	97.5812	81.0550
101	67.7360	**84.2622**	97.5812	81.0550
101	67.7360	84.2622	**97.5812**	81.0550
101	67.7360	84.2622	97.5812	**81.0550**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 16.403 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	18.434	24.329	37.236	30.010

102	18.859	24.890	38.095	30.702

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 36.775 kN
Governing Restoring Force : 677.904 kN
Minimum Sliding Ratio for the Critical Load Case : 18.434
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 73.730 kNm
Governing Resisting Moment : 2745.460 kNm
Minimum Overturning Ratio for the Critical Load Case : 37.236

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 27.864 kN
Governing Restoring Force : 677.904 kN
Minimum Sliding Ratio for the Critical Load Case : 24.329
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -91.485 kNm
Governing Resisting Moment : 2745.460 kNm
Minimum Overturning Ratio for the Critical Load Case : 30.010

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 644.996 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #203

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.352 m

Governing moment (M_u) = 585.356 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

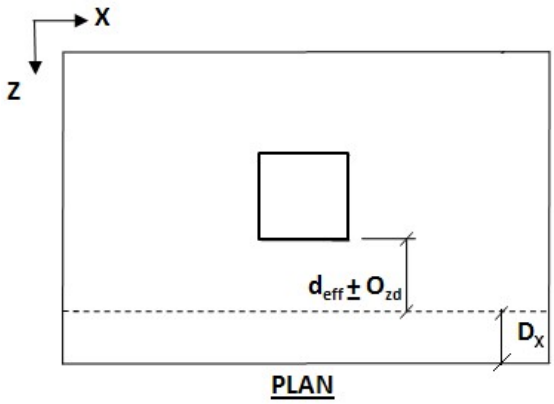
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 1728.352324 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 662.569 kN

Shear Stress(T_v) = 361.940874 kN/m2

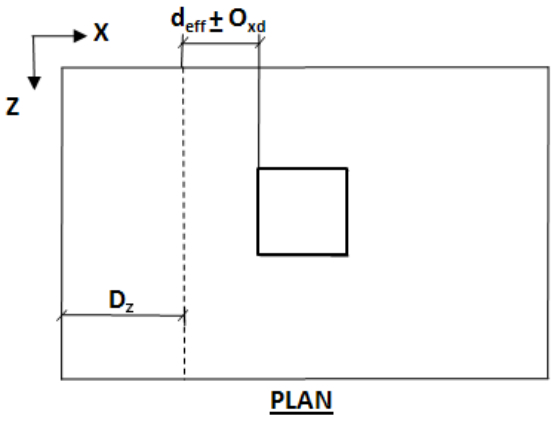
Percentage Of Steel(P_t) = 0.2558

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 368.398 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 624.087 kN

Shear Stress(T_v) = 340.919341 kN/m2

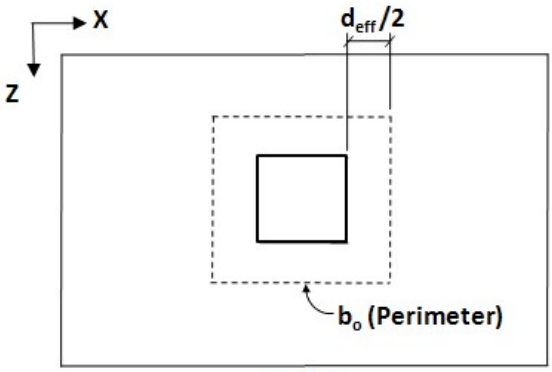
Percentage Of Steel(P_t) = 0.2832

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 384.979 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1745.228 kN

Shear Stress(T_v) = 1013.951 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

$K_s \times T_c$ = 1250.0000 kN/m2

$T_v <= K_s \times T_c$ hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.675 m

$l_{db} >= l_d$ hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

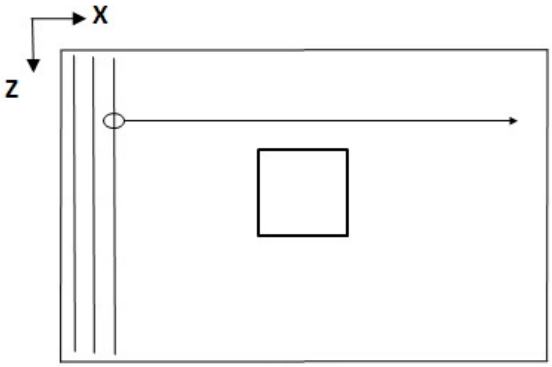
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.775 m

$l_{db} >= l_d$ hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 5416.898 mm2

Provided Area of Steel ($A_{st,Provided}$) = 5416.898 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 57.941 mm

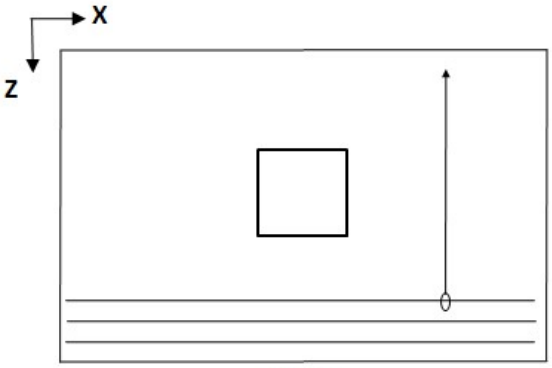
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 55.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #203

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4883.652 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4883.652 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 63.548 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

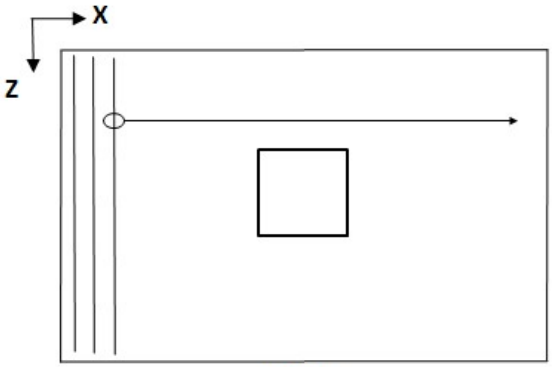
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 1725.300 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1978.020 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 57.140 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 64.656 mm

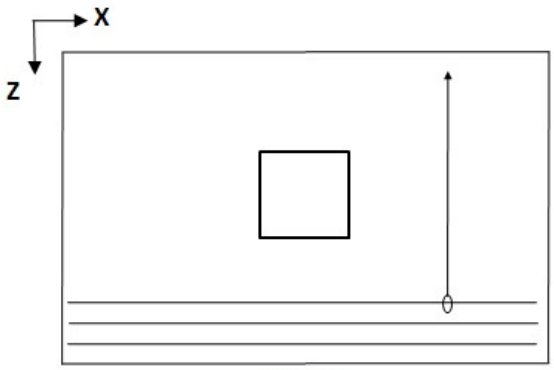
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 60 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1978.020 mm2

Calculated Area of Steel (A_{st}) = 4274.344 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4274.344 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 51.050 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 72.963 mm

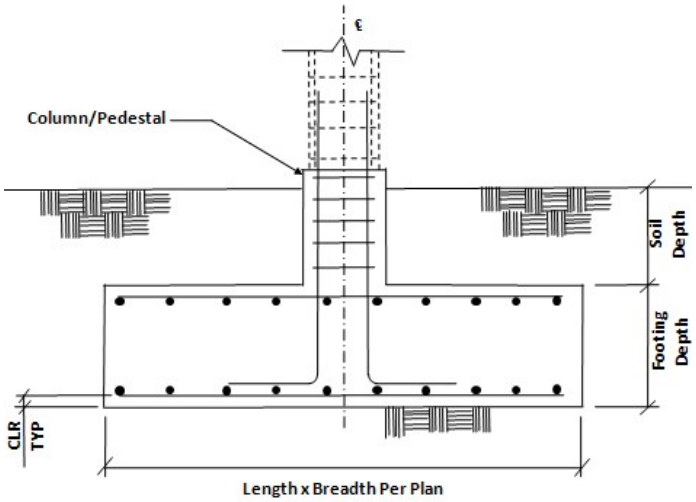
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

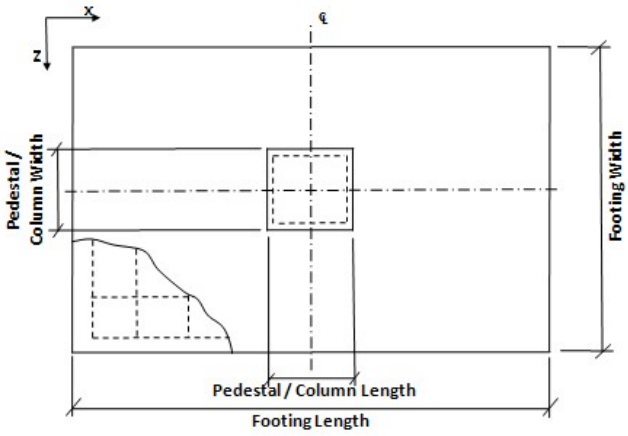
Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 70 mm o.c.

Isolated Footing 150



ELEVATION



PLAN

Input Values

Footing Geomtery

Design Type :	Calculate Dimension
Footing Thickness (Ft) :	305.000 mm
Footing Length - X (Fl) :	1000.000 mm
Footing Width - Z (Fw) :	1000.000 mm
Eccentricity along X (Oxd) :	0.000 mm
Eccentricity along Z (Ozd) :	0.000 mm

Column Dimensions

Column Shape :	Rectangular
Column Length - X (Pl) :	0.600 m
Column Width - Z (Pw) :	0.400 m

Pedestal

Include Pedestal?	No
Pedestal Shape :	N/A
Pedestal Height (Ph) :	N/A
Pedestal Length - X (Pl) :	N/A
Pedestal Width - Z (Pw) :	N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete :	25.000 kN/m3
Strength of Concrete :	25.000 N/mm2
Yield Strength of Steel :	415.000 N/mm2
Minimum Bar Size :	Ø6
Maximum Bar Size :	Ø32
Minimum Bar Spacing :	50.000 mm
Maximum Bar Spacing :	500.000 mm
Pedestal Clear Cover (P, CL) :	50.000 mm
Footing Clear Cover (F, CL) :	50.000 mm

Soil Properties

Soil Type :	Drained
Unit Weight :	22.000 kN/m3
Soil Bearing Capacity :	100.000 kN/m2
Soil Surcharge :	0.000 kN/m2
Depth of Soil above Footing :	0.000 mm
Cohesion :	0.000 kN/m2
Min Percentage of Slab :	0.000

Sliding and Overturning

Coefficient of Friction :	0.500
Factor of Safety Against Sliding :	1.500
Factor of Safety Against Overturning :	1.500

Load Combination/s- Service Stress Level	
Load Combination Number	**Load Combination Title**
101	1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL+1.000 x DL
102	0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL+0.800 x DL
Load Combination/s- Strength Level	
Load Combination Number	**Load Combination Title**

201	1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL+1.500 x DL
202	0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL+0.960 x DL
203	1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL+1.200 x DL
204	0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL+0.900 x DL

Applied Loads - Service Stress Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	1084.647	31.972	28.727	63.886	-77.389
102	867.717	25.577	22.982	51.109	-61.911

Applied Loads - Strength Level

LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	1626.970	47.957	43.091	95.829	-116.084
202	1041.261	30.693	27.578	61.331	-74.293
203	1301.576	38.366	34.472	76.663	-92.867
204	976.182	28.774	25.854	57.498	-69.650

Design Calculations

Footing Size

Initial Length (L_o) = 1.000 m

Initial Width (W_o) = 1.000 m

Uplift force due to buoyancy = 0.000 kN

Effect due to adhesion = 0.000 kN

Area from initial length and width, $A_o = L_o \times W_o$ = 1.000 m^2

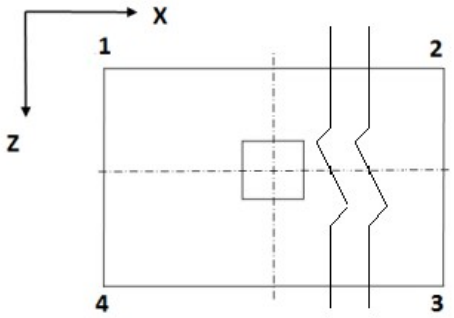
Min. area required from bearing pressure, $A_{min} = P / q_{max}$ = 10.923 m^2

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	3.850	m	Governing Load Case :	# 101
Width (W_2) =	3.850	m	Governing Load Case :	# 101
Depth (D_2) =	0.507	m	Governing Load Case :	# 101
Area (A_2) =	14.823	m^2		

Pressures at Four Corner



Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**64.0003**	82.3245	97.6012	79.2770	0.000
101	64.0003	**82.3245**	97.6012	79.2770	0.000
101	64.0003	82.3245	**97.6012**	79.2770	0.000
101	64.0003	82.3245	97.6012	**79.2770**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of adjusted Pressures at Four Corner

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**64.0003**	82.3245	97.6012	79.2770
101	64.0003	**82.3245**	97.6012	79.2770
101	64.0003	82.3245	**97.6012**	79.2770
101	64.0003	82.3245	97.6012	**79.2770**

Details of Out-of-Contact Area (If Any)

Governing load case = N/A
Plan area of footing = 14.823 sq.m
Area not in contact with soil = 0.000 sq.m
% of total area not in contact = 0.000%

Check For Stability Against Overturning And Sliding

-	Factor of safety against sliding		Factor of safety against overturning	
Load Case No.	Along X-Direction	Along Z-Direction	About X-Direction	About Z-Direction
101	18.730	20.846	31.735	26.457

102	19.172	21.337	32.484	27.081

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding X Direction

Critical Load Case for Sliding along X-Direction : 101
Governing Disturbing Force : 31.972 kN
Governing Restoring Force : 598.834 kN
Minimum Sliding Ratio for the Critical Load Case : 18.730
Critical Load Case for Overturning about X-Direction : 101
Governing Overturning Moment : 72.648 kNm
Governing Resisting Moment : 2305.470 kNm
Minimum Overturning Ratio for the Critical Load Case : 31.735

Critical Load Case And The Governing Factor Of Safety For Overturning and Sliding Z Direction

Critical Load Case for Sliding along Z-Direction : 101
Governing Disturbing Force : 28.727 kN
Governing Restoring Force : 598.834 kN
Minimum Sliding Ratio for the Critical Load Case : 20.846
Critical Load Case for Overturning about Z-Direction : 101
Governing Overturning Moment : -87.140 kNm
Governing Resisting Moment : 2305.470 kNm
Minimum Overturning Ratio for the Critical Load Case : 26.457

Moment Calculation

Check Trial Depth against moment (w.r.t. X Axis)

Critical Load Case = #204

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 404.866 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 842.080760 kNm

M_u <= M_{umax} hence, safe

Check Trial Depth against moment (w.r.t. Z Axis)

Critical Load Case = #204

Effective Depth = $D - (cc + 0.5 \times d_b)$ = 0.252 m

Governing moment (M_u) = 364.909 kNm

As Per IS 456 2000 ANNEX G G-1.1C

Limiting Factor1 (K_{umax}) = $\frac{700}{(1100 + 0.87 \times f_y)}$ = 0.479107

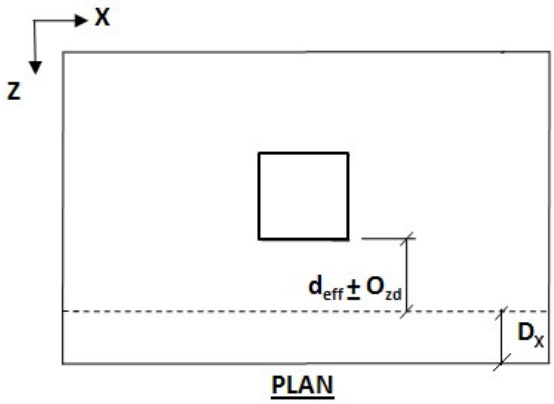
Limiting Factor2 (R_{umax}) = $0.36 \times f_{ck} \times k_{umax} \times (1 - 0.42 \times kumax)$ = 3444.291146 kN/m2

Limit Moment Of Resistance (M_{umax}) = $R_{umax} \times B \times d_e^2$ = 842.080760 kNm

M_u <= M_{umax} hence, safe

Shear Calculation

Check Trial Depth for one way shear (Along X Axis)
(Shear Plane Parallel to X Axis)



Critical Load Case = #201

D_X = 0.452 m

Shear Force(S) = 575.543 kN

Shear Stress(T_v) = 330.733983 kN/m2

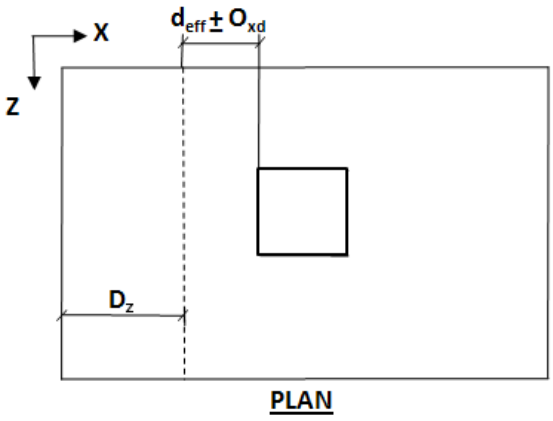
Percentage Of Steel(P_t) = 0.2224

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 346.559 kN/m2

T_v< T_c hence, safe

Check Trial Depth for one way shear (Along Z Axis)
(Shear Plane Parallel to Z Axis)



PLAN

Critical Load Case = #201

D_Z = 0.452 m

Shear Force(S) = 538.853 kN

Shear Stress(T_v) = 309.649969 kN/m2

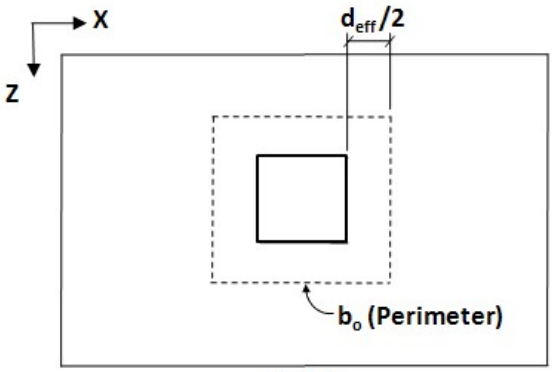
Percentage Of Steel(P_t) = 0.2478

As Per IS 456 2000 Clause 40 Table 19

Shear Strength Of Concrete(T_c) = 363.352 kN/m2

$T_v < T_c$ hence, safe

Check Trial Depth for two way shear



PLAN

Critical Load Case = #201

Shear Force(S) = 1528.589 kN

Shear Stress(T_v) = 888.086 kN/m2

As Per IS 456 2000 Clause 31.6.3.1

$K_s = \min[(0.5 + \beta), 1] = 1.000$

Shear Strength(T_c)= $0.25 \times \sqrt{f_{ck}}$ = 1250.0000 kN/m2

K_s x T_c = 1250.0000 kN/m2

T_v<= K_s x T_c hence, safe

Reinforcement Calculation

Calculation of Maximum Bar Size

Along X Axis

Bar diameter corresponding to max bar size (d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(B - b)}{2} - cc\right]$ = 1.575 m

l_{db} >=l_d hence, safe

Along Z Axis

Bar diameter corresponding to max bar size(d_b) = 32 mm

As Per IS 456 2000 Clause 26.2.1

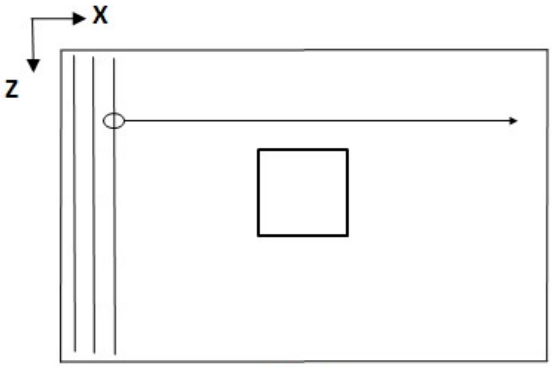
Development Length(l_d) = $\frac{d_b \times 0.87 \times f_y}{4 \times \tau_{bd}}$ = 1.289 m

Allowable Length(l_{db}) = $\left[\frac{(H - h)}{2} - cc\right]$ = 1.675 m

l_{db} >=l_d hence, safe

Bottom Reinforcement Design

Along Z Axis



PLAN

For moment w.r.t. X Axis (M_x)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #204

Minimum Area of Steel (A_{stmin}) = 1418.340 mm2

Calculated Area of Steel (A_{st}) = 4852.855 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4852.855 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 61.311 mm

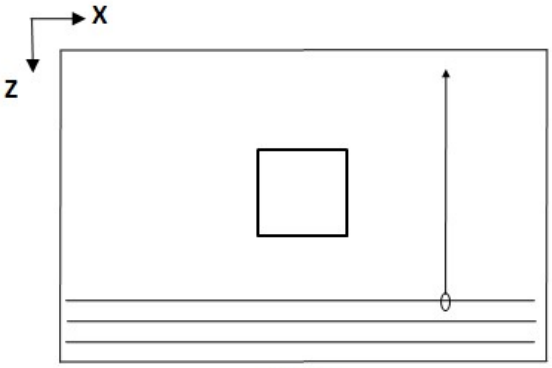
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 60.000 mm o.c.

Along X Axis



PLAN

For moment w.r.t. Z Axis (M_z)

As Per IS 456 2000 Clause 26.5.2.1

Critical Load Case = #204

Minimum Area of Steel (A_{stmin}) = 1418.340 mm2

Calculated Area of Steel (A_{st}) = 4331.798 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4331.798 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 68.000 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

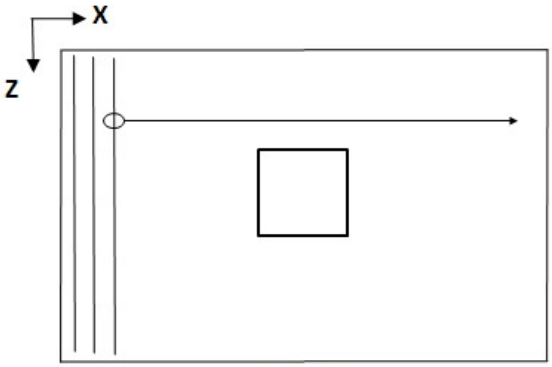
The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 65.000 mm o.c.

Top Reinforcement Design

Along Z Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1418.340 mm2

Calculated Area of Steel (A_{st}) = 1409.100 mm2

Provided Area of Steel ($A_{st,Provided}$) = 1418.340 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 48.529 kNm

Selected bar Size (d_b) = Ø6

Minimum spacing allowed (S_{min}) = 50.000 mm

Selected spacing (S) = 76.408 mm

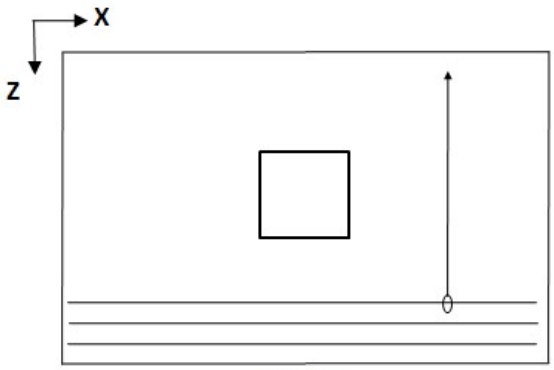
S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø6 @ 75 mm o.c.

Along X Axis



PLAN

Minimum Area of Steel (A_{stmin}) = 1418.340 mm2

Calculated Area of Steel (A_{st}) = 4331.798 mm2

Provided Area of Steel ($A_{st,Provided}$) = 4331.798 mm2

A_{stmin} <= $A_{st,Provided}$ Steel area is accepted

Governing Moment = 43.065 kNm

Selected bar Size (d_b) = Ø10

Minimum spacing allowed (S_{min}) = = 50.000 mm

Selected spacing (S) = 68.000 mm

S_{min} <= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

Ø10 @ 65 mm o.c.

Print Calculation Sheet